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Monteuuis et al.

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(54) **MISBEHAVIOR DETECTION SERVICE FOR SHARING CONNECTED AND SENSED OBJECTS**

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(21) Appl. No.: **17/931,058**

(57) **ABSTRACT**

(22) Filed: **Sep. 9, 2022**

Disclosed are systems, apparatuses, processes, and computer-readable media for wireless communications. For example, an example of a process includes determining, at the first network device, an estimated location of a second network device. The process may further include comparing, at the first network device, the estimated location with an expected location for the second network device. The process may include determining, at the first network device, whether the second network device is a misbehaving device based on the comparing. The process may further include generating, at the first network device, a report based on the determining of whether the second device is a misbehaving device.

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US 2024/0089903 A1 Mar. 14, 2024

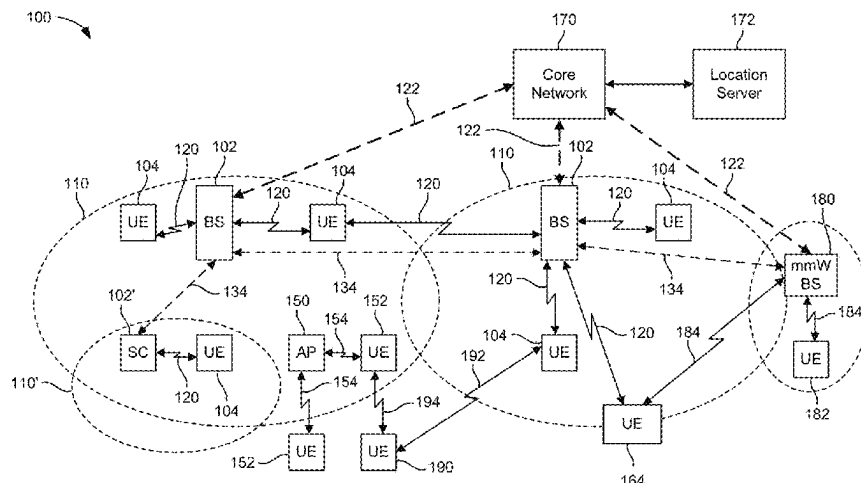
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H04W 4/40 (2018.01)
H04W 12/121 (2021.01)
H04W 64/00 (2009.01)

(52) **U.S. Cl.**
CPC **H04W 64/003** (2013.01); **H04W 4/40** (2018.02); **H04W 12/121** (2021.01)

(58) **Field of Classification Search**
CPC H04W 4/46; H04W 4/70; H04W 4/06; H04W 4/40; H04W 4/029; H04W 84/18; H04W 36/32; H04W 84/06; H04W 4/38; H04W 92/02

See application file for complete search history.

26 Claims, 26 Drawing Sheets



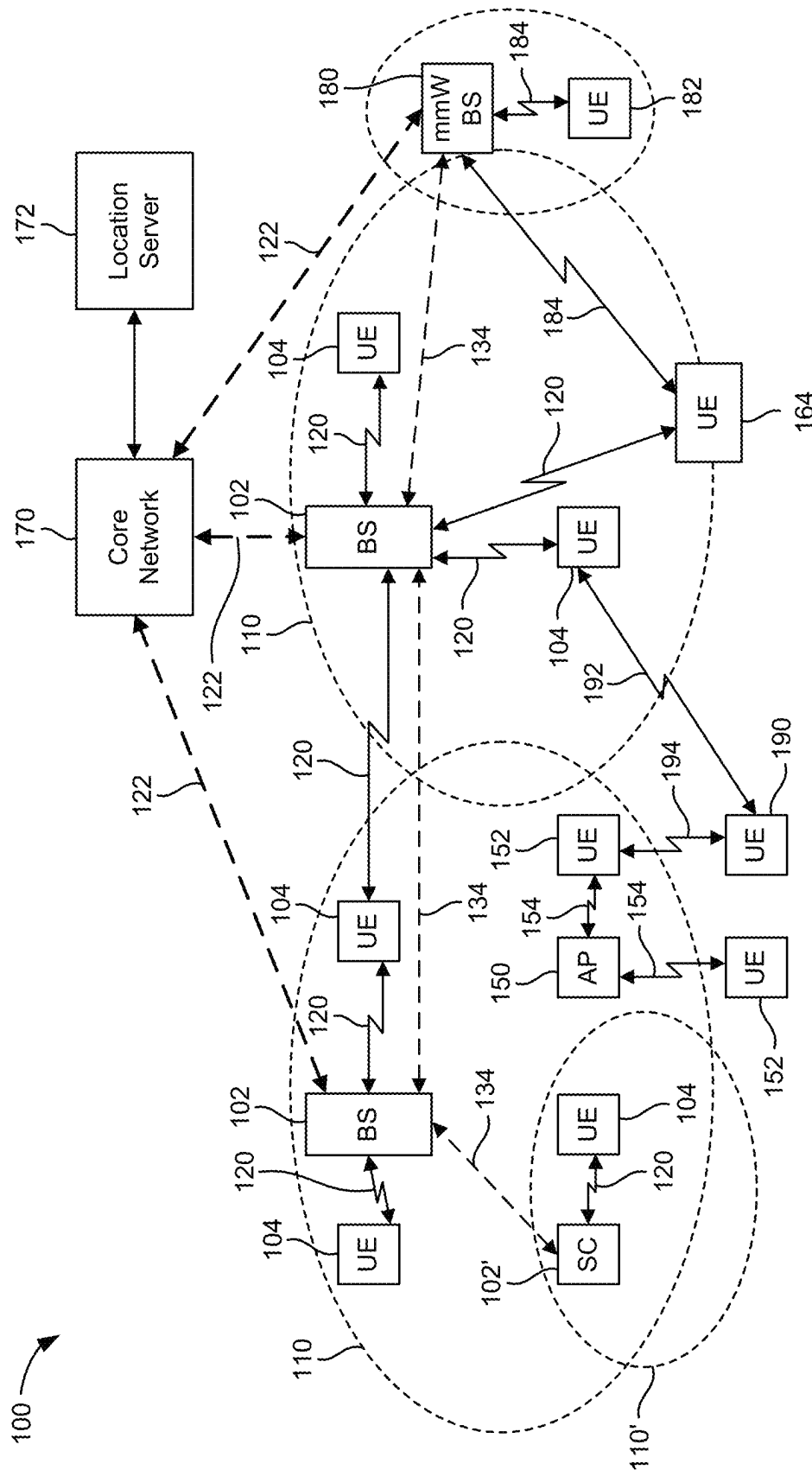


Fig. 1

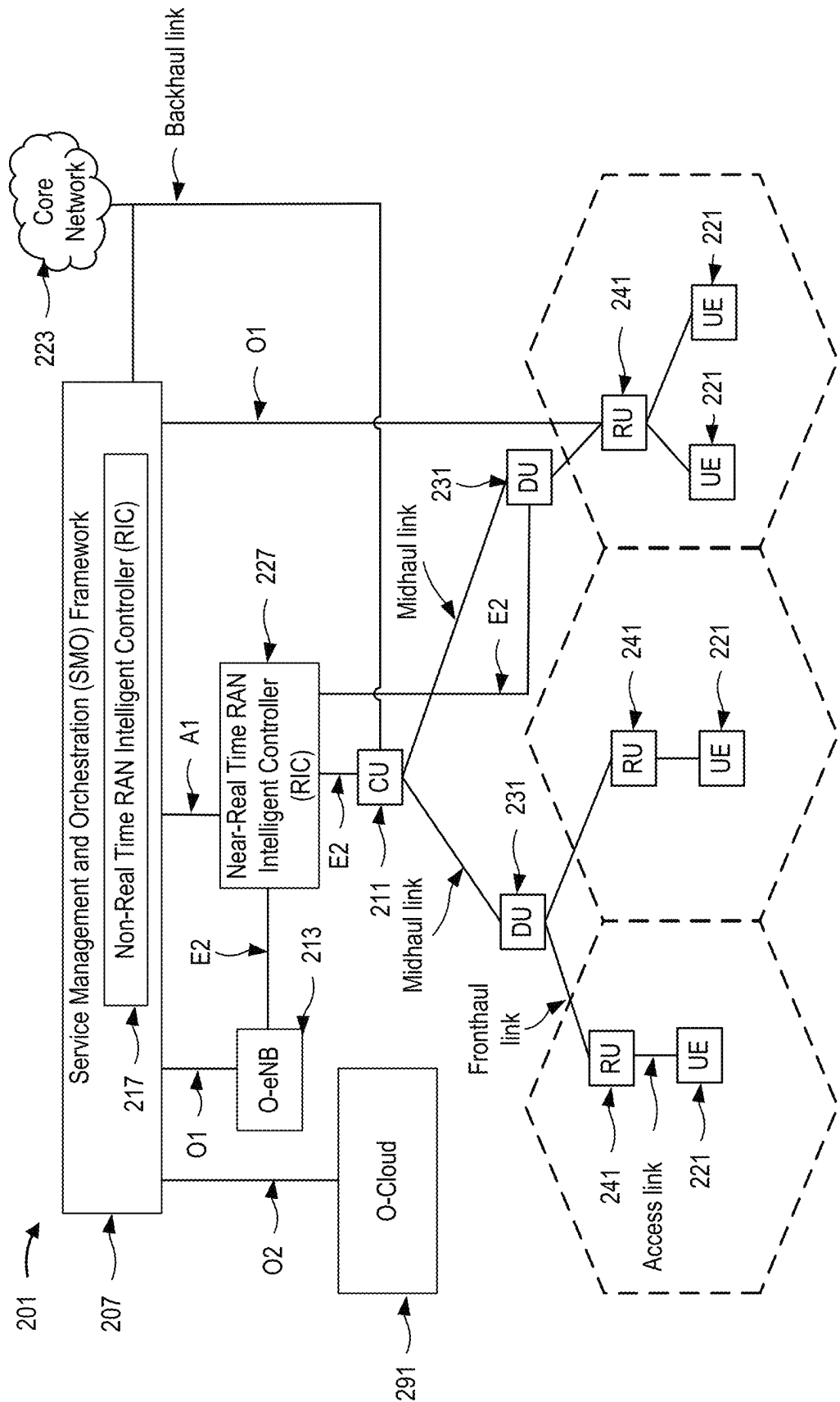


FIG. 2

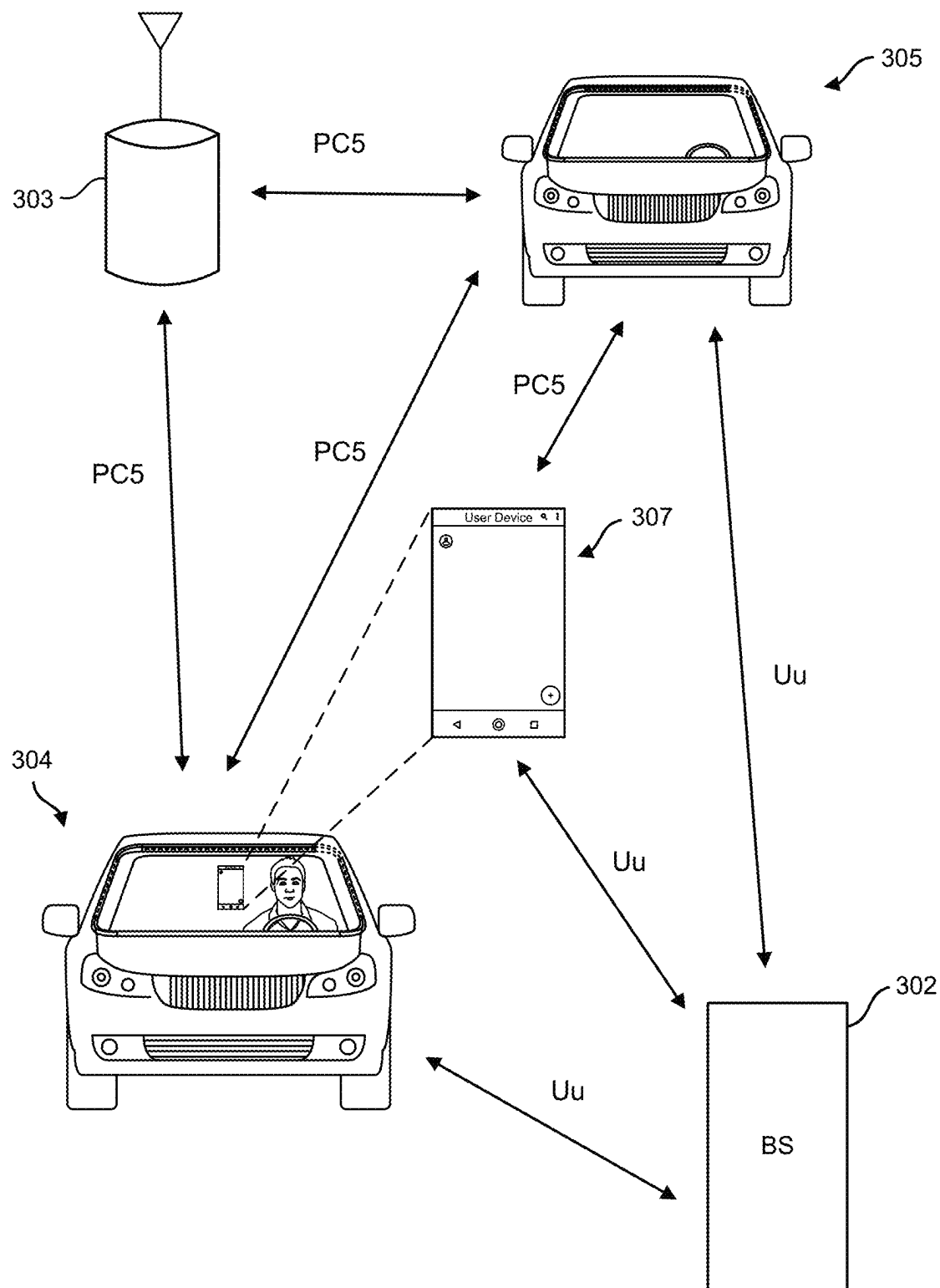
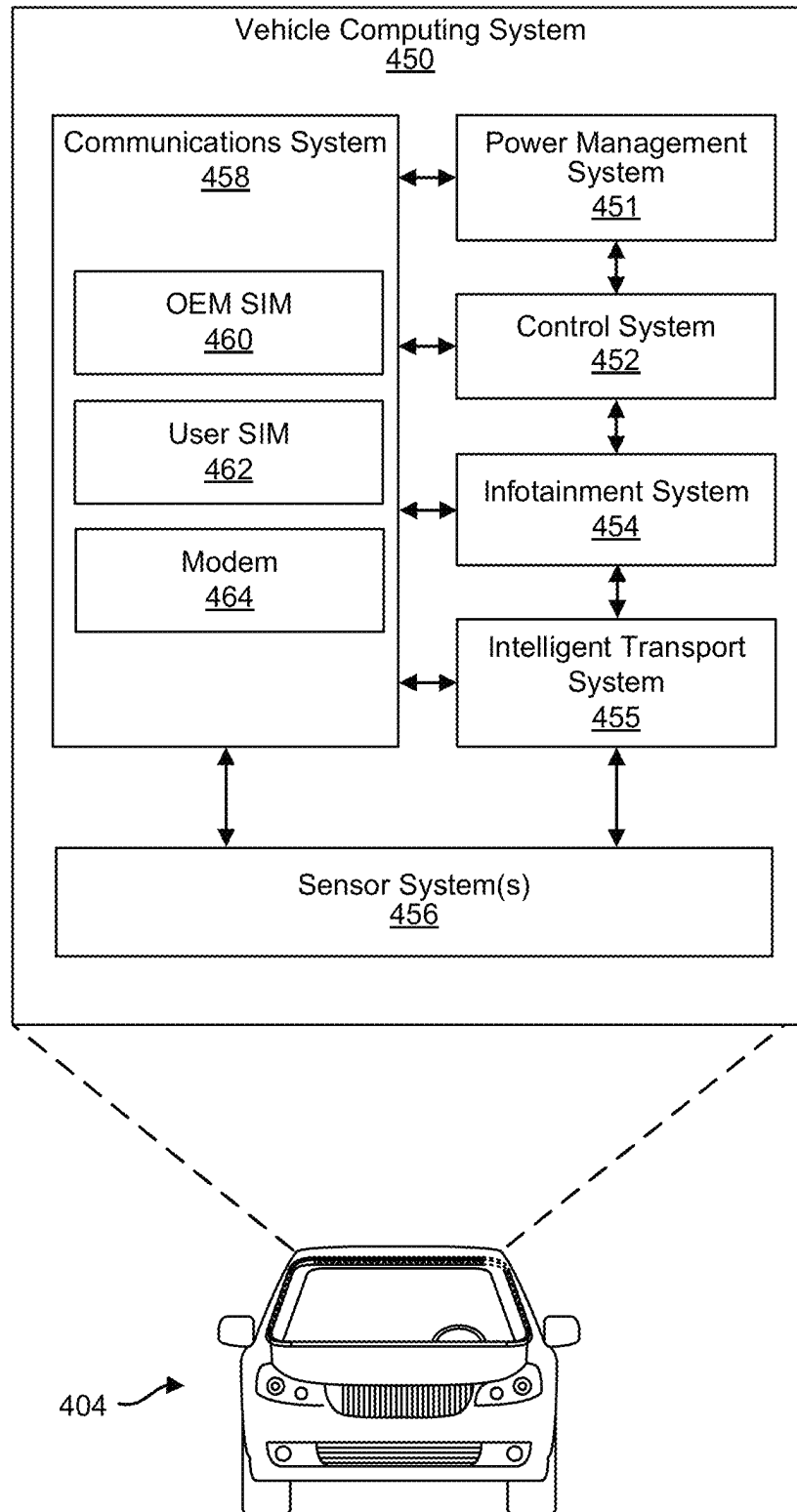
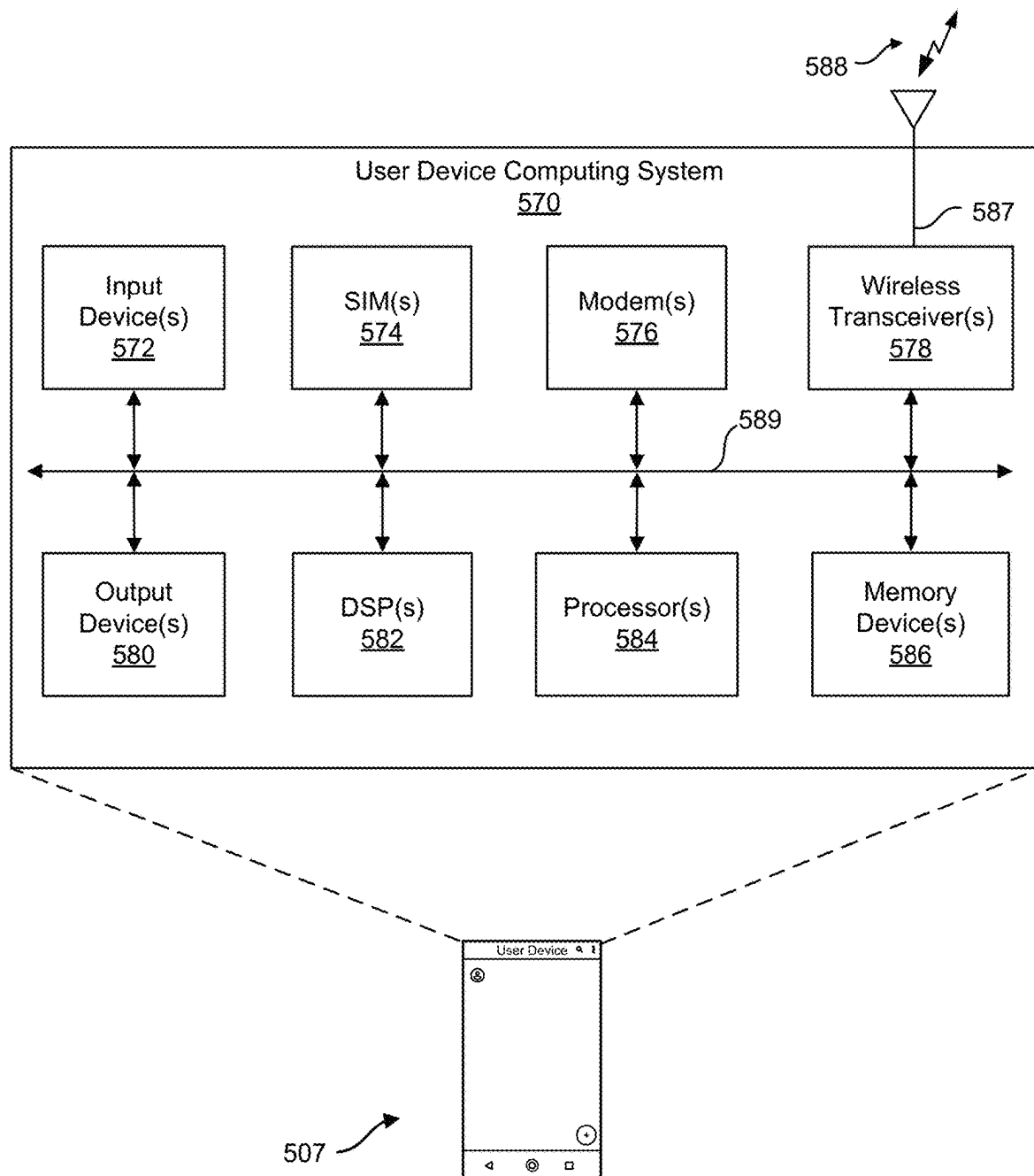


FIG. 3

**FIG. 4**

**FIG. 5**

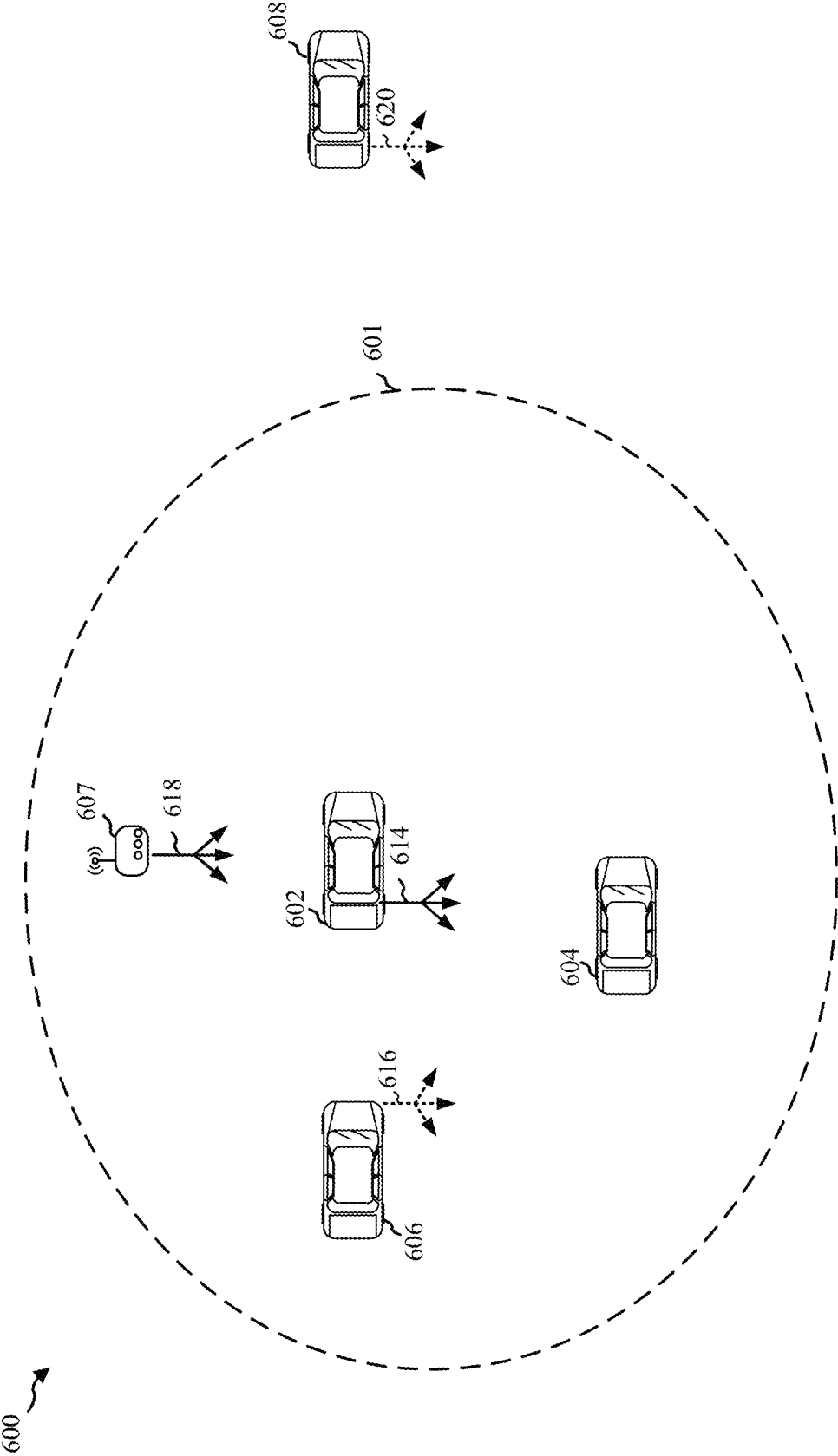


FIG. 6

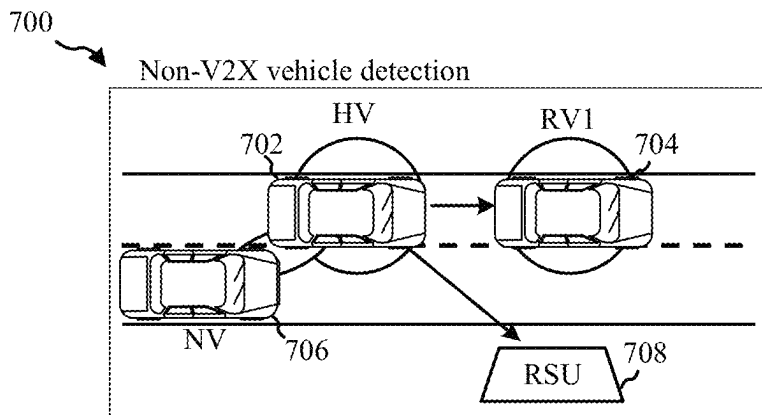


FIG. 7A

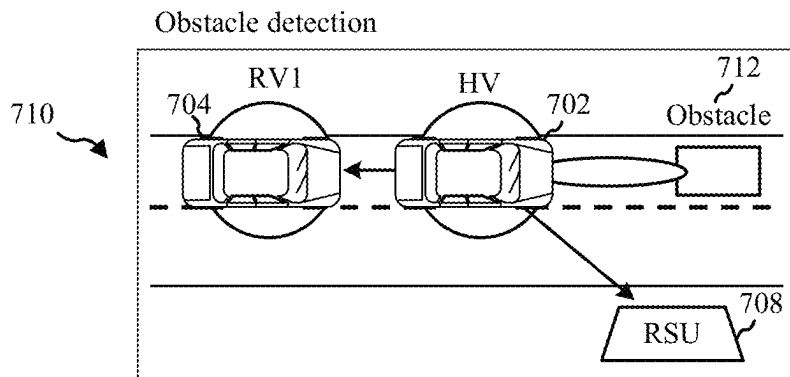


FIG. 7B

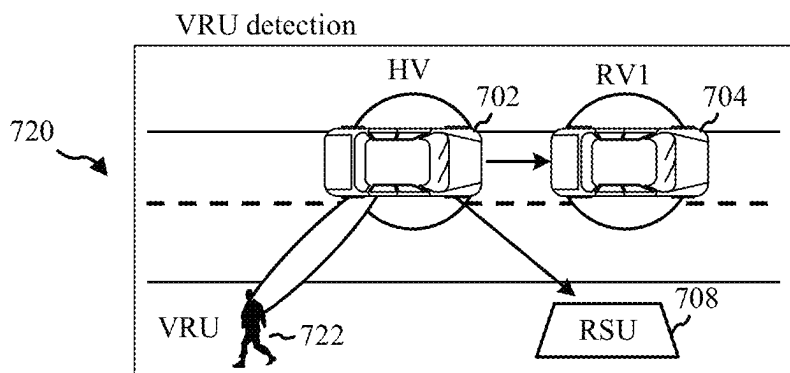


FIG. 7C

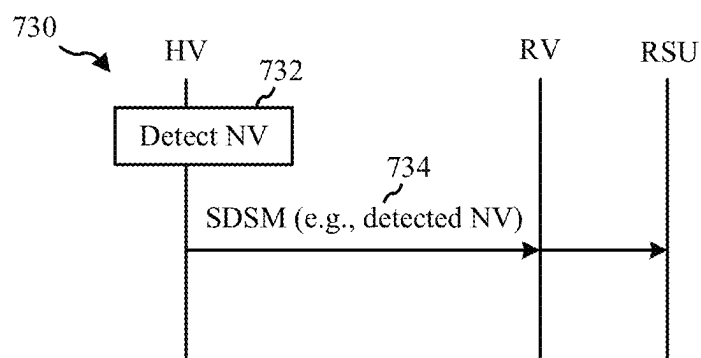


FIG. 7D

800 ↗

Data collection and retransmission (relay)

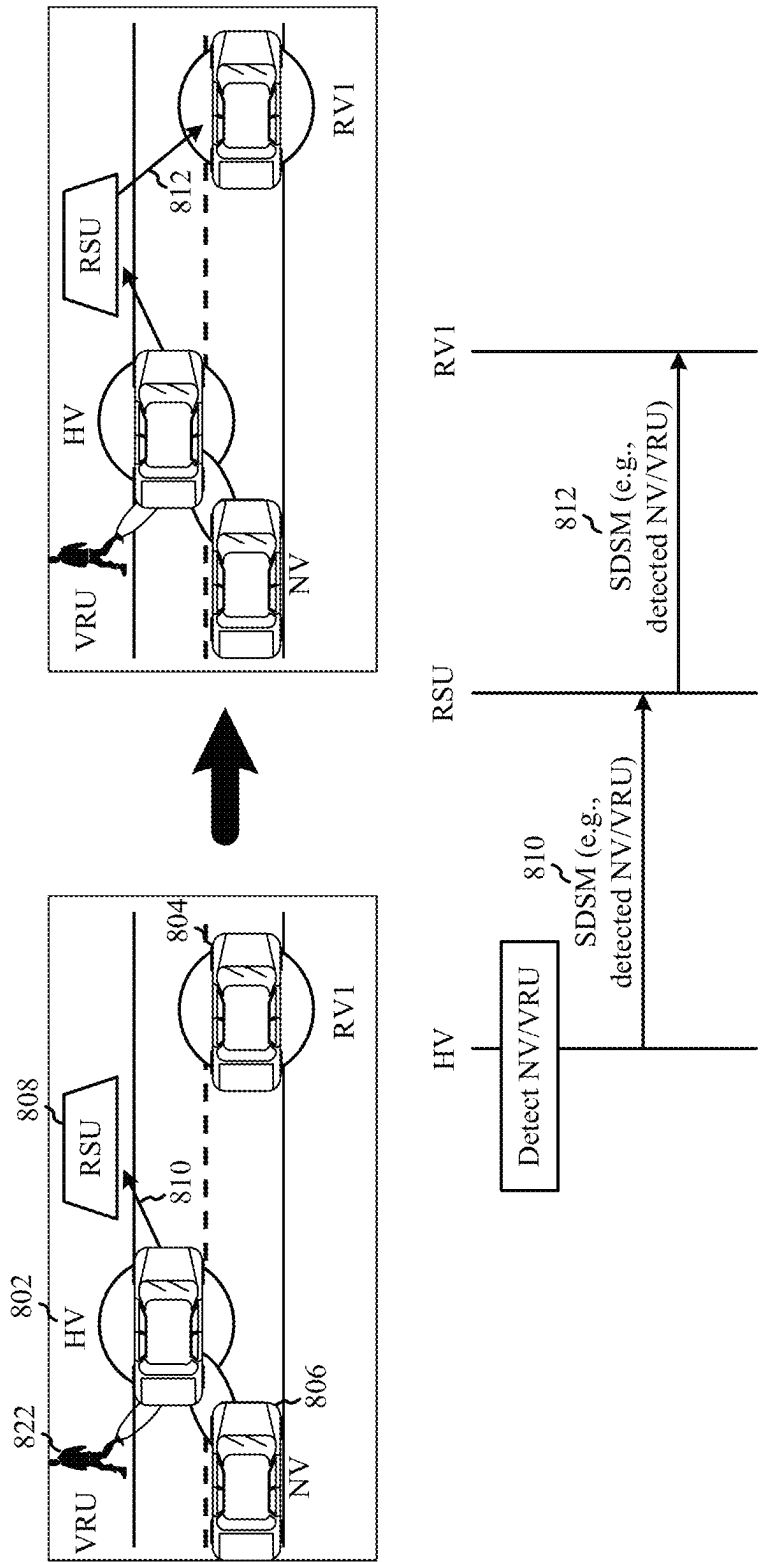


FIG. 8

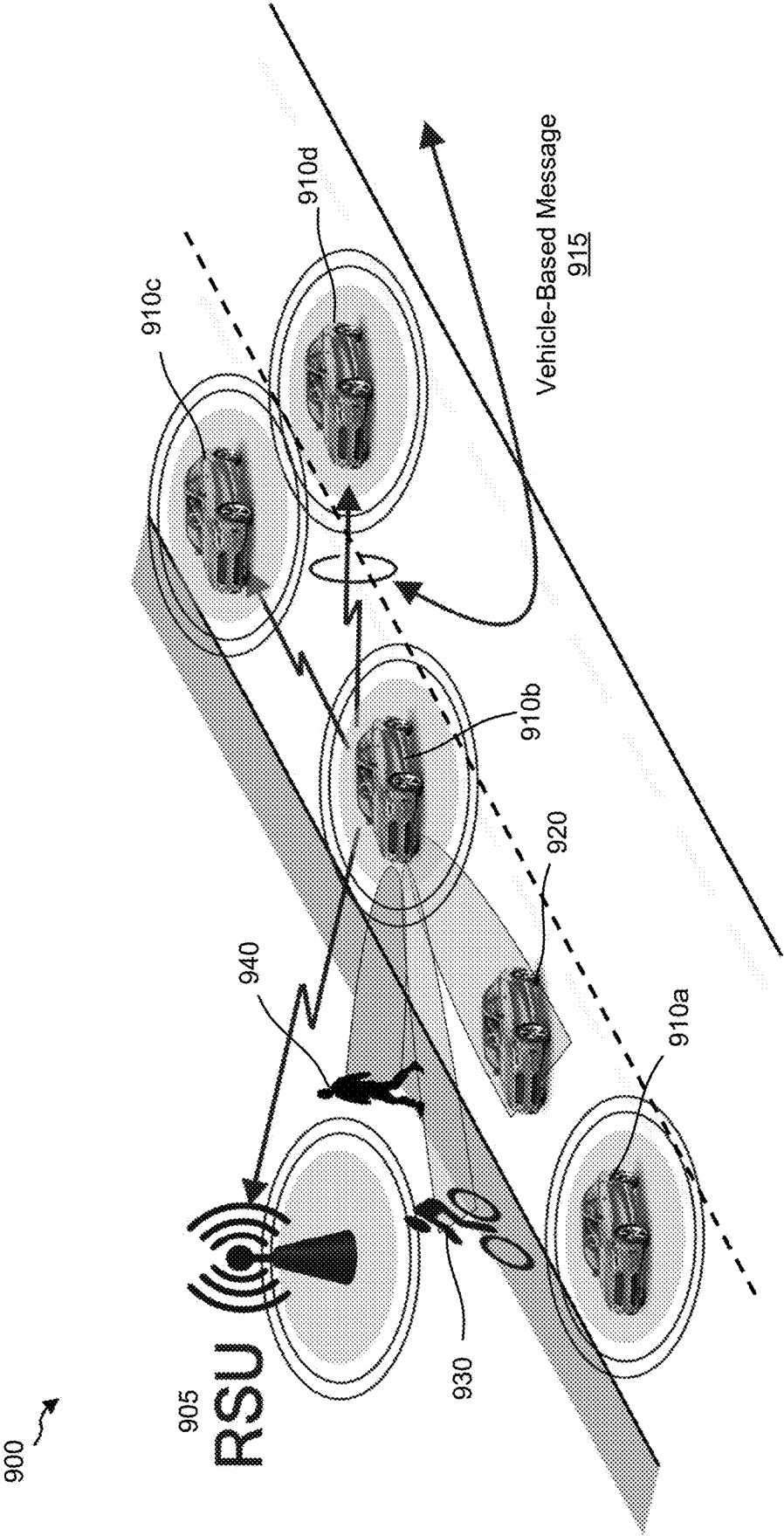


FIG. 9

1000 ↗

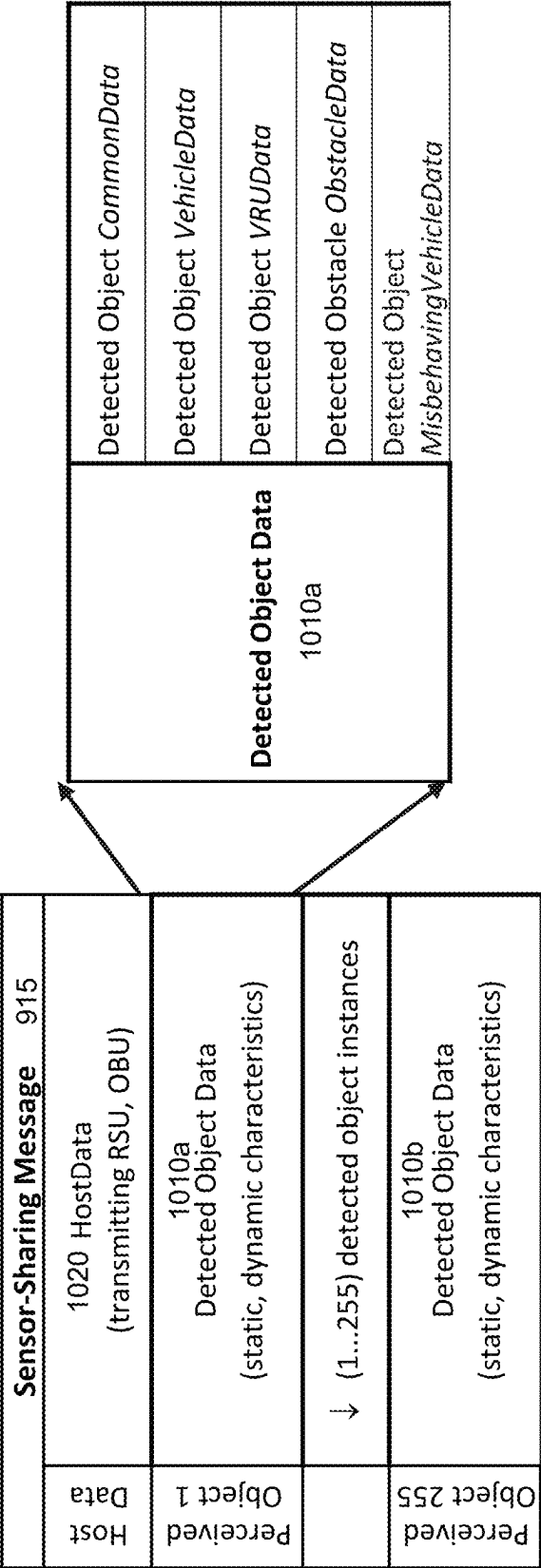


FIG. 10

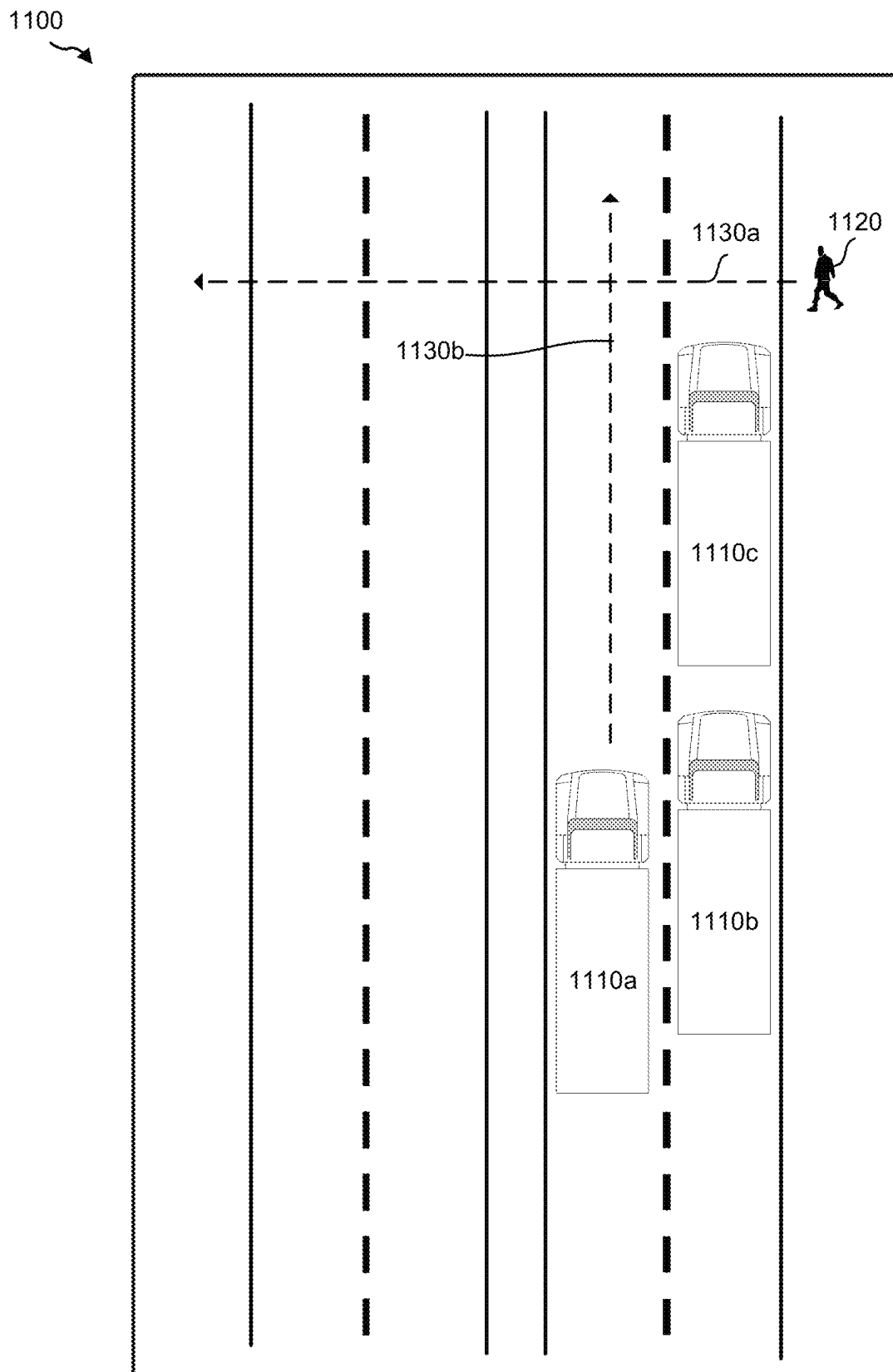


FIG. 11

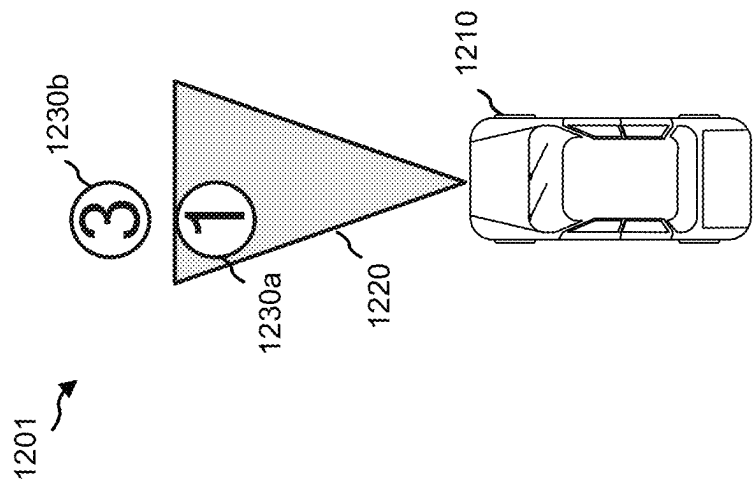


FIG. 12A

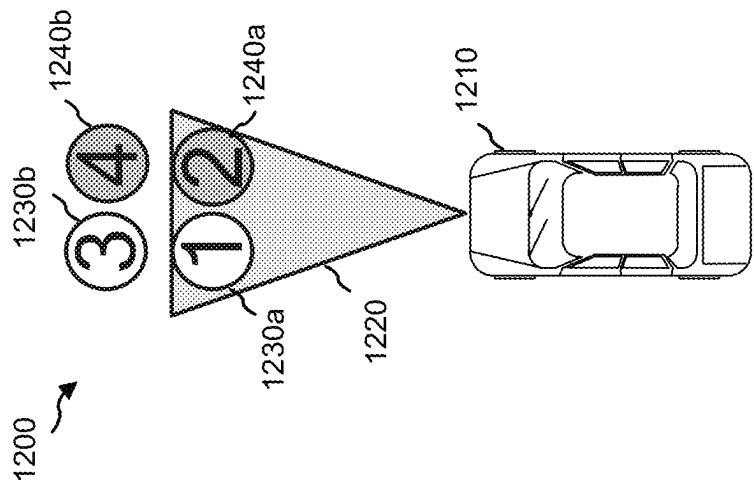


FIG. 12B

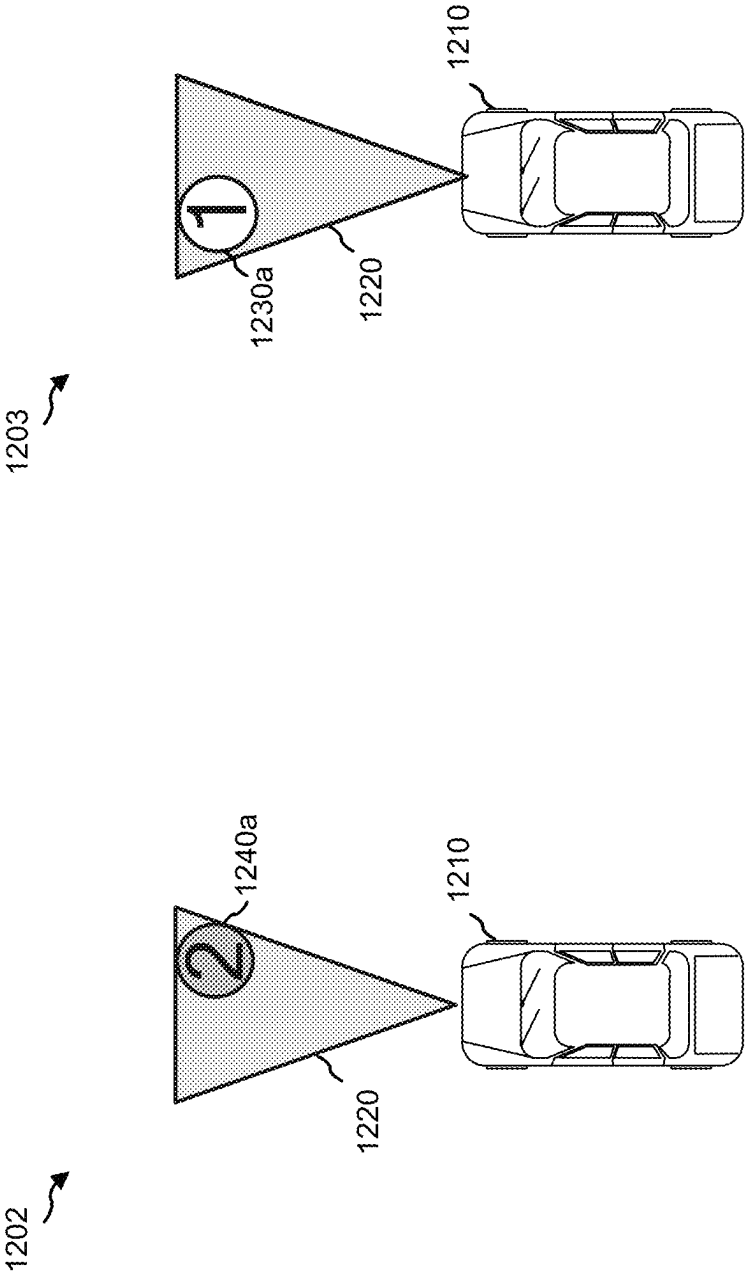


FIG. 12C

FIG. 12D

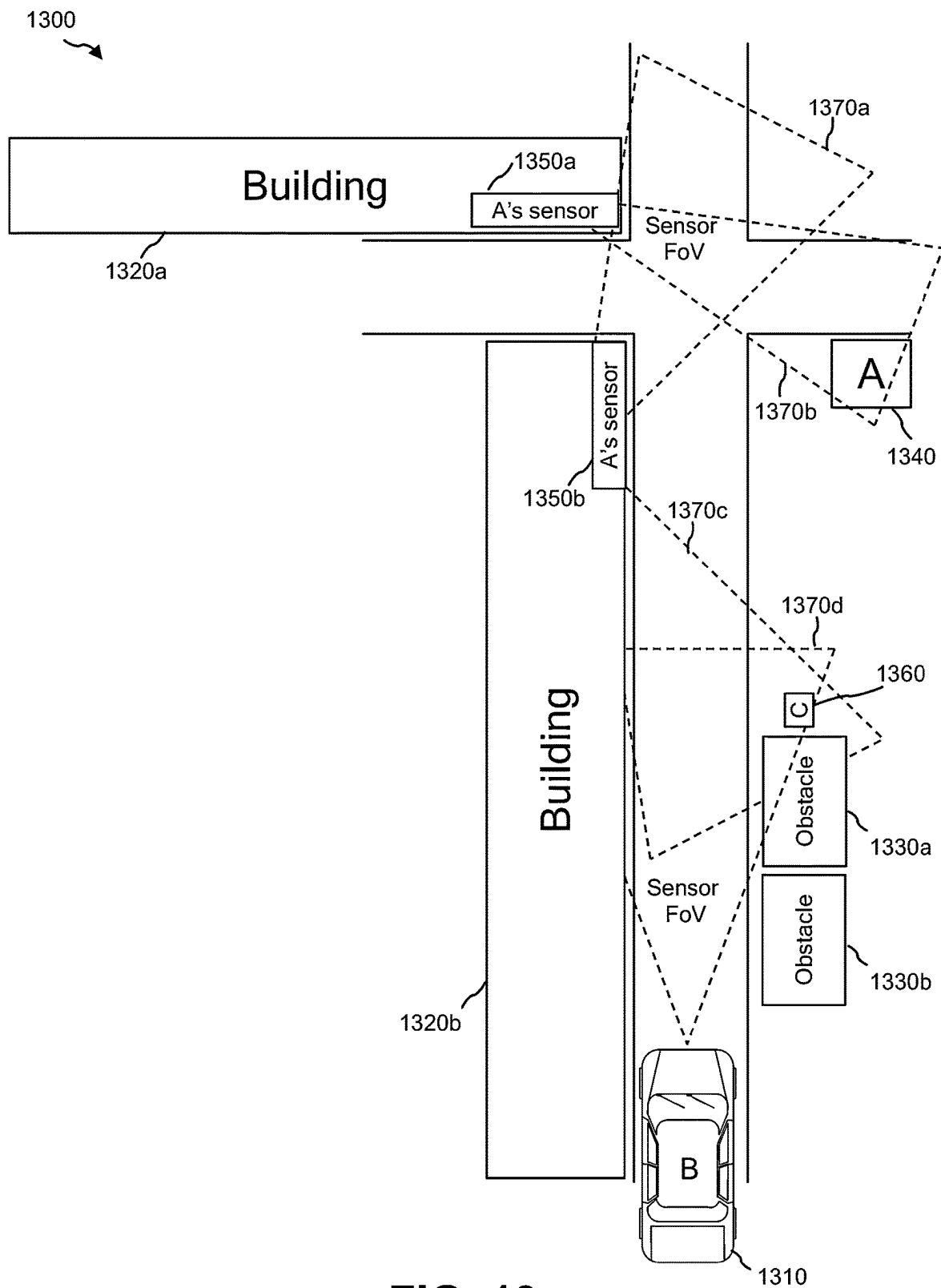


FIG. 13

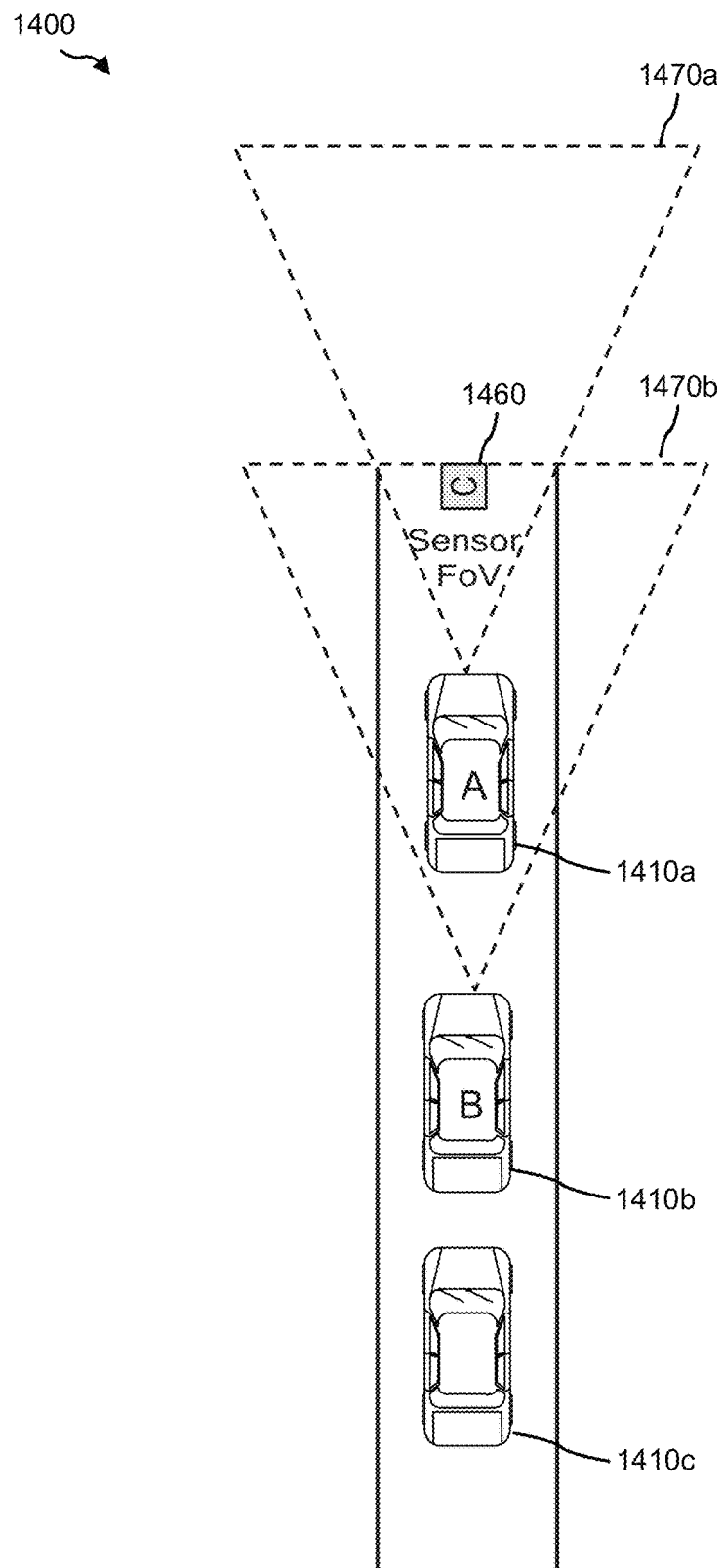


FIG. 14A

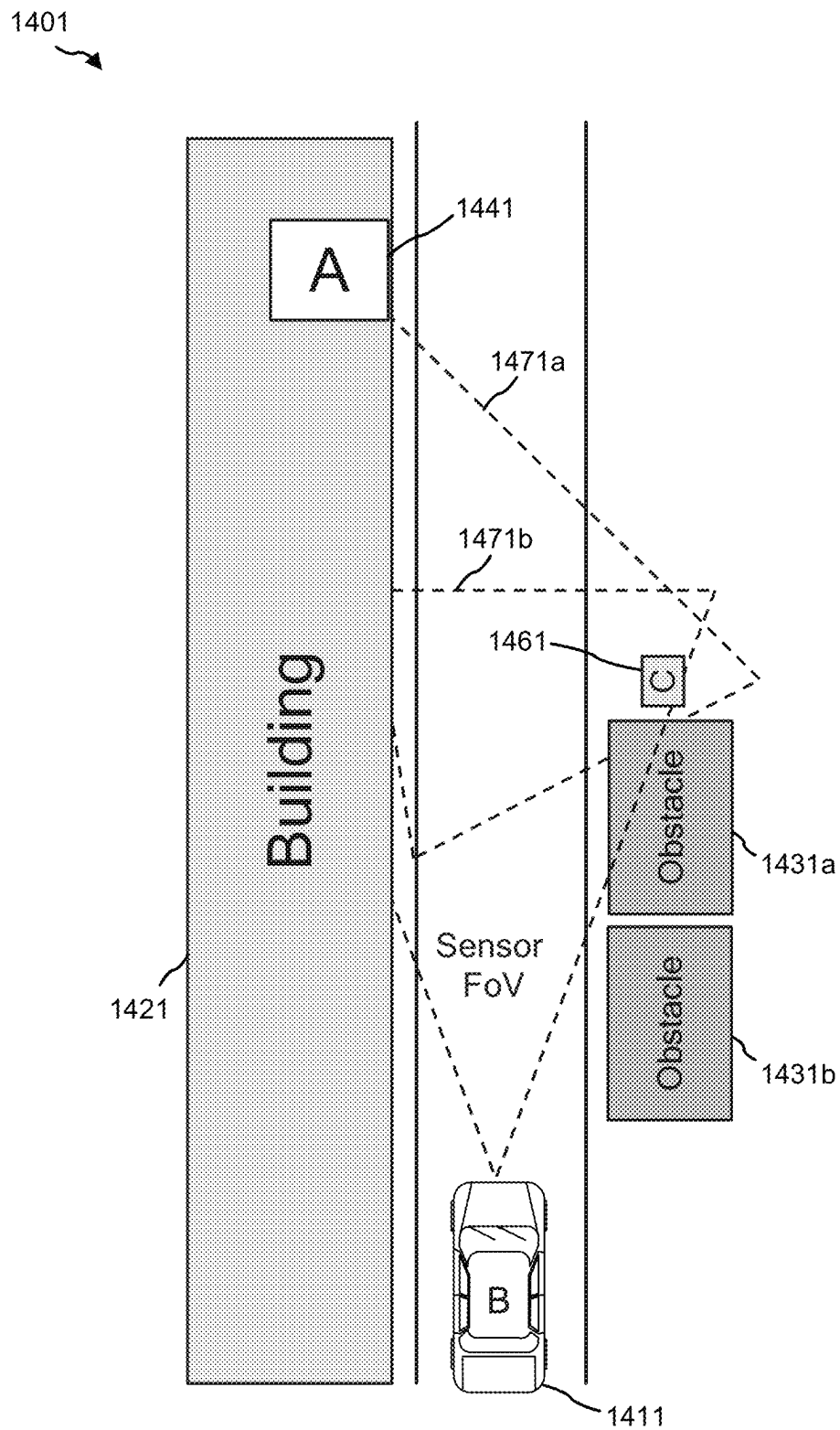


FIG. 14B

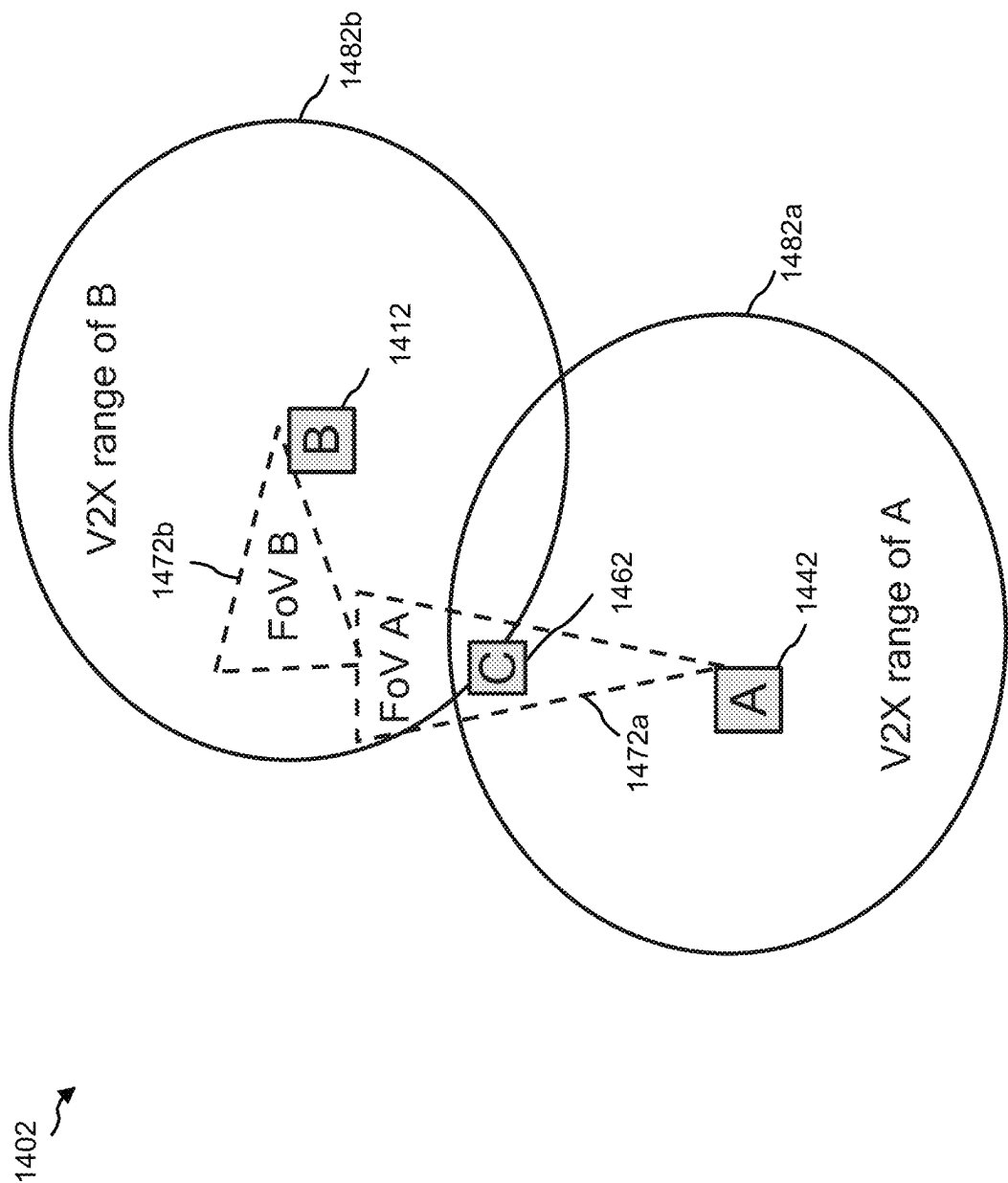


FIG. 14C

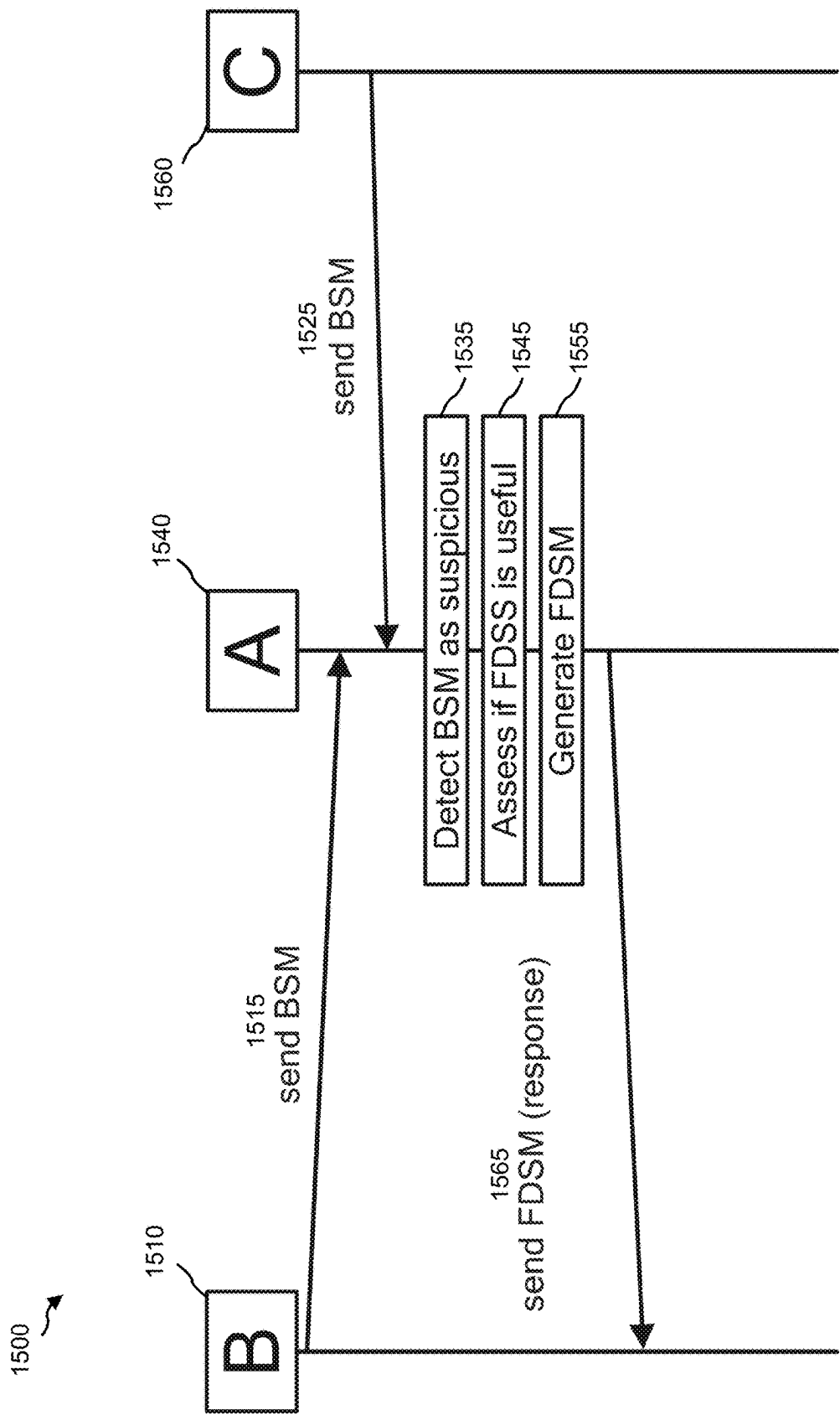


FIG. 15A

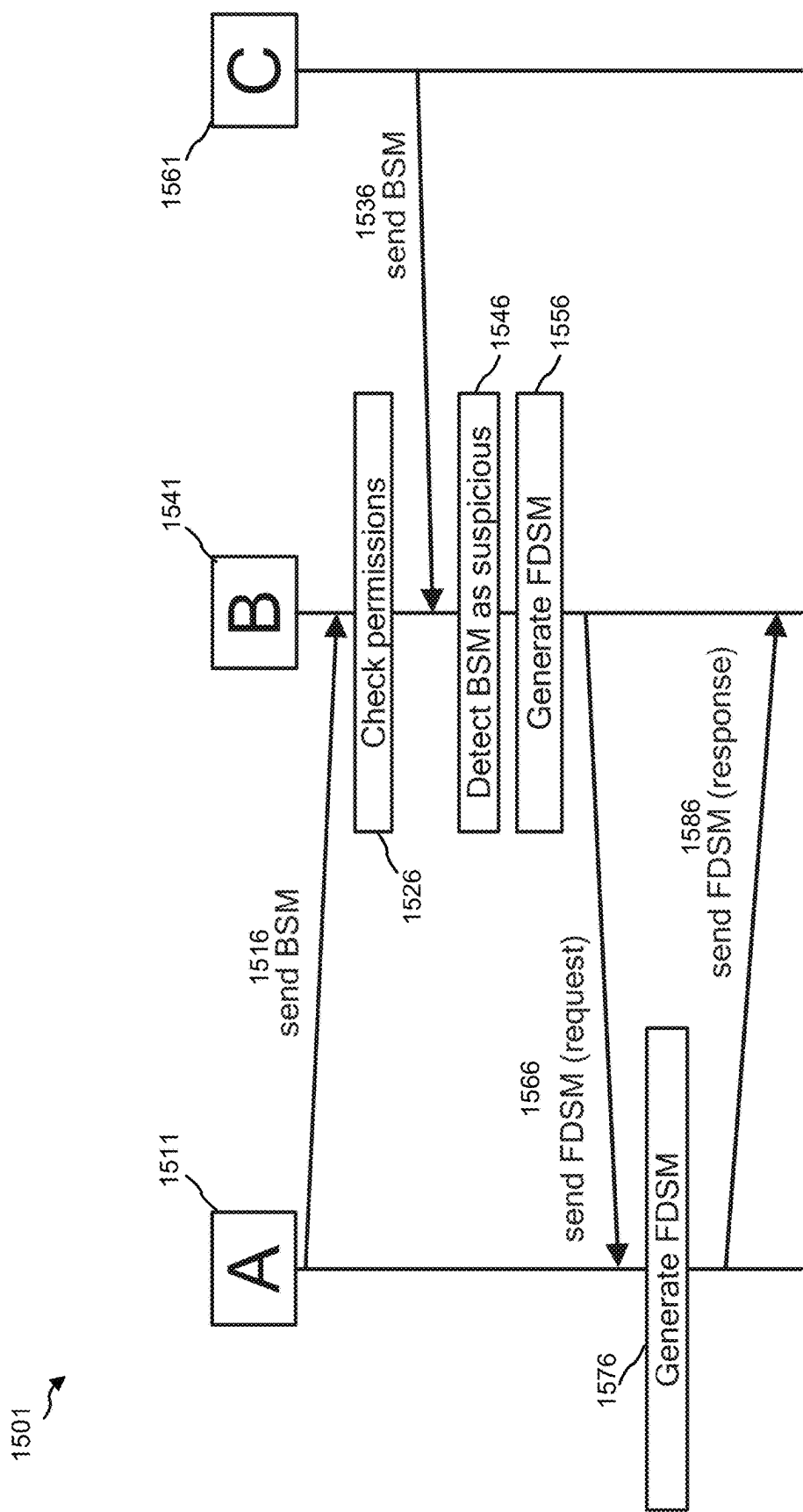


FIG. 15B

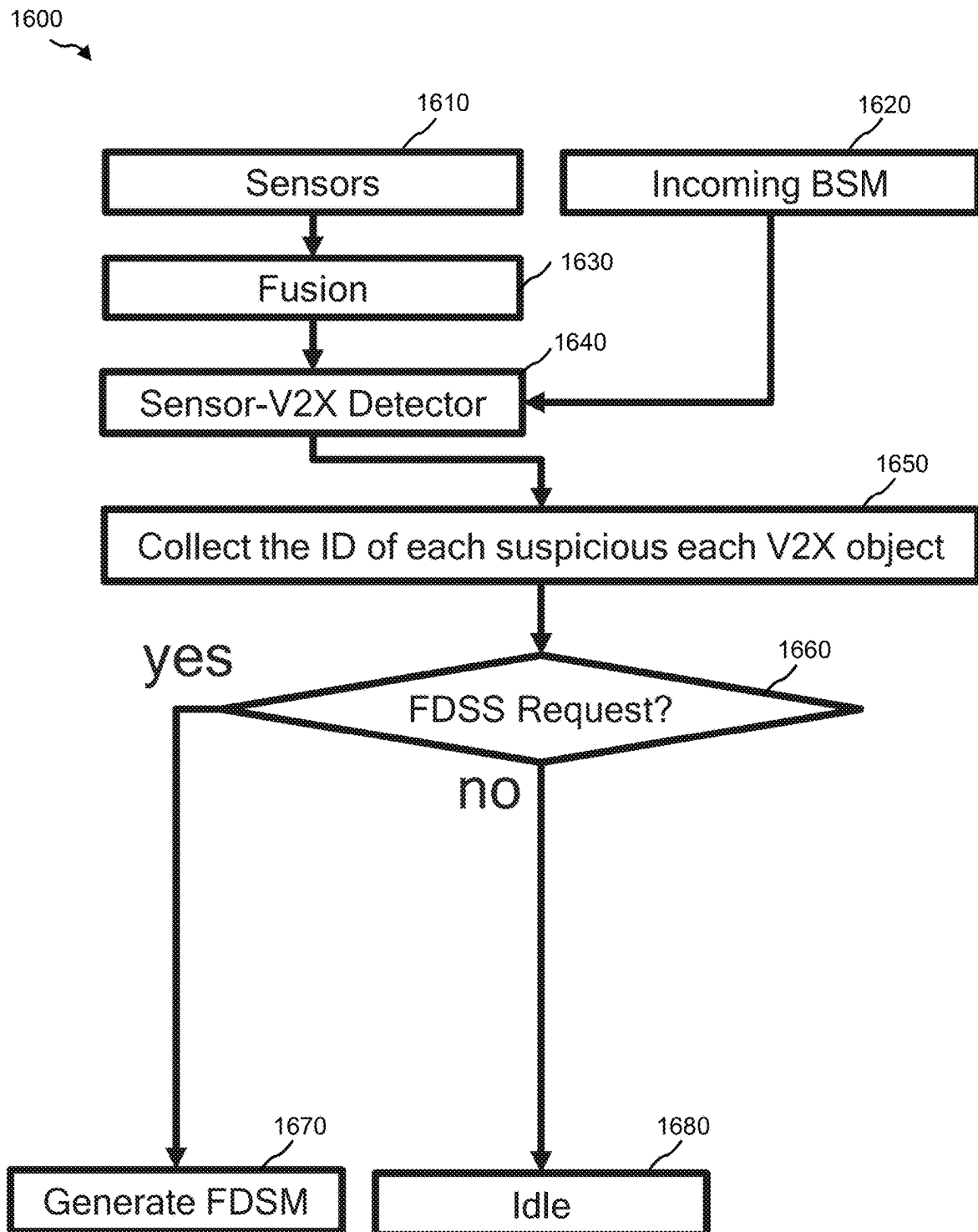
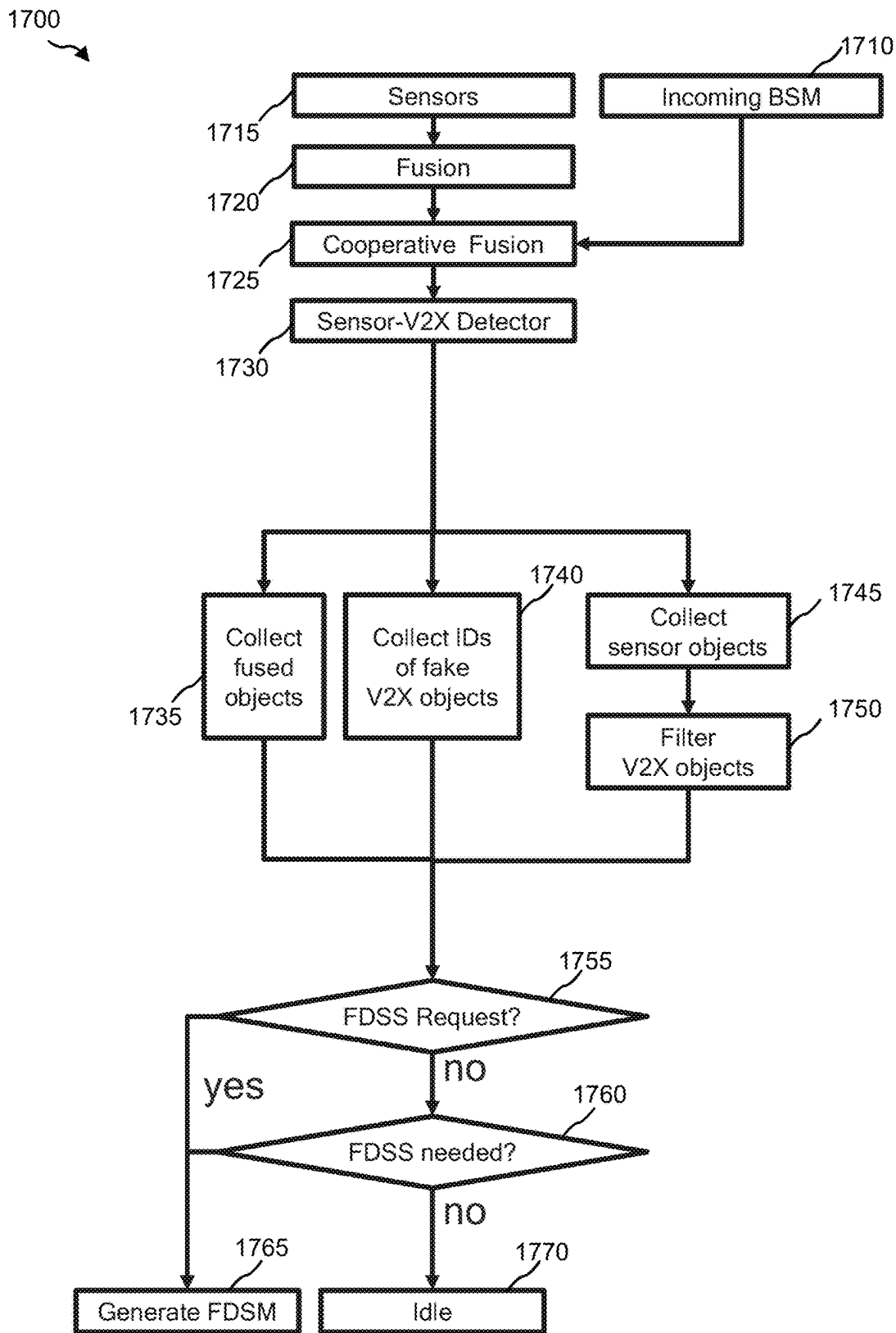
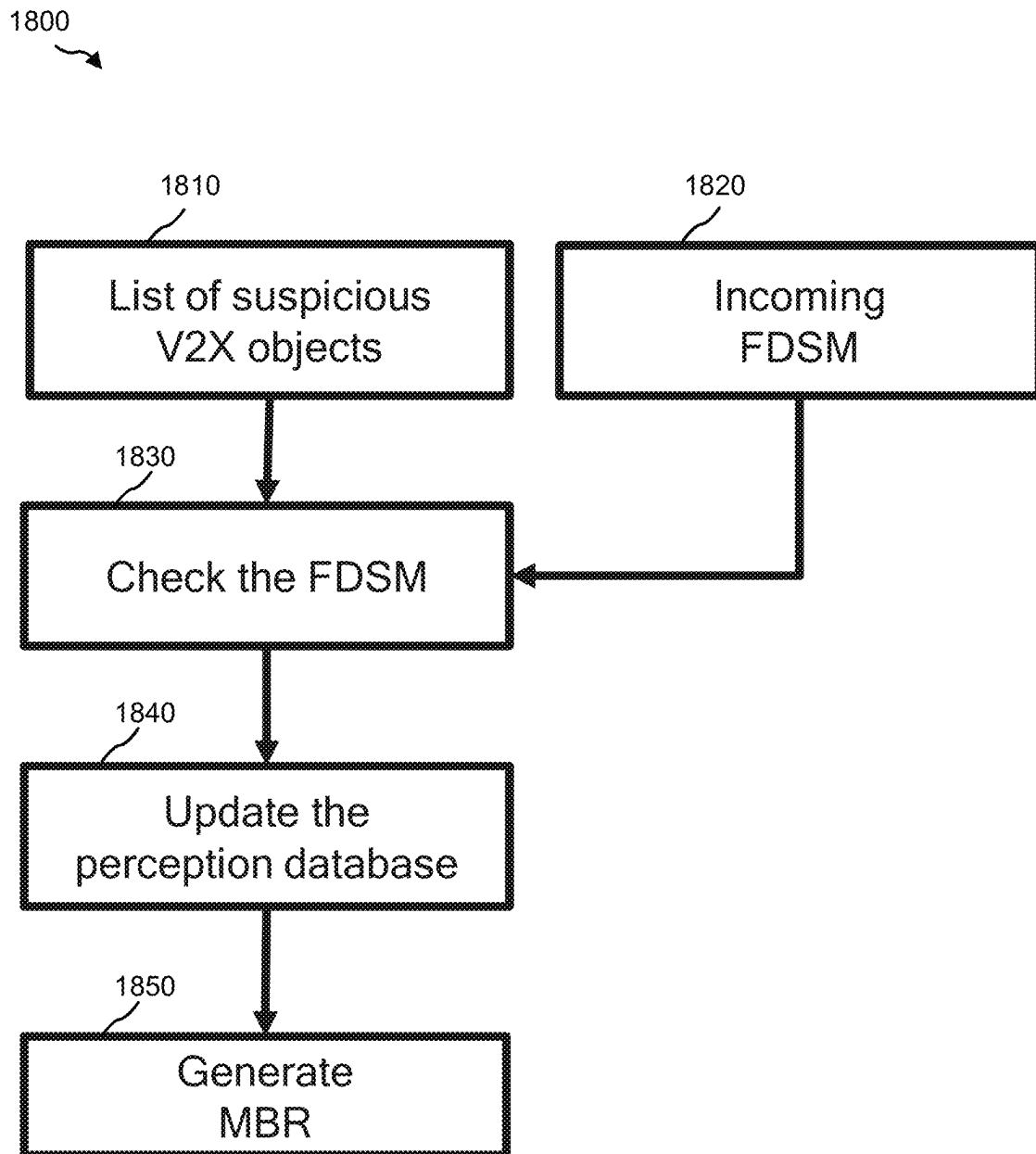


FIG. 16

**FIG. 17**

**FIG. 18**

1900 ↗

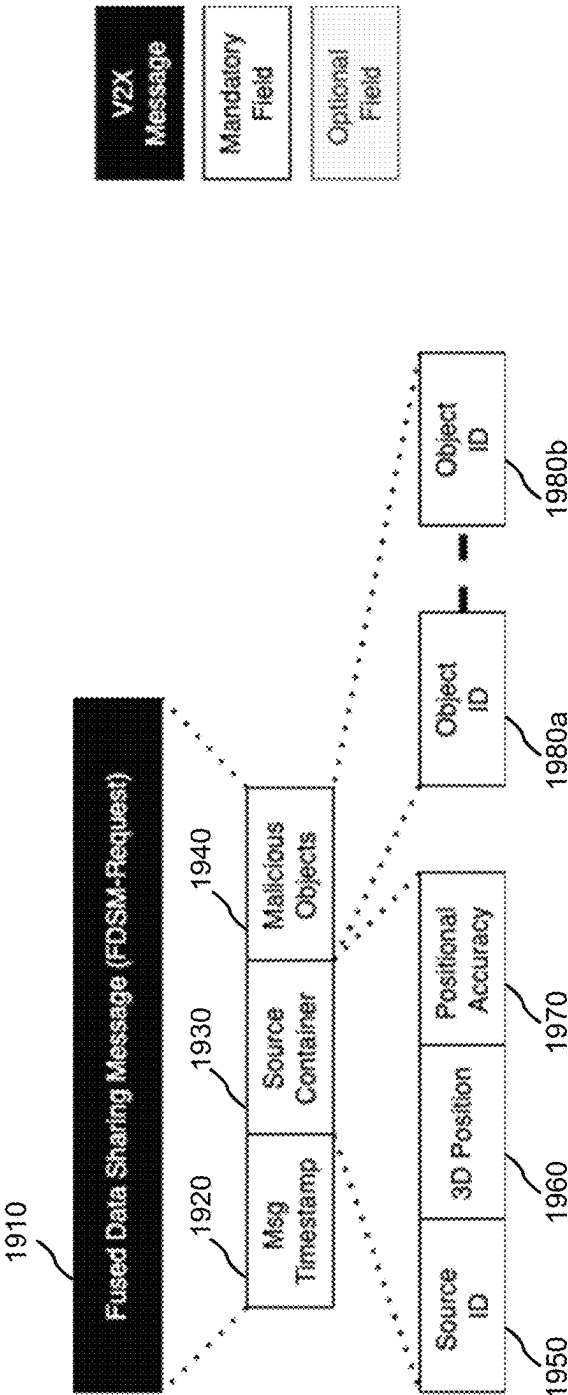


FIG. 19

2000 ↗

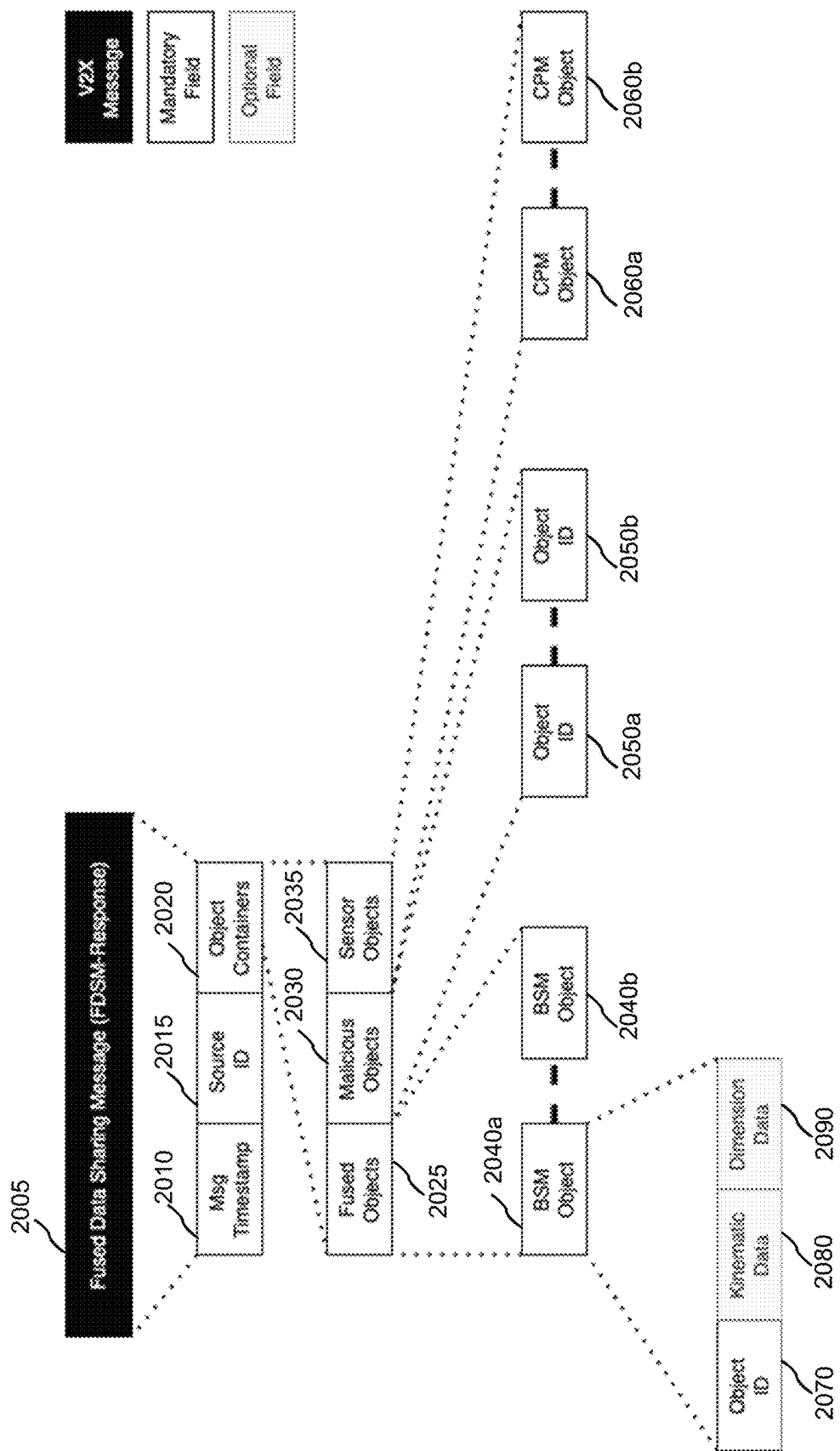
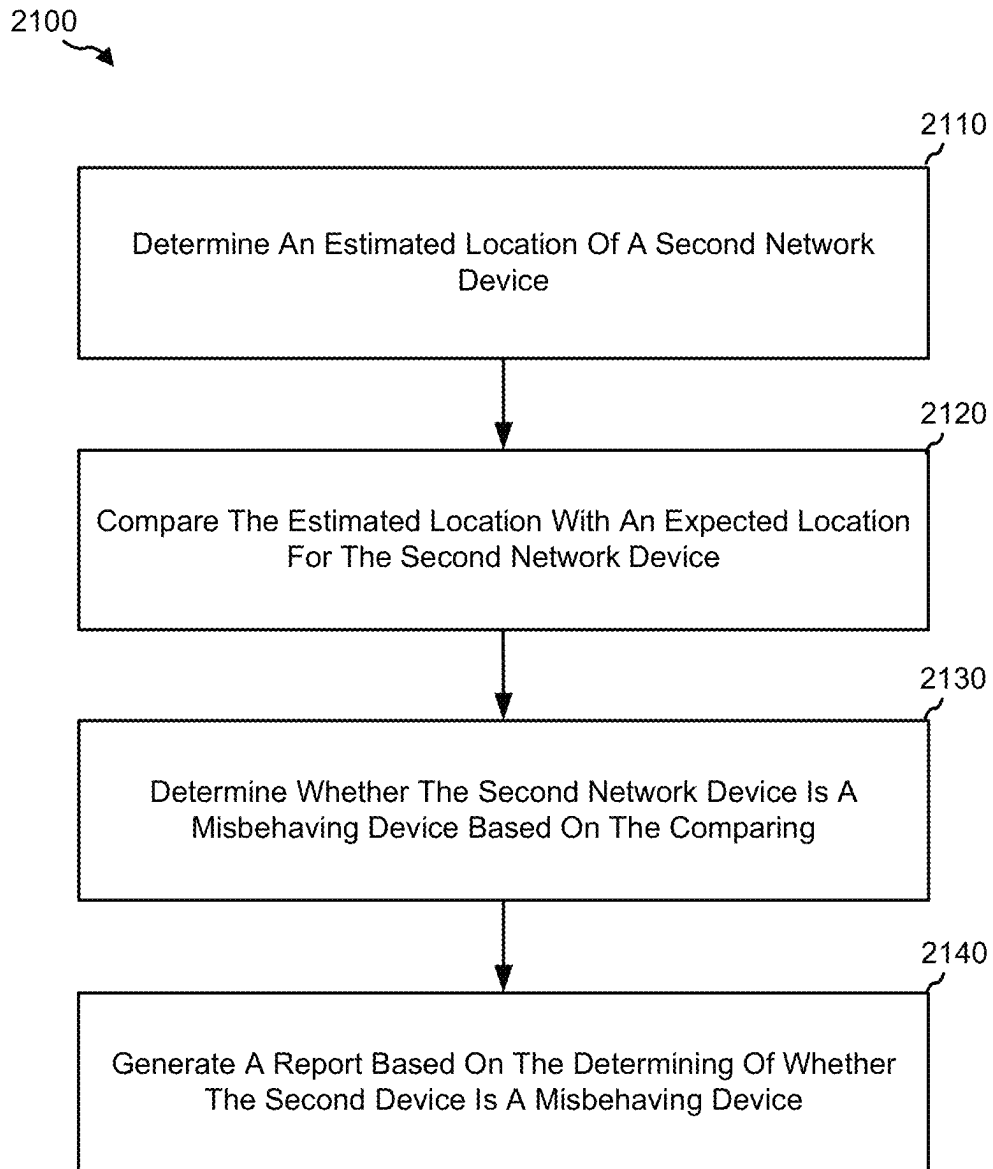


FIG. 20

**FIG. 21**

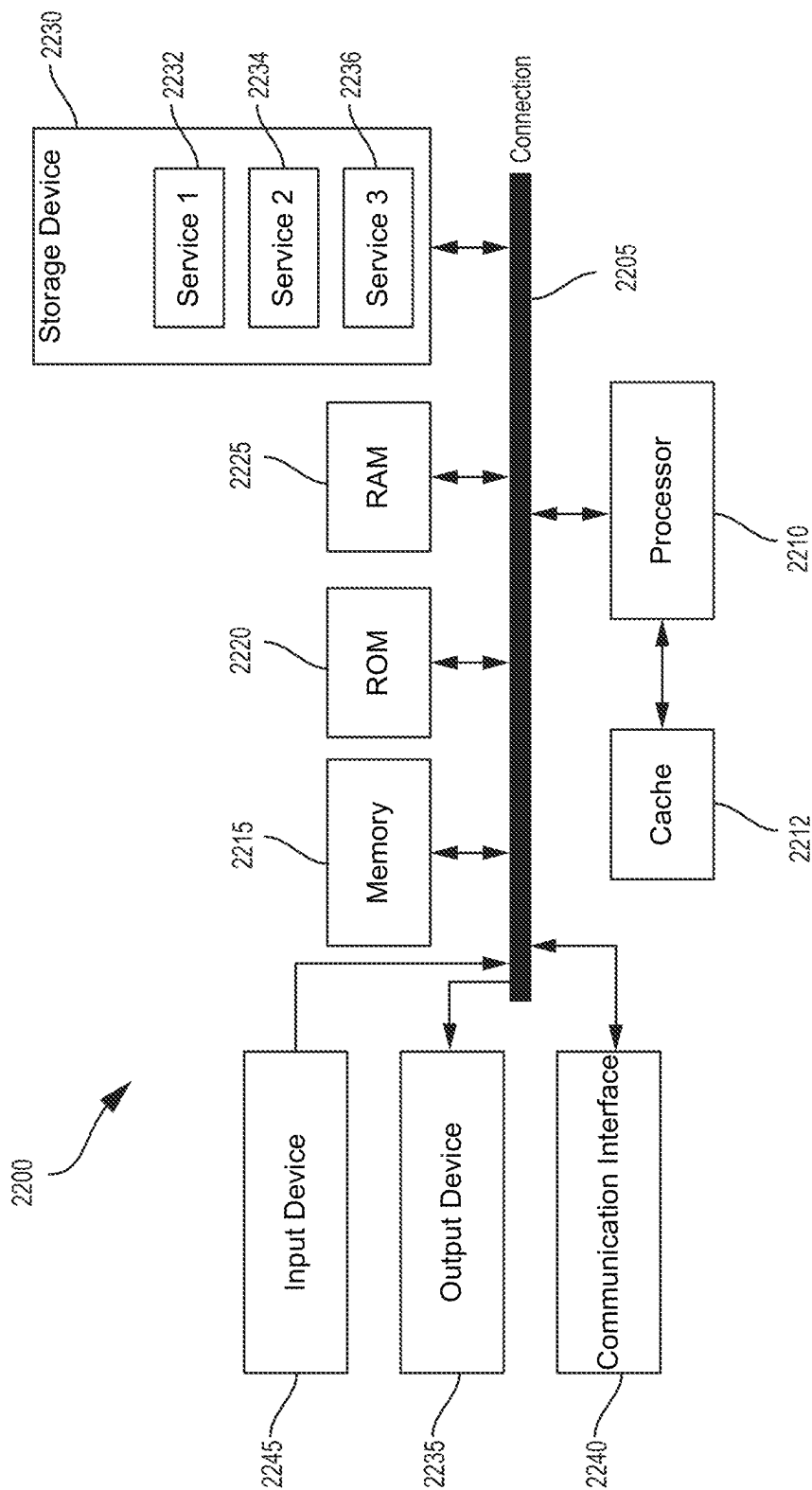


FIG. 22

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MISBEHAVIOR DETECTION SERVICE FOR SHARING CONNECTED AND SENSED OBJECTS

FIELD

The present disclosure generally relates to vehicle communications. For example, aspects of the present disclosure relate to a misbehavior detection service for sharing connected and sensed objects utilizing vehicle-to-everything (V2X) communications.

BACKGROUND

Wireless communication systems are widely deployed to provide various telecommunication services such as telephony, video, data, messaging, and broadcasts. Typical wireless communication systems may employ multiple-access technologies capable of supporting communication with multiple users by sharing available system resources. Examples of such multiple-access technologies include code division multiple access (CDMA) systems, time division multiple access (TDMA) systems, frequency division multiple access (FDMA) systems, orthogonal frequency division multiple access (OFDMA) systems, single-carrier frequency division multiple access (SC-FDMA) systems, and time division synchronous code division multiple access (TD-SCDMA) systems.

These multiple access technologies have been adopted in various telecommunication standards to provide a common protocol that enables different wireless devices to communicate on a municipal, national, regional, and even global level. An example telecommunication standard is 5G New Radio (NR). 5G NR is part of a continuous mobile broadband evolution promulgated by Third Generation Partnership Project (3GPP) to meet new requirements associated with latency, reliability, security, scalability (e.g., with Internet of Things (IoT)), and other requirements. 5G NR includes services associated with enhanced mobile broadband (eMBB), massive machine type communications (mMTC), and ultra-reliable low latency communications (URLLC). Some aspects of 5G NR may be based on the 4G Long Term Evolution (LTE) standard. Aspects of wireless communication may comprise direct communication between devices, such as in V2X, vehicle-to-vehicle (V2V), and/or device-to-device (D2D) communication. There exists a need for further improvements in V2X, V2V, and/or D2D technology. These improvements may also be applicable to other multi-access technologies and the telecommunication standards that employ these technologies.

SUMMARY

The following presents a simplified summary relating to one or more aspects disclosed herein. Thus, the following summary should not be considered an extensive overview relating to all contemplated aspects, nor should the following summary be considered to identify key or critical elements relating to all contemplated aspects or to delineate the scope associated with any particular aspect. Accordingly, the following summary has the sole purpose to present certain concepts relating to one or more aspects relating to the mechanisms disclosed herein in a simplified form to precede the detailed description presented below.

Disclosed are systems, apparatuses, methods and computer-readable media for a misbehavior detection service for sharing connected and sensed objects (e.g., utilizing V2X

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communications). According to at least one example, a method of wireless communication performed at a first network device is provided. The method includes: determining, at the first network device, an estimated location of a second network device; comparing, at the first network device, the estimated location with an expected location for the second network device; determining, at the first network device, whether the second network device is a misbehaving device based on the comparing; and generating, at the first network device, a report based on the determining of whether the second device is a misbehaving device.

In another example, an apparatus for performed at a first network device is provided that includes at least one memory and at least one processor coupled to the at least one memory. The at least one processor is configured to: determine an estimated location of a second network device; compare the estimated location with an expected location for the second network device; determine whether the second network device is a misbehaving device based on the comparing; and generate a report based on the determining of whether the second device is a misbehaving device.

In another example, a non-transitory computer-readable medium is provided that has stored thereon instructions that, when executed by one or more processors, cause the one or more processors to: determine an estimated location of a second network device; compare the estimated location with an expected location for the second network device; determine whether the second network device is a misbehaving device based on the comparing; and generate a report based on the determining of whether the second device is a misbehaving device.

In another example, an apparatus for performed at a first network device is provided. The apparatus includes: means for determining an estimated location of a second network device; means for comparing the estimated location with an expected location for the second network device; means for determining whether the second network device is a misbehaving device based on the comparing; and means for generating a report based on the determining of whether the second device is a misbehaving device.

In some aspects, the network devices or apparatuses described herein is, includes, or is part of, a vehicle (e.g., an automobile, truck, etc., or a component or system of an automobile, truck, etc.), a mobile device (e.g., a mobile telephone or so-called “smart phone” or other mobile device), a wearable device, an extended reality device (e.g., a virtual reality (VR) device, an augmented reality (AR) device, or a mixed reality (MR) device), a personal computer, a laptop computer, a server computer, a robotics device, or other device. In some aspects, the apparatus includes radio detection and ranging (radar) for capturing radio frequency (RF) signals. In some aspects, the apparatus includes one or more light detection and ranging (LIDAR) sensors, radar sensors, or other light-based sensors for capturing light-based (e.g., optical frequency) signals. In some aspects, the apparatus includes a camera or multiple cameras for capturing one or more images. In some aspects, the apparatus further includes a display for displaying one or more images, notifications, and/or other displayable data. In some aspects, the apparatuses described above can include one or more sensors, which can be used for determining a location of the apparatuses, a state of the apparatuses (e.g., a temperature, a humidity level, and/or other state), and/or for other purposes.

This summary is not intended to identify key or essential features of the claimed subject matter, nor is it intended for use in isolation to determine the scope of the claimed subject

matter. The subject matter should be understood by reference to appropriate portions of the entire specification of this patent, any or all drawings, and each claim.

Other objects and advantages associated with the aspects disclosed herein will be apparent to those skilled in the art based on the accompanying drawings and detailed description.

BRIEF DESCRIPTION OF THE DRAWINGS

Illustrative aspects of the present application are described in detail below with reference to the following figures:

FIG. 1 is a diagram illustrating an example wireless communications system, in accordance with some aspects of the present disclosure.

FIG. 2 is a diagram illustrating an example of a disaggregated base station architecture, which may be employed by the disclosed system for a misbehavior detection service for sharing connected and sensed objects, in accordance with some aspects of the present disclosure.

FIG. 3 is a diagram illustrating an example of various user equipment (UEs) communicating over direct communication interfaces (e.g., a cellular based PC5 sidelink interface, 802.11p defined DSRC interface, or other direct interface) and wide area network (Uu) interfaces, in accordance with some aspects of the present disclosure.

FIG. 4 is a block diagram illustrating an example of a computing system of a vehicle, in accordance with some aspects of the present disclosure.

FIG. 5 is a block diagram illustrating an example of a computing system of a user device, in accordance with some aspects of the present disclosure.

FIG. 6 is a diagram illustrating an example of devices involved in wireless communications (e.g., sidelink communications), in accordance with some aspects of the present disclosure.

FIGS. 7A-7D are diagrams illustrating examples of sensor-sharing for cooperative and automated driving systems, in accordance with some aspects of the present disclosure.

FIG. 8 is a diagram illustrating an example of sensor-sharing for cooperative and automated driving systems, in accordance with some aspects of the present disclosure.

FIG. 9 is a diagram illustrating an example of a system for sensor sharing in wireless communications (e.g., V2X communications), in accordance with some aspects of the present disclosure.

FIG. 10 is a diagram illustrating an example of a vehicle-based message (shown as a sensor-sharing message), in accordance with some aspects of the present disclosure.

FIG. 11 is a diagram illustrating an example of a vehicle configuration for a non-line of sight (NLoS) scenario, in accordance with some aspects of the present disclosure.

FIG. 12A is a diagram illustrating an example of a ground truth view from a receiver perspective, in accordance with some aspects of the present disclosure.

FIG. 12B is a diagram illustrating an example of a Basic Safety Message (BSM) view from a receiver perspective, in accordance with some aspects of the present disclosure.

FIG. 12C is a diagram illustrating an example of a Collective Perception Message (CPM) view from a receiver perspective, in accordance with some aspects of the present disclosure.

FIG. 12D is a diagram illustrating an example of a fused object view from a receiver perspective, in accordance with some aspects of the present disclosure.

FIG. 13 is a diagram illustrating an example of a system for a misbehavior detection service for sharing connected and sensed objects, where an equipped (e.g., V2X capable) network device with multiple sensors provides visual confirmation of another equipped network device, in accordance with some aspects of the present disclosure.

FIG. 14A is a diagram illustrating an example of a system for a misbehavior detection service for sharing connected and sensed objects, where a platoon leader provides visual confirmation of an equipped (e.g., V2X capable) network device, in accordance with some aspects of the present disclosure.

FIG. 14B is a diagram illustrating an example of a system for a misbehavior detection service for sharing connected and sensed objects, where an equipped (e.g., V2X capable) network device with a single sensor provides visual confirmation of another equipped network device (e.g., vehicle), in accordance with some aspects of the present disclosure.

FIG. 14C is a diagram illustrating an example of a system for a misbehavior detection service for sharing connected and sensed objects, where an equipped (e.g., V2X capable) network device (e.g., an RSU) located in one range provides visual confirmation of another equipped network device (e.g., an RSU) located within two ranges, in accordance with some aspects of the present disclosure.

FIG. 15A is a diagram illustrating an example of signaling for a feedback approach, in accordance with some aspects of the present disclosure.

FIG. 15B is a diagram illustrating an example of signaling for a request-response approach, in accordance with some aspects of the present disclosure.

FIG. 16 is a flow chart illustrating an example of a process for a fused data sharing detector (FDS) from an initial requester side perspective, in accordance with some aspects of the present disclosure.

FIG. 17 is a flow chart illustrating an example of a process for a FDS from a responder side perspective, in accordance with some aspects of the present disclosure.

FIG. 18 is a flow chart illustrating an example of a process for a FDS from a final responder side perspective, in accordance with some aspects of the present disclosure.

FIG. 19 is a diagram illustrating an example of a fused data sharing message (FDSM) request, in accordance with some aspects of the present disclosure.

FIG. 20 is a diagram illustrating an example of an FDSM response, in accordance with some aspects of the present disclosure.

FIG. 21 is a flow chart illustrating an example of a process for wireless communications, according to some aspects of the present disclosure.

FIG. 22 illustrates an example computing system, according to aspects of the disclosure.

DETAILED DESCRIPTION

Certain aspects of this disclosure are provided below for illustration purposes. Alternate aspects may be devised without departing from the scope of the disclosure. Additionally, well-known elements of the disclosure will not be described in detail or will be omitted so as not to obscure the relevant details of the disclosure. Some of the aspects described herein can be applied independently and some of them may be applied in combination as would be apparent to those of skill in the art. In the following description, for the purposes of explanation, specific details are set forth in order to provide a thorough understanding of aspects of the application. However, it will be apparent that various aspects

may be practiced without these specific details. The figures and description are not intended to be restrictive.

The ensuing description provides example aspects only, and is not intended to limit the scope, applicability, or configuration of the disclosure. Rather, the ensuing description of the example aspects will provide those skilled in the art with an enabling description for implementing an example aspect. It should be understood that various changes may be made in the function and arrangement of elements without departing from the spirit and scope of the application as set forth in the appended claims.

The terms “exemplary” and/or “example” are used herein to mean “serving as an example, instance, or illustration.” Any aspect described herein as “exemplary” and/or “example” is not necessarily to be construed as preferred or advantageous over other aspects. Likewise, the term “aspects of the disclosure” does not require that all aspects of the disclosure include the discussed feature, advantage or mode of operation.

Wireless communications systems are deployed to provide various telecommunication services, including telephony, video, data, messaging, broadcasts, among others. Wireless communications systems have developed through various generations. A fifth generation (5G) mobile standard calls for higher data transfer speeds, greater numbers of connections, and better coverage, among other improvements. The 5G standard (also referred to as “New Radio” or “NR”), according to the Next Generation Mobile Networks Alliance, is designed to provide data rates of several tens of megabits per second to each of tens of thousands of users.

Vehicles are an example of systems that can include wireless communications capabilities. For example, vehicles (e.g., automotive vehicles, autonomous vehicles, aircraft, maritime vessels, among others) can communicate with other vehicles and/or with other devices that have wireless communications capabilities. Wireless vehicle communication systems encompass vehicle-to-vehicle (V2V), vehicle-to-infrastructure (V2I), vehicle-to-network (V2N), and vehicle-to-pedestrian (V2P) communications, which are all collectively referred to as vehicle-to-everything (V2X) communications. V2X communications is a vehicular communication system that supports the wireless transfer of information from a vehicle to other entities (e.g., other vehicles, pedestrians with smart phones, equipped vulnerable road users (VRUs), such as bicyclists, and/or other traffic infrastructure) located within the traffic system that may affect the vehicle. The main purpose of the V2X technology is to improve road safety, fuel savings, and traffic efficiency.

In a V2X communication system, information is transmitted from vehicle sensors (and other sources) through wireless links to allow the information to be communicated to other vehicles, pedestrians, VRUs, and/or traffic infrastructure. The information may be transmitted using one or more vehicle-based messages, such as cellular-vehicle-to-everything (C-V2X) messages, which can include Sensor Data Sharing Messages (SDSMs), Basic Safety Messages (BSMs), Cooperative Awareness Messages (CAMs), Collective Perception Messages (CPMs), Decentralized Environmental Messages (DENMs), and/or other types of vehicle-based messages. By sharing this information with other vehicles, the V2X technology improves vehicle (and driver) awareness of potential dangers to help reduce collisions with other vehicles and entities. In addition, the V2X technology enhances traffic efficiency by providing traffic warnings to vehicles of potential upcoming road dangers and obstacles such that vehicles may choose alternative traffic routes.

As previously mentioned, the V2X technology includes V2V communications, which can also be referred to as peer-to-peer communications. V2V communications allows for vehicles to directly wireless communicate with each other while on the road. With V2V communications, vehicles can gain situational awareness by receiving information regarding upcoming road dangers (e.g., unforeseen oncoming vehicles, accidents, and road conditions) from the other vehicles.

The IEEE 802.11p Standard supports (uses) a dedicated short-range communications (DSRC) interface for V2X wireless communications. Characteristics of the IEEE 802.11p based DSRC interface include low latency and the use of the unlicensed 5.9 Gigahertz (GHz) frequency band. C-V2X was adopted as an alternative to using the IEEE 802.11p based DSRC interface for the wireless communications. The 5G Automotive Association (5GAA) supports the use of C-V2X technology. In some cases, the C-V2X technology uses Long-Term Evolution (LTE) as the underlying technology, and the C-V2X functionalities are based on the LTE technology. C-V2X includes a plurality of operational modes. One of the operational modes allows for direct wireless communication between vehicles over the LTE sidelink PC5 interface. Similar to the IEEE 802.11p based DSRC interface, the LTE C-V2X sidelink PC5 interface operates over the 5.9 GHz frequency band. Vehicle-based messages, such as BSMs and CAMs, which are application layer messages, are designed to be wirelessly broadcasted over the 802.11p based DSRC interface and the LTE C-V2X sidelink PC5 interface.

In one or more cases, a transmitting network device (e.g., a V2X-capable vehicle that generates and sends vehicle-based messages, such as BSMs) may be misbehaving (e.g., operating as a misbehaving vehicle) by sending (e.g., either purposely or not purposely) vehicle-based messages (e.g., BSMs) containing incorrect information. For example, a transmitting network device may be operating as a misbehaving vehicle if the information contained within its vehicle-based messages identifies an incorrect position (location) for the transmitting network device.

In some cases, a transmitting network device (e.g., a V2X-capable vehicle that generates and sends vehicle-based messages) may be misbehaving (e.g., operating as a misbehaving vehicle) by operating as an attacker by creating a non-visible, V2X ghost object to disturb traffic on the road. A ghost V2X object is a V2X object that is not located at the location of the transmitting network device. Ghost V2X objects are objects with no physical existence, such as a simulated vulnerable road user (VRU) or a simulated vehicle. A non-visible object is an object that is out of a sensor field-of-view (FoV) or not in the line-of-sight (LoS) (e.g., a non-line of sight scenario) of a receiving network device (e.g., a V2X-capable vehicle that receives vehicle-based messages).

A goal of an attacker can be to remove the benefits of sensors utilizing V2X communications. In some cases, an attacker may employ non-line of sight (NLoS) ghost objects. For example, when an object (e.g., a ghost object) is not located within a LoS of a vehicle, sensors associated with the vehicle cannot readily confirm the existence (or non-existence) of the object. Since the vehicle is unable to confirm the existence (or non-existence) of the object, the driver of the vehicle may be forced to decelerate (e.g., slow down) in order to ensure that the vehicle does not collide with the object, which may not actually exist and merely be a ghost object. This affect of causing the driver to perform unnecessary maneuvers (e.g., decelerating) can cause frus-

tration to the driver as well as a disruption to traffic, which may result in a traffic jam and/or a vehicle collision.

Currently, detectors can be inefficient and/or insufficient to be able to detect an attack. For example, current detectors can have a limited range, cannot operate with V2X-capable objects that are not in the sensorial line of sight (e.g., in non-line of sight), and/or cannot be utilized by vehicles without sensors. In some cases, current detectors may not be able to detect an attack if a mobility pattern of a ghost object (e.g., ghost vehicle) is perfect or close to perfect. Current detectors using sensors to detect ghost attacks (e.g., ghost V2X attacks) can be inefficient in cluttered road environments. For example, camera-based detectors cannot detect objects that are NLoS. For another example, received signal strength indication (RSSI)-based detectors can be affected by signal attenuation that is caused by obstacles. Current detectors utilizing sensors to detect ghost attacks (e.g., ghost V2X attacks) are only effective if the network device (e.g., vehicle) itself has sensors.

In addition, a collection perception service (CPS) is not able to verify the physical existence of ghost objects. As such, the CPS will need to keep track of all ghost objects (e.g., all ghost objects indicated by vehicle-based messages, such as BSMs). Also, the CPS cannot provide V2X data redundancy from a receiving network device (e.g., a receiving vehicle) perspective. For example, a Collection Perception Message (CPM) cannot contain V2X objects perceived by a transmitting network device's (e.g., transmitting vehicle's) sensors. A CPM can contain non-V2X objects perceived by a transmitting network device's (e.g., transmitting vehicle's) sensors.

An Intelligent transportation system-station (ITS-S) approach using an ITS-S can detect a simulated V2X object (e.g., ghost object). Sensors of an ITS-S can determine the location (position) of the simulated V2X object (e.g., ghost object) when the simulated V2X object (e.g., ghost object) is located within the FoV of the ITS-S. The ITS-S can use the determined location of the object to verify the location of the object indicated in a vehicle-based message (e.g., BSM) received from the object. However, this ITS-S approach has several limitations including, but not limited to, the approach requires an ITS-S equipped with sensors, the approach cannot detect objects (e.g., V2X objects) that are in NLoS of the ITS-S (e.g., obstacles block the FoV of the sensors of the ITS-S), and this approach can have a limited detection range.

Systems, apparatuses, (e.g., network devices), methods (also referred to as processes), and computer-readable media (collectively referred to herein as "systems and techniques") are provided for optimizing situational awareness of vehicle misbehavior, which can lead to a disruption in traffic. The systems and techniques can provide a misbehavior detection service for sharing connected and sensed objects (e.g., a V2X security service for sharing V2X-sensor objects). In some cases, the disclosed service can utilize a LoS of a V2X-capable network device (e.g., V2X-capable base station) to confirm the existence of ghost objects that are located in a NLoS of other V2X-capable network devices (e.g., V2X-capable base stations and/or vehicles). The service can allow for the sharing of sensor-V2X detector outputs (e.g., outputs from sensors on V2X-capable network devices) to V2X-capable network devices (e.g., V2X-capable vehicles) without sensors. The service can also allow for communication (e.g., V2X communication) to network devices (e.g., V2X-capable vehicles) regarding V2X-sensor objects (e.g., physical objects perceived by sensors on V2X-capable network devices local to the objects). In addition,

the service can provide data redundancy for verification of the consistency between V2X objects (e.g., ghost objects indicated in vehicle-based messages transmitted via V2X communications) and V2X-sensor objects (e.g., physical objects perceived by sensors on V2X-capable network devices local to the objects).

Additional aspects of the present disclosure are described in more detail below.

As used herein, the terms "user equipment" (UE) and "network entity" are not intended to be specific or otherwise limited to any particular radio access technology (RAT), unless otherwise noted. In general, a UE may be any wireless communication device (e.g., a mobile phone, router, tablet computer, laptop computer, and/or tracking device, etc.), wearable (e.g., smartwatch, smart-glasses, wearable ring, and/or an extended reality (XR) device such as a virtual reality (VR) headset, an augmented reality (AR) headset or glasses, or a mixed reality (MR) headset), vehicle (e.g., automobile, motorcycle, bicycle, etc.), and/or Internet of Things (IoT) device, etc., used by a user to communicate over a wireless communications network. A UE may be mobile or may (e.g., at certain times) be stationary, and may communicate with a radio access network (RAN). As used herein, the term "UE" may be referred to interchangeably as an "access terminal" or "AT," a "client device," a "wireless device," a "subscriber device," a "subscriber terminal," a "subscriber station," a "user terminal" or "UT," a "mobile device," a "mobile terminal," a "mobile station," or variations thereof. Generally, UEs can communicate with a core network via a RAN, and through the core network the UEs can be connected with external networks such as the Internet and with other UEs. Of course, other mechanisms of connecting to the core network and/or the Internet are also possible for the UEs, such as over wired access networks, wireless local area network (WLAN) networks (e.g., based on IEEE 802.11 communication standards, etc.) and so on.

In some cases, a network entity can be implemented in an aggregated or monolithic base station or server architecture, or alternatively, in a disaggregated base station or server architecture, and may include one or more of a central unit (CU), a distributed unit (DU), a radio unit (RU), a Near-Real Time (Near-RT) RAN Intelligent Controller (RIC), or a Non-Real Time (Non-RT) RIC. In some cases, a network entity can include a server device, such as a Multi-access Edge Compute (MEC) device. A base station or server (e.g., with an aggregated/monolithic base station architecture or disaggregated base station architecture) may operate according to one of several RATs in communication with UEs, road side units (RSUs), and/or other devices depending on the network in which it is deployed, and may be alternatively referred to as an access point (AP), a network node, a NodeB (NB), an evolved NodeB (eNB), a next generation eNB (ng-eNB), a New Radio (NR) Node B (also referred to as a gNB or gNodeB), etc. A base station may be used primarily to support wireless access by UEs, including supporting data, voice, and/or signaling connections for the supported UEs. In some systems, a base station may provide edge node signaling functions while in other systems it may provide additional control and/or network management functions. A communication link through which UEs can send signals to a base station is called an uplink (UL) channel (e.g., a reverse traffic channel, a reverse control channel, an access channel, etc.). A communication link through which the base station can send signals to UEs is called a downlink (DL) or forward link channel (e.g., a paging channel, a control channel, a broadcast channel, or a forward traffic channel,

etc.). The term traffic channel (TCH), as used herein, can refer to either an uplink, reverse or downlink, and/or a forward traffic channel.

The term “network entity” or “base station” (e.g., with an aggregated/monolithic base station architecture or disaggregated base station architecture) may refer to a single physical TRP or to multiple physical TRPs that may or may not be co-located. For example, where the term “network entity” or “base station” refers to a single physical TRP, the physical TRP may be an antenna of the base station corresponding to a cell (or several cell sectors) of the base station. Where the term “network entity” or “base station” refers to multiple co-located physical TRPs, the physical TRPs may be an array of antennas (e.g., as in a multiple-input multiple-output (MIMO) system or where the base station employs beamforming) of the base station. Where the term “base station” refers to multiple non-co-located physical TRPs, the physical TRPs may be a distributed antenna system (DAS) (a network of spatially separated antennas connected to a common source via a transport medium) or a remote radio head (RRH) (a remote base station connected to a serving base station). Alternatively, the non-co-located physical TRPs may be the serving base station receiving the measurement report from the UE and a neighbor base station whose reference radio frequency (RF) signals (or simply “reference signals”) the UE is measuring. Because a TRP is the point from which a base station transmits and receives wireless signals, as used herein, references to transmission from or reception at a base station are to be understood as referring to a particular TRP of the base station.

In some implementations that support positioning of UEs, a network entity or base station may not support wireless access by UEs (e.g., may not support data, voice, and/or signaling connections for UEs), but may instead transmit reference signals to UEs to be measured by the UEs, and/or may receive and measure signals transmitted by the UEs. Such a base station may be referred to as a positioning beacon (e.g., when transmitting signals to UEs) and/or as a location measurement unit (e.g., when receiving and measuring signals from UEs).

A roadside unit (RSU) is a device that can transmit and receive messages over a communications link or interface (e.g., a cellular-based sidelink or PC5 interface, an 802.11 or WiFi™ based Dedicated Short Range Communication (DSRC) interface, and/or other interface) to and from one or more UEs, other RSUs, and/or base stations. An example of messages that can be transmitted and received by an RSU includes vehicle-to-everything (V2X) messages, which are described in more detail below. RSUs can be located on various transportation infrastructure systems, including roads, bridges, parking lots, toll booths, and/or other infrastructure systems. In some examples, an RSU can facilitate communication between UEs (e.g., vehicles, pedestrian user devices, and/or other UEs) and the transportation infrastructure systems. In some implementations, a RSU can be in communication with a server, base station, and/or other system that can perform centralized management functions.

An RSU can communicate with a communications system of a UE. For example, an intelligent transport system (ITS) of a UE (e.g., a vehicle and/or other UE) can be used to generate and sign messages for transmission to an RSU and to validate messages received from an RSU. An RSU can communicate (e.g., over a PC5 interface, DSRC interface, etc.) with vehicles traveling along a road, bridge, or other infrastructure system in order to obtain traffic-related data (e.g., time, speed, location, etc. of the vehicle). In some cases, in response to obtaining the traffic-related data, the

RSU can determine or estimate traffic congestion information (e.g., a start of traffic congestion, an end of traffic congestion, etc.), a travel time, and/or other information for a particular location. In some examples, the RSU can communicate with other RSUs (e.g., over a PC5 interface, DSRC interface, etc.) in order to determine the traffic-related data. The RSU can transmit the information (e.g., traffic congestion information, travel time information, and/or other information) to other vehicles, pedestrian UEs, and/or other UEs. For example, the RSU can broadcast or otherwise transmit the information to any UE (e.g., vehicle, pedestrian UE, etc.) that is in a coverage range of the RSU.

A radio frequency signal or “RF signal” comprises an electromagnetic wave of a given frequency that transports information through the space between a transmitter and a receiver. As used herein, a transmitter may transmit a single “RF signal” or multiple “RF signals” to a receiver. However, the receiver may receive multiple “RF signals” corresponding to each transmitted RF signal due to the propagation characteristics of RF signals through multipath channels. The same transmitted RF signal on different paths between the transmitter and receiver may be referred to as a “multipath” RF signal. As used herein, an RF signal may also be referred to as a “wireless signal” or simply a “signal” where it is clear from the context that the term “signal” refers to a wireless signal or an RF signal.

According to various aspects, FIG. 1 illustrates an exemplary wireless communications system **100**. The wireless communications system **100** (which may also be referred to as a wireless wide area network (WWAN)) can include various base stations **102** and various UEs **104**. In some aspects, the base stations **102** may also be referred to as “network entities” or “network nodes.” One or more of the base stations **102** can be implemented in an aggregated or monolithic base station architecture. Additionally or alternatively, one or more of the base stations **102** can be implemented in a disaggregated base station architecture, and may include one or more of a central unit (CU), a distributed unit (DU), a radio unit (RU), a Near-Real Time (Near-RT) RAN Intelligent Controller (MC), or a Non-Real Time (Non-RT) RIC. The base stations **102** can include macro cell base stations (high power cellular base stations) and/or small cell base stations (low power cellular base stations). In an aspect, the macro cell base station may include eNBs and/or ng-eNBs where the wireless communications system **100** corresponds to a long term evolution (LTE) network, or gNBs where the wireless communications system **100** corresponds to a NR network, or a combination of both, and the small cell base stations may include femtocells, picocells, microcells, etc.

The base stations **102** may collectively form a RAN and interface with a core network **170** (e.g., an evolved packet core (EPC) or a 5G core (5GC)) through backhaul links **122**, and through the core network **170** to one or more location servers **172** (which may be part of core network **170** or may be external to core network **170**). In addition to other functions, the base stations **102** may perform functions that relate to one or more of transferring user data, radio channel ciphering and deciphering, integrity protection, header compression, mobility control functions (e.g., handover, dual connectivity), inter-cell interference coordination, connection setup and release, load balancing, distribution for non-access stratum (NAS) messages, NAS node selection, synchronization, RAN sharing, multimedia broadcast multicast service (MBMS), subscriber and equipment trace, RAN information management (RIM), paging, positioning, and delivery of warning messages. The base stations **102** may

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communicate with each other directly or indirectly (e.g., through the EPC or 5GC) over backhaul links **134**, which may be wired and/or wireless.

The base stations **102** may wirelessly communicate with the UEs **104**. Each of the base stations **102** may provide communication coverage for a respective geographic coverage area **110**. In an aspect, one or more cells may be supported by a base station **102** in each coverage area **110**. A “cell” is a logical communication entity used for communication with a base station (e.g., over some frequency resource, referred to as a carrier frequency, component carrier, carrier, band, or the like), and may be associated with an identifier (e.g., a physical cell identifier (PCI), a virtual cell identifier (VCI), a cell global identifier (CGI)) for distinguishing cells operating via the same or a different carrier frequency. In some cases, different cells may be configured according to different protocol types (e.g., machine-type communication (MTC), narrowband IoT (NB-IoT), enhanced mobile broadband (eMBB), or others) that may provide access for different types of UEs. Because a cell is supported by a specific base station, the term “cell” may refer to either or both of the logical communication entity and the base station that supports it, depending on the context. In addition, because a TRP is typically the physical transmission point of a cell, the terms “cell” and “TRP” may be used interchangeably. In some cases, the term “cell” may also refer to a geographic coverage area of a base station (e.g., a sector), insofar as a carrier frequency can be detected and used for communication within some portion of geographic coverage areas **110**.

While neighboring macro cell base station **102** geographic coverage areas **110** may partially overlap (e.g., in a handover region), some of the geographic coverage areas **110** may be substantially overlapped by a larger geographic coverage area **110**. For example, a small cell base station **102'** may have a coverage area **110'** that substantially overlaps with the coverage area **110** of one or more macro cell base stations **102**. A network that includes both small cell and macro cell base stations may be known as a heterogeneous network. A heterogeneous network may also include home eNBs (HeNBs), which may provide service to a restricted group known as a closed subscriber group (CSG).

The communication links **120** between the base stations **102** and the UEs **104** may include uplink (also referred to as reverse link) transmissions from a UE **104** to a base station **102** and/or downlink (also referred to as forward link) transmissions from a base station **102** to a UE **104**. The communication links **120** may use MIMO antenna technology, including spatial multiplexing, beamforming, and/or transmit diversity. The communication links **120** may be through one or more carrier frequencies. Allocation of carriers may be asymmetric with respect to downlink and uplink (e.g., more or less carriers may be allocated for downlink than for uplink).

The wireless communications system **100** may further include a WLAN AP **150** in communication with WLAN stations (STAs) **152** via communication links **154** in an unlicensed frequency spectrum (e.g., 5 Gigahertz (GHz)). When communicating in an unlicensed frequency spectrum, the WLAN STAs **152** and/or the WLAN AP **150** may perform a clear channel assessment (CCA) or listen before talk (LBT) procedure prior to communicating in order to determine whether the channel is available. In some examples, the wireless communications system **100** can include devices (e.g., UEs, etc.) that communicate with one or more UEs **104**, base stations **102**, APs **150**, etc. utilizing

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the ultra-wideband (UWB) spectrum. The UWB spectrum can range from 3.1 to 10.5 GHz.

The small cell base station **102'** may operate in a licensed and/or an unlicensed frequency spectrum. When operating in an unlicensed frequency spectrum, the small cell base station **102'** may employ LTE or NR technology and use the same 5 GHz unlicensed frequency spectrum as used by the WLAN AP **150**. The small cell base station **102'**, employing LTE and/or 5G in an unlicensed frequency spectrum, may boost coverage to and/or increase capacity of the access network. NR in unlicensed spectrum may be referred to as NR-U. LTE in an unlicensed spectrum may be referred to as LTE-U, licensed assisted access (LAA), or MulteFire.

The wireless communications system **100** may further include a millimeter wave (mmW) base station **180** that may operate in mmW frequencies and/or near mmW frequencies in communication with a UE **182**. The mmW base station **180** may be implemented in an aggregated or monolithic base station architecture, or alternatively, in a disaggregated base station architecture (e.g., including one or more of a CU, a DU, a RU, a Near-RT RIC, or a Non-RT RIC). Extremely high frequency (EHF) is part of the RF in the electromagnetic spectrum. EHF has a range of 30 GHz to 300 GHz and a wavelength between 1 millimeter and 10 millimeters. Radio waves in this band may be referred to as a millimeter wave. Near mmW may extend down to a frequency of 3 GHz with a wavelength of 100 millimeters. The super high frequency (SHF) band extends between 3 GHz and 30 GHz, also referred to as centimeter wave. Communications using the mmW and/or near mmW radio frequency band have high path loss and a relatively short range. The mmW base station **180** and the UE **182** may utilize beamforming (transmit and/or receive) over an mmW communication link **184** to compensate for the extremely high path loss and short range. Further, it will be appreciated that in alternative configurations, one or more base stations **102** may also transmit using mmW or near mmW and beamforming. Accordingly, it will be appreciated that the foregoing illustrations are merely examples and should not be construed to limit the various aspects disclosed herein.

Transmit beamforming is a technique for focusing an RF signal in a specific direction. Traditionally, when a network node or entity (e.g., a base station) broadcasts an RF signal, it broadcasts the signal in all directions (omni-directionally). With transmit beamforming, the network node determines where a given target device (e.g., a UE) is located (relative to the transmitting network node) and projects a stronger downlink RF signal in that specific direction, thereby providing a faster (in terms of data rate) and stronger RF signal for the receiving device(s). To change the directionality of the RF signal when transmitting, a network node can control the phase and relative amplitude of the RF signal at each of the one or more transmitters that are broadcasting the RF signal. For example, a network node may use an array of antennas (referred to as a “phased array” or an “antenna array”) that creates a beam of RF waves that can be “steered” to point in different directions, without actually moving the antennas. Specifically, the RF current from the transmitter is fed to the individual antennas with the correct phase relationship so that the radio waves from the separate antennas add together to increase the radiation in a desired direction, while canceling to suppress radiation in undesired directions.

Transmit beams may be quasi-collocated, meaning that they appear to the receiver (e.g., a UE) as having the same parameters, regardless of whether or not the transmitting antennas of the network node themselves are physically

collocated. In NR, there are four types of quasi-collocation (QCL) relations. Specifically, a QCL relation of a given type means that certain parameters about a second reference RF signal on a second beam can be derived from information about a source reference RF signal on a source beam. Thus, if the source reference RF signal is QCL Type A, the receiver can use the source reference RF signal to estimate the Doppler shift, Doppler spread, average delay, and delay spread of a second reference RF signal transmitted on the same channel. If the source reference RF signal is QCL Type B, the receiver can use the source reference RF signal to estimate the Doppler shift and Doppler spread of a second reference RF signal transmitted on the same channel. If the source reference RF signal is QCL Type C, the receiver can use the source reference RF signal to estimate the Doppler shift and average delay of a second reference RF signal transmitted on the same channel. If the source reference RF signal is QCL Type D, the receiver can use the source reference RF signal to estimate the spatial receive parameter of a second reference RF signal transmitted on the same channel.

In receiving beamforming, the receiver uses a receive beam to amplify RF signals detected on a given channel. For example, the receiver can increase the gain setting and/or adjust the phase setting of an array of antennas in a particular direction to amplify (e.g., to increase the gain level of) the RF signals received from that direction. Thus, when a receiver is said to beamform in a certain direction, it means the beam gain in that direction is high relative to the beam gain along other directions, or the beam gain in that direction is the highest compared to the beam gain of other beams available to the receiver. This results in a stronger received signal strength, (e.g., reference signal received power (RSRP), reference signal received quality (RSRQ), signal-to-interference-plus-noise ratio (SINR), etc.) of the RF signals received from that direction.

Receive beams may be spatially related. A spatial relation means that parameters for a transmit beam for a second reference signal can be derived from information about a receive beam for a first reference signal. For example, a UE may use a particular receive beam to receive one or more reference downlink reference signals (e.g., positioning reference signals (PRS), tracking reference signals (TRS), phase tracking reference signal (PTRS), cell-specific reference signals (CRS), channel state information reference signals (CSI-RS), primary synchronization signals (PSS), secondary synchronization signals (SSS), synchronization signal blocks (SSBs), etc.) from a network node or entity (e.g., a base station). The UE can then form a transmit beam for sending one or more uplink reference signals (e.g., uplink positioning reference signals (UL-PRS), sounding reference signal (SRS), demodulation reference signals (DMRS), PTRS, etc.) to that network node or entity (e.g., a base station) based on the parameters of the receive beam.

Note that a “downlink” beam may be either a transmit beam or a receive beam, depending on the entity forming it. For example, if a network node or entity (e.g., a base station) is forming the downlink beam to transmit a reference signal to a UE, the downlink beam is a transmit beam. If the UE is forming the downlink beam, however, it is a receive beam to receive the downlink reference signal. Similarly, an “uplink” beam may be either a transmit beam or a receive beam, depending on the entity forming it. For example, if a network node or entity (e.g., a base station) is forming the uplink beam, it is an uplink receive beam, and if a UE is forming the uplink beam, it is an uplink transmit beam.

In 5G, the frequency spectrum in which wireless network nodes or entities (e.g., base stations **102/180**, UEs **104/182**) operate is divided into multiple frequency ranges, FR1 (from 450 to 6000 Megahertz (MHz)), FR2 (from 24250 to 52600 MHz), FR3 (above 52600 MHz), and FR4 (between FR1 and FR2). In a multi-carrier system, such as 5G, one of the carrier frequencies is referred to as the “primary carrier” or “anchor carrier” or “primary serving cell” or “PCell,” and the remaining carrier frequencies are referred to as “secondary carriers” or “secondary serving cells” or “SCells.” In carrier aggregation, the anchor carrier is the carrier operating on the primary frequency (e.g., FR1) utilized by a UE **104/182** and the cell in which the UE **104/182** either performs the initial radio resource control (RRC) connection establishment procedure or initiates the RRC connection re-establishment procedure. The primary carrier carries all common and UE-specific control channels, and may be a carrier in a licensed frequency (however, this is not always the case). A secondary carrier is a carrier operating on a second frequency (e.g., FR2) that may be configured once the RRC connection is established between the UE **104** and the anchor carrier and that may be used to provide additional radio resources. In some cases, the secondary carrier may be a carrier in an unlicensed frequency. The secondary carrier may contain only necessary signaling information and signals, for example, those that are UE-specific may not be present in the secondary carrier, since both primary uplink and downlink carriers are typically UE-specific. This means that different UEs **104/182** in a cell may have different downlink primary carriers. The same is true for the uplink primary carriers. The network is able to change the primary carrier of any UE **104/182** at any time. This is done, for example, to balance the load on different carriers. Because a “serving cell” (whether a PCell or an SCell) corresponds to a carrier frequency and/or component carrier over which some base station is communicating, the term “cell,” “serving cell,” “component carrier,” “carrier frequency,” and the like can be used interchangeably.

For example, still referring to FIG. 1, one of the frequencies utilized by the macro cell base stations **102** may be an anchor carrier (or “PCell”) and other frequencies utilized by the macro cell base stations **102** and/or the mmW base station **180** may be secondary carriers (“SCells”). In carrier aggregation, the base stations **102** and/or the UEs **104** may use spectrum up to Y MHz (e.g., 5, 10, 15, 20, 100 MHz) bandwidth per carrier up to a total of Yx MHz (x component carriers) for transmission in each direction. The component carriers may or may not be adjacent to each other on the frequency spectrum. Allocation of carriers may be asymmetric with respect to the downlink and uplink (e.g., more or less carriers may be allocated for downlink than for uplink). The simultaneous transmission and/or reception of multiple carriers enables the UE **104/182** to significantly increase its data transmission and/or reception rates. For example, two 20 MHz aggregated carriers in a multi-carrier system would theoretically lead to a two-fold increase in data rate (i.e., 40 MHz), compared to that attained by a single 20 MHz carrier.

In order to operate on multiple carrier frequencies, a base station **102** and/or a UE **104** is equipped with multiple receivers and/or transmitters. For example, a UE **104** may have two receivers, “Receiver 1” and “Receiver 2,” where “Receiver 1” is a multi-band receiver that can be tuned to band (i.e., carrier frequency) ‘X’ or band ‘Y,’ and “Receiver 2” is a one-band receiver tuneable to band ‘Z’ only. In this example, if the UE **104** is being served in band ‘X,’ band ‘X’ would be referred to as the PCell or the active carrier

frequency, and “Receiver 1” would need to tune from band ‘X’ to band ‘Y’ (an SCell) in order to measure band ‘Y’ (and vice versa). In contrast, whether the UE 104 is being served in band ‘X’ or band ‘Y,’ because of the separate “Receiver 2,” the UE 104 can measure band ‘Z’ without interrupting the service on band ‘X’ or band ‘Y.’

The wireless communications system 100 may further include a UE 164 that may communicate with a macro cell base station 102 over a communication link 120 and/or the mmW base station 180 over an mmW communication link 184. For example, the macro cell base station 102 may support a PCell and one or more SCells for the UE 164 and the mmW base station 180 may support one or more SCells for the UE 164.

The wireless communications system 100 may further include one or more UEs, such as UE 190, that connects indirectly to one or more communication networks via one or more device-to-device (D2D) peer-to-peer (P2P) links (referred to as “sidelinks”). In the example of FIG. 1, UE 190 has a D2D P2P link 192 with one of the UEs 104 connected to one of the base stations 102 (e.g., through which UE 190 may indirectly obtain cellular connectivity) and a D2D P2P link 194 with WLAN STA 152 connected to the WLAN AP 150 (through which UE 190 may indirectly obtain WLAN-based Internet connectivity). In an example, the D2D P2P links 192 and 194 may be supported with any well-known D2D RAT, such as LTE Direct (LTE-D), Wi-Fi Direct (Wi-Fi-D), Bluetooth®, and so on.

FIG. 2 is a diagram illustrating an example of a disaggregated base station architecture, which may be employed by the disclosed system for a misbehavior detection service for sharing connected and sensed objects, in accordance with some examples. Deployment of communication systems, such as 5G NR systems, may be arranged in multiple manners with various components or constituent parts. In a 5G NR system, or network, a network node, a network entity, a mobility element of a network, a radio access network (RAN) node, a core network node, a network element, or a network equipment, such as a base station (BS), or one or more units (or one or more components) performing base station functionality, may be implemented in an aggregated or disaggregated architecture. For example, a BS (such as a Node B (NB), evolved NB (eNB), NR BS, 5G NB, AP, a transmit receive point (TRP), or a cell, etc.) may be implemented as an aggregated base station (also known as a standalone BS or a monolithic BS) or a disaggregated base station.

An aggregated base station may be configured to utilize a radio protocol stack that is physically or logically integrated within a single RAN node. A disaggregated base station may be configured to utilize a protocol stack that is physically or logically distributed among two or more units (such as one or more central or centralized units (CUs), one or more distributed units (DUs), or one or more radio units (RUs)). In some aspects, a CU may be implemented within a RAN node, and one or more DUs may be co-located with the CU, or alternatively, may be geographically or virtually distributed throughout one or multiple other RAN nodes. The DUs may be implemented to communicate with one or more RUs. Each of the CU, DU and RU also can be implemented as virtual units, i.e., a virtual central unit (VCU), a virtual distributed unit (VDU), or a virtual radio unit (VRU).

Base station-type operation or network design may consider aggregation characteristics of base station functionality. For example, disaggregated base stations may be utilized in an integrated access backhaul (IAB) network, an open radio access network (O-RAN (such as the network con-

figuration sponsored by the O-RAN Alliance)), or a virtualized radio access network (vRAN, also known as a cloud radio access network (C-RAN)). Disaggregation may include distributing functionality across two or more units at various physical locations, as well as distributing functionality for at least one unit virtually, which can enable flexibility in network design. The various units of the disaggregated base station, or disaggregated RAN architecture, can be configured for wired or wireless communication with at least one other unit.

As previously mentioned, FIG. 2 shows a diagram illustrating an example disaggregated base station 201 architecture. The disaggregated base station 201 architecture may include one or more central units (CUs) 211 that can communicate directly with a core network 223 via a backhaul link, or indirectly with the core network 223 through one or more disaggregated base station units (such as a Near-Real Time (Near-RT) RAN Intelligent Controller (RIC) 227 via an E2 link, or a Non-Real Time (Non-RT) RIC 217 associated with a Service Management and Orchestration (SMO) Framework 207, or both). A CU 211 may communicate with one or more distributed units (DUs) 231 via respective midhaul links, such as an F1 interface. The DUs 231 may communicate with one or more radio units (RUs) 241 via respective fronthaul links. The RUs 241 may communicate with respective UEs 221 via one or more RF access links. In some implementations, the UE 221 may be simultaneously served by multiple RUs 241.

Each of the units, i.e., the CUs 211, the DUs 231, the RUs 241, as well as the Near-RT RICs 227, the Non-RT RICs 217 and the SMO Framework 207, may include one or more interfaces or be coupled to one or more interfaces configured to receive or transmit signals, data, or information (collectively, signals) via a wired or wireless transmission medium. Each of the units, or an associated processor or controller providing instructions to the communication interfaces of the units, can be configured to communicate with one or more of the other units via the transmission medium. For example, the units can include a wired interface configured to receive or transmit signals over a wired transmission medium to one or more of the other units. Additionally, the units can include a wireless interface, which may include a receiver, a transmitter or transceiver (such as an RF transceiver), configured to receive or transmit signals, or both, over a wireless transmission medium to one or more of the other units.

In some aspects, the CU 211 may host one or more higher layer control functions. Such control functions can include radio resource control (RRC), packet data convergence protocol (PDCP), service data adaptation protocol (SDAP), or the like. Each control function can be implemented with an interface configured to communicate signals with other control functions hosted by the CU 211. The CU 211 may be configured to handle user plane functionality (i.e., Central Unit-User Plane (CU-UP)), control plane functionality (i.e., Central Unit-Control Plane (CU-CP)), or a combination thereof. In some implementations, the CU 211 can be logically split into one or more CU-UP units and one or more CU-CP units. The CU-UP unit can communicate bidirectionally with the CU-CP unit via an interface, such as the E1 interface when implemented in an O-RAN configuration. The CU 211 can be implemented to communicate with the DU 131, as necessary, for network control and signaling.

The DU 231 may correspond to a logical unit that includes one or more base station functions to control the operation of one or more RUs 241. In some aspects, the DU 231 may host one or more of a radio link control (RLC) layer, a

medium access control (MAC) layer, and one or more high physical (PHY) layers (such as modules for forward error correction (FEC) encoding and decoding, scrambling, modulation and demodulation, or the like) depending, at least in part, on a functional split, such as those defined by the 3rd Generation Partnership Project (3GPP). In some aspects, the DU 231 may further host one or more low PHY layers. Each layer (or module) can be implemented with an interface configured to communicate signals with other layers (and modules) hosted by the DU 231, or with the control functions hosted by the CU 211.

Lower-layer functionality can be implemented by one or more RUs 241. In some deployments, an RU 241, controlled by a DU 231, may correspond to a logical node that hosts RF processing functions, or low-PHY layer functions (such as performing fast Fourier transform (FFT), inverse FFT (iFFT), digital beamforming, physical random access channel (PRACH) extraction and filtering, or the like), or both, based at least in part on the functional split, such as a lower layer functional split. In such an architecture, the RU(s) 241 can be implemented to handle over the air (OTA) communication with one or more UEs 221. In some implementations, real-time and non-real-time aspects of control and user plane communication with the RU(s) 241 can be controlled by the corresponding DU 231. In some scenarios, this configuration can enable the DU(s) 231 and the CU 211 to be implemented in a cloud-based RAN architecture, such as a vRAN architecture.

The SMO Framework 207 may be configured to support RAN deployment and provisioning of non-virtualized and virtualized network elements. For non-virtualized network elements, the SMO Framework 207 may be configured to support the deployment of dedicated physical resources for RAN coverage requirements which may be managed via an operations and maintenance interface (such as an O1 interface). For virtualized network elements, the SMO Framework 207 may be configured to interact with a cloud computing platform (such as an open cloud (O-Cloud) 291) to perform network element life cycle management (such as to instantiate virtualized network elements) via a cloud computing platform interface (such as an O2 interface). Such virtualized network elements can include, but are not limited to, CUs 211, DUs 231, RUs 241 and Near-RT RICs 227. In some implementations, the SMO Framework 207 can communicate with a hardware aspect of a 4G RAN, such as an open eNB (O-eNB) 213, via an O1 interface. Additionally, in some implementations, the SMO Framework 207 can communicate directly with one or more RUs 241 via an O1 interface. The SMO Framework 207 also may include a Non-RT RIC 217 configured to support functionality of the SMO Framework 207.

The Non-RT RIC 217 may be configured to include a logical function that enables non-real-time control and optimization of RAN elements and resources, Artificial Intelligence/Machine Learning (AI/ML) workflows including model training and updates, or policy-based guidance of applications/features in the Near-RT RIC 227. The Non-RT RIC 217 may be coupled to or communicate with (such as via an AI interface) the Near-RT RIC 227. The Near-RT MC 227 may be configured to include a logical function that enables near-real-time control and optimization of RAN elements and resources via data collection and actions over an interface (such as via an E2 interface) connecting one or more CUs 211, one or more DUs 231, or both, as well as an O-eNB 213, with the Near-RT MC 227.

In some implementations, to generate AI/ML models to be deployed in the Near-RT RIC 227, the Non-RT MC 217 may

receive parameters or external enrichment information from external servers. Such information may be utilized by the Near-RT RIC 227 and may be received at the SMO Framework 207 or the Non-RT MC 217 from non-network data sources or from network functions. In some examples, the Non-RT MC 217 or the Near-RT MC 227 may be configured to tune RAN behavior or performance. For example, the Non-RT MC 217 may monitor long-term trends and patterns for performance and employ AI/ML models to perform corrective actions through the SMO Framework 207 (such as reconfiguration via O1) or via creation of RAN management policies (such as A1 policies).

FIG. 3 illustrates examples of different communication mechanisms used by various UEs. In one example of side-link communications, FIG. 3 illustrates a vehicle 304, a vehicle 305, and an RSU 303 communicating with each other using PC5, DSRC, or other device to device direct signaling interfaces. In addition, the vehicle 304 and the vehicle 305 may communicate with a base station 302 (shown as BS 302) using a network (Uu) interface. The base station 302 can include a gNB in some examples. FIG. 3 also illustrates a user device 307 communicating with the base station 302 using a network (Uu) interface. As described below, functionalities can be transferred from a vehicle (e.g., vehicle 304) to a user device (e.g., user device 307) based on one or more characteristics or factors (e.g., temperature, humidity, etc.). In one illustrative example, V2X functionality can be transitioned from the vehicle 304 to the user device 307, after which the user device 307 can communicate with other vehicles (e.g., vehicle 305) over a PC5 interface (or other device to device direct interface, such as a DSRC interface), as shown in FIG. 3.

While FIG. 3 illustrates a particular number of vehicles (e.g., two vehicles 304 and 305) communicating with each other and/or with RSU 303, BS 302, and/or user device 307, the present disclosure is not limited thereto. For instance, tens or hundreds of such vehicles may be communicating with one another and/or with RSU 303, BS 302, and/or user device 307. At any given point in time, each such vehicle, RSU 303, BS 302, and/or user device 307 may transmit various types of information as messages to other nearby vehicles resulting in each vehicle (e.g., vehicles 304 and/or 305), RSU 303, BS 302, and/or user device 307 receiving hundreds or thousands of messages from other nearby vehicles, RSUs, base stations, and/or other UEs per second.

While PC5 interfaces are shown in FIG. 3, the various UEs (e.g., vehicles, user devices, etc.) and RSU(s) can communicate directly using any suitable type of direct interface, such as an 802.11 DSRC interface, a Bluetooth™ interface, and/or other interface. For example, a vehicle can communicate with a user device over a direct communications interface (e.g., using PC5 and/or DSRC), a vehicle can communicate with another vehicle over the direct communications interface, a user device can communicate with another user device over the direct communications interface, a UE (e.g., a vehicle, user device, etc.) can communicate with an RSU over the direct communications interface, an RSU can communicate with another RSU over the direct communications interface, and the like.

FIG. 4 is a block diagram illustrating an example a vehicle computing system 450 of a vehicle 404. The vehicle 404 is an example of a UE that can communicate with a network (e.g., an eNB, a gNB, a positioning beacon, a location measurement unit, and/or other network entity) over a Uu interface and with other UEs using V2X communications over a PC5 interface (or other device to device direct interface, such as a DSRC interface). As shown, the vehicle

computing system 450 can include at least a power management system 451, a control system 452, an infotainment system 454, an intelligent transport system (ITS) 455, one or more sensor systems 456, and a communications system 458. In some cases, the vehicle computing system 450 can include or can be implemented using any type of processing device or system, such as one or more central processing units (CPUs), digital signal processors (DSPs), application specific integrated circuits (ASICs), field programmable gate arrays (FPGAs), application processors (APs), graphics processing units (GPUs), vision processing units (VPUs), Neural Network Signal Processors (NSPs), microcontrollers, dedicated hardware, any combination thereof, and/or other processing device or system.

The control system 452 can be configured to control one or more operations of the vehicle 404, the power management system 451, the computing system 450, the infotainment system 454, the ITS 455, and/or one or more other systems of the vehicle 404 (e.g., a braking system, a steering system, a safety system other than the ITS 455, a cabin system, and/or other system). In some examples, the control system 452 can include one or more electronic control units (ECUs). An ECU can control one or more of the electrical systems or subsystems in a vehicle. Examples of specific ECUs that can be included as part of the control system 452 include an engine control module (ECM), a powertrain control module (PCM), a transmission control module (TCM), a brake control module (BCM), a central control module (CCM), a central timing module (CTM), among others. In some cases, the control system 452 can receive sensor signals from the one or more sensor systems 456 and can communicate with other systems of the vehicle computing system 450 to operate the vehicle 404.

The vehicle computing system 450 also includes a power management system 451. In some implementations, the power management system 451 can include a power management integrated circuit (PMIC), a standby battery, and/or other components. In some cases, other systems of the vehicle computing system 450 can include one or more PMICs, batteries, and/or other components. The power management system 451 can perform power management functions for the vehicle 404, such as managing a power supply for the computing system 450 and/or other parts of the vehicle. For example, the power management system 451 can provide a stable power supply in view of power fluctuations, such as based on starting an engine of the vehicle. In another example, the power management system 451 can perform thermal monitoring operations, such as by checking ambient and/or transistor junction temperatures. In another example, the power management system 451 can perform certain functions based on detecting a certain temperature level, such as causing a cooling system (e.g., one or more fans, an air conditioning system, etc.) to cool certain components of the vehicle computing system 450 (e.g., the control system 452, such as one or more ECUs), shutting down certain functionalities of the vehicle computing system 450 (e.g., limiting the infotainment system 454, such as by shutting off one or more displays, disconnecting from a wireless network, etc.), among other functions.

The vehicle computing system 450 further includes a communications system 458. The communications system 458 can include both software and hardware components for transmitting signals to and receiving signals from a network (e.g., a gNB or other network entity over a Uu interface) and/or from other UEs (e.g., to another vehicle or UE over a PC5 interface, WiFi interface (e.g., DSRC), Bluetooth™ interface, and/or other wireless and/or wired interface). For

example, the communications system 458 is configured to transmit and receive information wirelessly over any suitable wireless network (e.g., a 3G network, 4G network, 5G network, WiFi network, Bluetooth™ network, and/or other network). The communications system 458 includes various components or devices used to perform the wireless communication functionalities, including an original equipment manufacturer (OEM) subscriber identity module (referred to as a SIM or SIM card) 460, a user SIM 462, and a modem 464. While the vehicle computing system 450 is shown as having two SIMs and one modem, the computing system 450 can have any number of SIMs (e.g., one SIM or more than two SIMs) and any number of modems (e.g., one modem, two modems, or more than two modems) in some implementations.

A SIM is a device (e.g., an integrated circuit) that can securely store an international mobile subscriber identity (IMSI) number and a related key (e.g., an encryption-decryption key) of a particular subscriber or user. The IMSI and key can be used to identify and authenticate the subscriber on a particular UE. The OEM SIM 460 can be used by the communications system 458 for establishing a wireless connection for vehicle-based operations, such as for conducting emergency-calling (eCall) functions, communicating with a communications system of the vehicle manufacturer (e.g., for software updates, etc.), among other operations. The OEM SIM 460 can be important for the OEM SIM to support critical services, such as eCall for making emergency calls in the event of a car accident or other emergency. For instance, eCall can include a service that automatically dials an emergency number (e.g., “9-1-1” in the United States, “1-1-2” in Europe, etc.) in the event of a vehicle accident and communicates a location of the vehicle to the emergency services, such as a police department, fire department, etc.

The user SIM 462 can be used by the communications system 458 for performing wireless network access functions in order to support a user data connection (e.g., for conducting phone calls, messaging, infotainment related services, among others). In some cases, a user device of a user can connect with the vehicle computing system 450 over an interface (e.g., over PC5, Bluetooth™, WiFi™ (e.g., DSRC), a universal serial bus (USB) port, and/or other wireless or wired interface). Once connected, the user device can transfer wireless network access functionality from the user device to communications system 458 the vehicle, in which case the user device can cease performance of the wireless network access functionality (e.g., during the period in which the communications system 458 is performing the wireless access functionality). The communications system 458 can begin interacting with a base station to perform one or more wireless communication operations, such as facilitating a phone call, transmitting and/or receiving data (e.g., messaging, video, audio, etc.), among other operations. In such cases, other components of the vehicle computing system 450 can be used to output data received by the communications system 458. For example, the infotainment system 454 (described below) can display video received by the communications system 458 on one or more displays and/or can output audio received by the communications system 458 using one or more speakers.

A modem is a device that modulates one or more carrier wave signals to encode digital information for transmission, and demodulates signals to decode the transmitted information. The modem 464 (and/or one or more other modems of the communications system 458) can be used for communication of data for the OEM SIM 460 and/or the user SIM

462. In some examples, the modem **464** can include a 4G (or LTE) modem and another modem (not shown) of the communications system **458** can include a 5G (or NR) modem. In some examples, the communications system **458** can include one or more Bluetooth™ modems (e.g., for Bluetooth™ Low Energy (BLE) or other type of Bluetooth communications), one or more WiFi™ modems (e.g., for DSRC communications and/or other WiFi communications), wideband modems (e.g., an ultra-wideband (UWB) modem), any combination thereof, and/or other types of modems.

In some cases, the modem **464** (and/or one or more other modems of the communications system **458**) can be used for performing V2X communications (e.g., with other vehicles for V2V communications, with other devices for D2D communications, with infrastructure systems for V2I communications, with pedestrian UEs for V2P communications, etc.). In some examples, the communications system **458** can include a V2X modem used for performing V2X communications (e.g., sidelink communications over a PC5 interface or DSRC interface), in which case the V2X modem can be separate from one or more modems used for wireless network access functions (e.g., for network communications over a network/Uu interface and/or sidelink communications other than V2X communications).

In some examples, the communications system **458** can be or can include a telematics control unit (TCU). In some implementations, the TCU can include a network access device (NAD) (also referred to in some cases as a network control unit or NCU). The NAD can include the modem **464**, any other modem not shown in FIG. 4, the OEM SIM **460**, the user SIM **462**, and/or other components used for wireless communications. In some examples, the communications system **458** can include a Global Navigation Satellite System (GNSS). In some cases, the GNSS can be part of the one or more sensor systems **456**, as described below. The GNSS can provide the ability for the vehicle computing system **450** to perform one or more location services, navigation services, and/or other services that can utilize GNSS functionality.

In some cases, the communications system **458** can further include one or more wireless interfaces (e.g., including one or more transceivers and one or more baseband processors for each wireless interface) for transmitting and receiving wireless communications, one or more wired interfaces (e.g., a serial interface such as a universal serial bus (USB) input, a lightening connector, and/or other wired interface) for performing communications over one or more hardwired connections, and/or other components that can allow the vehicle **404** to communicate with a network and/or other UEs.

The vehicle computing system **450** can also include an infotainment system **454** that can control content and one or more output devices of the vehicle **404** that can be used to output the content. The infotainment system **454** can also be referred to as an in-vehicle infotainment (IVI) system or an In-car entertainment (ICE) system. The content can include navigation content, media content (e.g., video content, music or other audio content, and/or other media content), among other content. The one or more output devices can include one or more graphical user interfaces, one or more displays, one or more speakers, one or more extended reality devices (e.g., a VR, AR, and/or MR headset), one or more haptic feedback devices (e.g., one or more devices configured to vibrate a seat, steering wheel, and/or other part of the vehicle **404**), and/or other output device.

In some examples, the computing system **450** can include the intelligent transport system (ITS) **455**. In some examples, the ITS **455** can be used for implementing V2X communications. For example, an ITS stack of the ITS **455** can generate V2X messages based on information from an application layer of the ITS. In some cases, the application layer can determine whether certain conditions have been met for generating messages for use by the ITS **455** and/or for generating messages that are to be sent to other vehicles (for V2V communications), to pedestrian UEs (for V2P communications), and/or to infrastructure systems (for V2I communications). In some cases, the communications system **458** and/or the ITS **455** can obtain car access network (CAN) information (e.g., from other components of the vehicle via a CAN bus). In some examples, the communications system **458** (e.g., a TCU NAD) can obtain the CAN information via the CAN bus and can send the CAN information to a PHY/MAC layer of the ITS **455**. The ITS **455** can provide the CAN information to the ITS stack of the ITS **455**. The CAN information can include vehicle related information, such as a heading of the vehicle, speed of the vehicle, breaking information, among other information. The CAN information can be continuously or periodically (e.g., every 1 millisecond (ms), every 10 ms, or the like) provided to the ITS **455**.

The conditions used to determine whether to generate messages can be determined using the CAN information based on safety-related applications and/or other applications, including applications related to road safety, traffic efficiency, infotainment, business, and/or other applications. In one illustrative example, the ITS **455** can perform lane change assistance or negotiation. For instance, using the CAN information, the ITS **455** can determine that a driver of the vehicle **404** is attempting to change lanes from a current lane to an adjacent lane (e.g., based on a blinker being activated, based on the user veering or steering into an adjacent lane, etc.). Based on determining the vehicle **404** is attempting to change lanes, the ITS **455** can determine a lane-change condition has been met that is associated with a message to be sent to other vehicles that are nearby the vehicle in the adjacent lane. The ITS **455** can trigger the ITS stack to generate one or more messages for transmission to the other vehicles, which can be used to negotiate a lane change with the other vehicles. Other examples of applications include forward collision warning, automatic emergency breaking, lane departure warning, pedestrian avoidance or protection (e.g., when a pedestrian is detected near the vehicle **404**, such as based on V2P communications with a UE of the user), traffic sign recognition, among others.

The ITS **455** can use any suitable protocol to generate messages (e.g., V2X messages). Examples of protocols that can be used by the ITS **455** include one or more Society of Automotive Engineering (SAE) standards, such as SAE J2735, SAE J2945, SAE J3161, and/or other standards, which are hereby incorporated by reference in their entirety and for all purposes.

A security layer of the ITS **455** can be used to securely sign messages from the ITS stack that are sent to and verified by other UEs configured for V2X communications, such as other vehicles, pedestrian UEs, and/or infrastructure systems. The security layer can also verify messages received from such other UEs. In some implementations, the signing and verification processes can be based on a security context of the vehicle. In some examples, the security context may include one or more encryption-decryption algorithms, a public and/or private key used to generate a signature using an encryption-decryption algorithm, and/or other informa-

tion. For example, each ITS message generated by the ITS 455 can be signed by the security layer of the ITS 455. The signature can be derived using a public key and an encryption-decryption algorithm. A vehicle, pedestrian UE, and/or infrastructure system receiving a signed message can verify the signature to make sure the message is from an authorized vehicle. In some examples, the one or more encryption-decryption algorithms can include one or more symmetric encryption algorithms (e.g., advanced encryption standard (AES), data encryption standard (DES), and/or other symmetric encryption algorithm), one or more asymmetric encryption algorithms using public and private keys (e.g., Rivest-Shamir-Adleman (RSA) and/or other asymmetric encryption algorithm), and/or other encryption-decryption algorithm.

In some examples, the ITS 455 can determine certain operations (e.g., V2X-based operations) to perform based on messages received from other UEs. The operations can include safety-related and/or other operations, such as operations for road safety, traffic efficiency, infotainment, business, and/or other applications. In some examples, the operations can include causing the vehicle (e.g., the control system 452) to perform automatic functions, such as automatic braking, automatic steering (e.g., to maintain a heading in a particular lane), automatic lane change negotiation with other vehicles, among other automatic functions. In one illustrative example, a message can be received by the communications system 458 from another vehicle (e.g., over a PC5 interface, a DSRC interface, or other device to device direct interface) indicating that the other vehicle is coming to a sudden stop. In response to receiving the message, the ITS stack can generate a message or instruction and can send the message or instruction to the control system 452, which can cause the control system 452 to automatically break the vehicle 404 so that it comes to a stop before making impact with the other vehicle. In other illustrative examples, the operations can include triggering display of a message alerting a driver that another vehicle is in the lane next to the vehicle, a message alerting the driver to stop the vehicle, a message alerting the driver that a pedestrian is in an upcoming cross-walk, a message alerting the driver that a toll booth is within a certain distance (e.g., within 1 mile) of the vehicle, among others.

In some examples, the ITS 455 can receive a large number of messages from the other UEs (e.g., vehicles, RSUs, etc.), in which case the ITS 455 will authenticate (e.g., decode and decrypt) each of the messages and/or determine which operations to perform. Such a large number of messages can lead to a large computational load for the vehicle computing system 450. In some cases, the large computational load can cause a temperature of the computing system 450 to increase. Rising temperatures of the components of the computing system 450 can adversely affect the ability of the computing system 450 to process the large number of incoming messages. One or more functionalities can be transitioned from the vehicle 404 to another device (e.g., a user device, a RSU, etc.) based on a temperature of the vehicle computing system 450 (or component thereof) exceeding or approaching one or more thermal levels. Transitioning the one or more functionalities can reduce the computational load on the vehicle 404, helping to reduce the temperature of the components. A thermal load balancer can be provided that enable the vehicle computing system 450 to perform thermal based load balancing to control a processing load depending on the temperature of the computing system 450 and processing capacity of the vehicle computing system 450.

The computing system 450 further includes one or more sensor systems 456 (e.g., a first sensor system through an Nth sensor system, where N is a value equal to or greater than 0). When including multiple sensor systems, the sensor system(s) 456 can include different types of sensor systems that can be arranged on or in different parts the vehicle 404. The sensor system(s) 456 can include one or more camera sensor systems, LIDAR sensor systems, radio detection and ranging (RADAR) sensor systems, Electromagnetic Detection and Ranging (EmDAR) sensor systems, Sound Navigation and Ranging (SONAR) sensor systems, Sound Detection and Ranging (SODAR) sensor systems, Global Navigation Satellite System (GNSS) receiver systems (e.g., one or more Global Positioning System (GPS) receiver systems), accelerometers, gyroscopes, inertial measurement units (IMUs), infrared sensor systems, laser rangefinder systems, ultrasonic sensor systems, infrasonic sensor systems, microphones, any combination thereof, and/or other sensor systems. It should be understood that any number of sensors or sensor systems can be included as part of the computing system 450 of the vehicle 404.

While the vehicle computing system 450 is shown to include certain components and/or systems, one of ordinary skill will appreciate that the vehicle computing system 450 can include more or fewer components than those shown in FIG. 4. For example, the vehicle computing system 450 can also include one or more input devices and one or more output devices (not shown). In some implementations, the vehicle computing system 450 can also include (e.g., as part of or separate from the control system 452, the infotainment system 454, the communications system 458, and/or the sensor system(s) 456) at least one processor and at least one memory having computer-executable instructions that are executed by the at least one processor. The at least one processor is in communication with and/or electrically connected to (referred to as being “coupled to” or “communicatively coupled”) the at least one memory. The at least one processor can include, for example, one or more microcontrollers, one or more central processing units (CPUs), one or more field programmable gate arrays (FPGAs), one or more graphics processing units (GPUs), one or more application processors (e.g., for running or executing one or more software applications), and/or other processors. The at least one memory can include, for example, read-only memory (ROM), random access memory (RAM) (e.g., static RAM (SRAM)), electrically erasable programmable read-only memory (EEPROM), flash memory, one or more buffers, one or more databases, and/or other memory. The computer-executable instructions stored in or on the at least memory can be executed to perform one or more of the functions or operations described herein.

FIG. 5 illustrates an example of a computing system 570 of a user device 507. The user device 507 is an example of a UE that can be used by an end-user. For example, the user device 507 can include a mobile phone, router, tablet computer, laptop computer, tracking device, wearable device (e.g., a smart watch, glasses, an XR device, etc.), Internet of Things (IoT) device, and/or other device used by a user to communicate over a wireless communications network. The computing system 570 includes software and hardware components that can be electrically or communicatively coupled via a bus 589 (or may otherwise be in communication, as appropriate). For example, the computing system 570 includes one or more processors 584. The one or more processors 584 can include one or more CPUs, ASICs, FPGAs, APs, GPUs, VPUs, NSPs, microcontrollers, dedicated hardware, any combination thereof, and/or other pro-

cessing device or system. The bus 589 can be used by the one or more processors 584 to communicate between cores and/or with the one or more memory devices 586.

The computing system 570 may also include one or more memory devices 586, one or more digital signal processors (DSPs) 582, one or more SIMs 574, one or more modems 576, one or more wireless transceivers 578, an antenna 587, one or more input devices 572 (e.g., a camera, a mouse, a keyboard, a touch sensitive screen, a touch pad, a keypad, a microphone, and/or the like), and one or more output devices 580 (e.g., a display, a speaker, a printer, and/or the like).

The one or more wireless transceivers 578 can receive wireless signals (e.g., signal 588) via antenna 587 from one or more other devices, such as other user devices, vehicles (e.g., vehicle 404 of FIG. 4 described above), network devices (e.g., base stations such as eNBs and/or gNBs, WiFi routers, etc.), cloud networks, and/or the like. In some examples, the computing system 570 can include multiple antennae. The wireless signal 588 may be transmitted via a wireless network. The wireless network may be any wireless network, such as a cellular or telecommunications network (e.g., 3G, 4G, 5G, etc.), wireless local area network (e.g., a WiFi network), a Bluetooth™ network, and/or other network. In some examples, the one or more wireless transceivers 578 may include an RF front end including one or more components, such as an amplifier, a mixer (also referred to as a signal multiplier) for signal down conversion, a frequency synthesizer (also referred to as an oscillator) that provides signals to the mixer, a baseband filter, an analog-to-digital converter (ADC), one or more power amplifiers, among other components. The RF front-end can generally handle selection and conversion of the wireless signals 588 into a baseband or intermediate frequency and can convert the RF signals to the digital domain.

In some cases, the computing system 570 can include a coding-decoding device (or CODEC) configured to encode and/or decode data transmitted and/or received using the one or more wireless transceivers 578. In some cases, the computing system 570 can include an encryption-decryption device or component configured to encrypt and/or decrypt data (e.g., according to the AES and/or DES standard) transmitted and/or received by the one or more wireless transceivers 578.

The one or more SIMs 574 can each securely store an IMSI number and related key assigned to the user of the user device 507. As noted above, the IMSI and key can be used to identify and authenticate the subscriber when accessing a network provided by a network service provider or operator associated with the one or more SIMs 574. The one or more modems 576 can modulate one or more signals to encode information for transmission using the one or more wireless transceivers 578. The one or more modems 576 can also demodulate signals received by the one or more wireless transceivers 578 in order to decode the transmitted information. In some examples, the one or more modems 576 can include a 4G (or LTE) modem, a 5G (or NR) modem, a modem configured for V2X communications, and/or other types of modems. The one or more modems 576 and the one or more wireless transceivers 578 can be used for communicating data for the one or more SIMs 574.

The computing system 570 can also include (and/or be in communication with) one or more non-transitory machine-readable storage media or storage devices (e.g., one or more memory devices 586), which can include, without limitation, local and/or network accessible storage, a disk drive, a drive array, an optical storage device, a solid-state storage

device such as a RAM and/or a ROM, which can be programmable, flash-updateable and/or the like. Such storage devices may be configured to implement any appropriate data storage, including without limitation, various file systems, database structures, and/or the like.

In various aspects, functions may be stored as one or more computer-program products (e.g., instructions or code) in memory device(s) 586 and executed by the one or more processor(s) 584 and/or the one or more DSPs 582. The computing system 570 can also include software elements (e.g., located within the one or more memory devices 586), including, for example, an operating system, device drivers, executable libraries, and/or other code, such as one or more application programs, which may comprise computer programs implementing the functions provided by various aspects, and/or may be designed to implement methods and/or configure systems, as described herein.

FIG. 6 illustrates an example 600 of wireless communication between devices based on sidelink communication, such as V2X or other D2D communication. The communication may be based on a slot structure. For example, transmitting UE 602 may transmit a transmission 614, e.g., comprising a control channel and/or a corresponding data channel, that may be received by receiving UEs 604, 606, 608. At least one UE may comprise an autonomous vehicle or an unmanned aerial vehicle. A control channel may include information for decoding a data channel and may also be used by receiving device to avoid interference by refraining from transmitting on the occupied resources during a data transmission. The number of TTIs, as well as the RBs that will be occupied by the data transmission, may be indicated in a control message from the transmitting device. The UEs 602, 604, 606, 608 may each be capable of operating as a transmitting device in addition to operating as a receiving device. Thus, UEs 606, 608 are illustrated as transmitting transmissions 616, 620. The transmissions 614, 616, 620 (and 618 by RSU 607) may be broadcast or multicast to nearby devices. For example, UE 614 may transmit communication intended for receipt by other UEs within a range 601 of UE 614. Additionally/alternatively, RSU 607 may receive communication from and/or transmit communication 618 to UEs 602, 604, 606, 608. UE 602, 604, 606, 608 or RSU 607 may comprise a detection component. UE 602, 604, 606, 608 or RSU 607 may also comprise a BSM or mitigation component.

In wireless communications, such as V2X communications, V2X entities may perform sensor sharing with other V2X entities for cooperative and automated driving. For example, with reference to diagram 700 of FIG. 7A, the host vehicle (HV) 702 may detect a number of items within its environment. For example, the HV 702 may detect the presence of the non-V2X entity (NV) 706 at block 732. The HV 702 may inform other entities, such as a first remote vehicle (RV1) 704 or a road side unit (RSU) 708, about the presence of the NV 706, if the RV1 704 and/or the RSU 708, by themselves, are unable to detect the NV 706. The HV 702 informing the RV1 704 and/or the RSU 708 about the NV 706 is a sharing of sensor information. With reference to diagram 710 of FIG. 7B, the HV 702 may detect a physical obstacle 712, such as a pothole, debris, or an object that may be an obstruction in the path of the HV 702 and/or RV1 704 that has not yet been detected by RV1 704 and/or RSU 708. The HV 702 may inform the RV1 704 and/or the RSU 708 of the obstacle 712, such that the obstacle 712 may be avoided. With reference to diagram 720 of FIG. 7C, the HV 702 may detect the presence of a vulnerable road user (VRU) 722 and may share the detection of the VRU 722 with the RV1 704

and the RSU 708, in instances where the RSU 708 and/or RV1 704 may not be able to detect the VRU 722. With reference to diagram 730 of FIG. 7D, the HV, upon detection of a nearby entity (e.g., NV, VRU, obstacle) may transmit a sensor data sharing message (SDSM) 734 to the RV and/or the RSU to share the detection of the entity. The SDSM 734 may be a broadcast message such that any receiving device within the vicinity of the HV may receive the message. In some instances, the shared information may be relayed to other entities, such as RVs. For example, with reference to diagram 800 of FIG. 8, the HV 802 may detect the presence of the NV 806 and/or the VRU 822. The HV 802 may broadcast the SDSM 810 to the RSU 808 to report the detection of NV 806 and/or VRU 822. The RSU 808 may relay the SDSM 810 received from the HV 802 to remote vehicles such that the remote vehicles are aware of the presence of the NV 806 and/or VRU 822. For example, the RSU 808 may transmit an SDSM 812 to the RV1 804, where the SDSM 812 includes information related to the detection of NV 806 and/or VRU 822.

FIG. 9 is a diagram illustrating an example of a system 900 for sensor sharing in wireless communications (e.g., V2X communications), in accordance with some aspects of the present disclosure. In FIG. 9, the system 900 is shown to include a plurality of equipped (e.g., V2X capable) network devices. The plurality of equipped network devices includes vehicles (e.g., automobiles) 910a, 910b, 910c, 910d, and an RSU 905. Also shown are a plurality of non-equipped network devices, which include a non-equipped vehicle 920, a VRU (e.g., a bicyclist) 930, and a pedestrian 940. The system 900 may comprise more or less equipped network devices and/or more or less non-equipped network devices, than as shown in FIG. 9. In addition, the system 900 may comprise more or less different types of equipped network devices (e.g., which may include equipped UEs) and/or more or less different types of non-equipped network devices (e.g., which may include non-equipped UEs) than as shown in FIG. 9. In addition, in one or more examples, the equipped network devices may be equipped with heterogeneous capability, which may include, but is not limited to, C-V2X/DSRC capability, 4G/5G cellular connectivity, GPS capability, camera capability, radar capability, and/or LIDAR capability.

The plurality of equipped network devices may be capable of performing V2X communications. In addition, at least some of the equipped network devices are configured to transmit and receive sensing signals for radar (e.g., RF sensing signals) and/or LIDAR (e.g., optical sensing signals) to detect nearby vehicles and/or objects. Additionally or alternatively, in some cases, at least some of the equipped network devices are configured to detect nearby vehicles and/or objects using one or more cameras (e.g., by processing images captured by the one or more cameras to detect the vehicles/objects). In one or more examples, vehicles 910a, 910b, 910c, 910d and RSU 905 may be configured to transmit and receive sensing signals of some kind (e.g., radar and/or LIDAR sensing signals).

In some examples, some of the equipped network devices may have higher capability sensors (e.g., GPS receivers, cameras, RF antennas, and/or optical lasers and/or optical sensors) than other equipped network devices of the system 900. For example, vehicle 910b may be a luxury vehicle and, as such, have more expensive, higher capability sensors than other vehicles that are economy vehicles. In one illustrative example, vehicle 910b may have one or more higher capability LIDAR sensors (e.g., high capability optical lasers and optical sensors) than the other equipped network devices in

the system 900. In one illustrative example, a LIDAR of vehicle 910b may be able to detect a VRU (e.g., cyclist) 930 and/or a pedestrian 940 with a large degree of confidence (e.g., a seventy percent degree of confidence). In another example, vehicle 910b may have higher capability radar (e.g., high capability RF antennas) than the other equipped network devices in the system 900. For instance, the radar of vehicle 910b may be able to detect the VRU (e.g., cyclist) 930 and/or pedestrian 940 with a degree of confidence (e.g., an eight five eighty-five percent degree of confidence). In another example, vehicle 910b may have higher capability camera (e.g., with higher resolution capabilities, higher frame rate capabilities, better lens, etc.) than the other equipped network devices in the system 900.

During operation of the system 900, the equipped network devices (e.g., RSU 905 and/or at least one of the vehicles 910a, 910b, 910c, 910d) may transmit and/or receive sensing signals (e.g., RF and/or optical signals) to sense and detect vehicles (e.g., vehicles 910a, 910b, 910c, 910d, and 920) and/or objects (e.g., VRU 930 and pedestrian 940) located within and surrounding the road. The equipped network devices (e.g., RSU 905 and/or at least one of the vehicles 910a, 910b, 910c, 910d) may then use the sensing signals to determine characteristics (e.g., motion, dimensions, type, heading, and speed) of the detected vehicles and/or objects. The equipped network devices (e.g., RSU 905 and/or at least one of the vehicles 910a, 910b, 910c, 910d) may generate at least one vehicle-based message 915 (e.g., a V2X message, such as a Sensor Data Sharing Message (SDSM), a Basic Safety Message (BSM), a Cooperative Awareness Message (CAM), Collective Perception Messages (CPMs), and/or other type of message) including information related to the determined characteristics of the detected vehicles and/or objects.

The vehicle-based message 915 may include information related to the detected vehicle or object (e.g., a position of the vehicle or object, an accuracy of the position, a speed of the vehicle or object, a direction in which the vehicle or object is traveling, and/or other information related to the vehicle or object), traffic conditions (e.g., low speed and/or dense traffic, high speed traffic, information related to an accident, etc.), weather conditions (e.g., rain, snow, etc.), message type (e.g., an emergency message, a non-emergency or "regular" message), etc.), road topology (line-of-sight (LOS) or non-LOS (NLOS), etc.), any combination, thereof, and/or other information. In some examples, the vehicle-based message 915 may also include information regarding the equipped network device's preference to receive vehicle-based messages from other certain equipped network devices. In some cases, the vehicle-based message 915 may include the current capabilities of the equipped network device (e.g., vehicles 910a, 910b, 910c, 910d), such as the equipped network device's sensing capabilities (which can affect the equipped network device's accuracy in sensing vehicles and/or objects), processing capabilities, the equipped network device's thermal status (which can affect the vehicle's ability to process data), and the equipped network device's state of health.

In some aspects, the vehicle-based message 915 may include a dynamic neighbor list (also referred to as a Local Dynamic Map (LDM) or a dynamic surrounding map) for each of the equipped network devices (e.g., vehicles 910a, 910b, 910c, 910d and RSU 905). For example, each dynamic neighbor list can include a listing of all of the vehicles and/or objects that are located within a specific predetermined distance (or radius of distance) away from a corresponding equipped network device. In some cases,

each dynamic neighbor list includes a mapping, which may include roads and terrain topology, of all of the vehicles and/or objects that are located within a specific predetermined distance (or radius of distance) away from a corresponding equipped network device.

In some implementations, the vehicle-based message **915** may include a specific use case or safety warning, such as a do-not-pass warning (DNPW) or a forward collision warning (FCW), related to the current conditions of the equipped network device (e.g., vehicles **910a**, **910b**, **910c**, **910d**). In some examples, the vehicle-based message **915** may be in the form of a standard Basic Safety Message (BSM), a Cooperative Awareness Message (CAM), a Collective Perception Message (CPM), a Sensor Data Sharing Message (SDSM) (e.g., SAE J3224 SDSM), and/or other format.

FIG. **10** is a diagram **1000** illustrating an example of a vehicle-based message (e.g., vehicle-based message **915** of FIG. **9**), in accordance with some aspects of the present disclosure. The vehicle-based message **915** is shown as a sensor-sharing message (e.g., an SDSM), but can include a BSM, a CAM, a CPM, or other vehicle-based message as noted herein. In FIG. **10**, the vehicle-based message **915** is shown to include HostData **1020** and Detected Object Data **1010a**, **1010b**. The HostData **1020** of the vehicle-based message **915** may include information related to the transmitting device (e.g., the transmitting equipped network entity, such as RSU **905** or an onboard unit (OBU), such as on vehicles **910a**, **910b**, **910c**, **910d**) of the vehicle-based message **915**. The Detected Object Data **1010a**, **1010b** of the vehicle-based message **915** may include information related to the detected vehicle or object (e.g., static or dynamic characteristics related to the detected vehicle or object, and/or other information related to the detected vehicle or object). The Detected Object Data **1010a**, **1010b** may specifically include Detected Object CommonData, Detected Object VehicleData, Detected Object VRUDData, Detected Obstacle ObstacleData, and Detected Object MisbehavingVehicleData.

These vehicle-based messages **915** are beneficial because they can provide an awareness and understanding to the equipped network devices (e.g., vehicles **910a**, **910b**, **910c**, **910d** of FIG. **9**) of upcoming potential road dangers (e.g., unforeseen oncoming vehicles, accidents, and road conditions).

As previously mentioned, a transmitting network device (e.g., a V2X-capable vehicle that generates and sends vehicle-based messages, such as BSMs) may be misbehaving (e.g., operating as a misbehaving vehicle) by sending, either purposely or not purposely, vehicle-based messages (e.g., BSMs) containing incorrect information. A transmitting network device may be operating as a misbehaving vehicle, for example, if information contained within its vehicle-based messages (e.g., BSMs) identifies an incorrect position (or location) for the transmitting network device.

A transmitting network device (e.g., a V2X-capable vehicle that generates and sends vehicle-based messages) may be misbehaving (e.g., operating as a misbehaving vehicle) by operating as an attacker by creating a non-visible V2X ghost object to disturb road traffic. A ghost V2X object is a V2X object that is not at the location of the transmitting network device. Ghost V2X objects are objects without a physical existence, such as a simulated vulnerable road user (VRU) or a simulated vehicle. A non-visible object is an object that is located outside of a sensor field-of-view (FoV) or not in the line-of-sight (LoS) (e.g., a non-line of sight scenario) of a receiving network device (e.g., a V2X-capable vehicle that receives vehicle-based messages).

A purpose of an attacker may be to remove any benefits from utilizing sensors utilizing V2X communications. An attacker may exploit non-line of sight (NLoS) ghost objects. For example, when an object (e.g., a ghost object) is not located within LoS of a vehicle, sensors associated with the vehicle cannot confirm the existence (or non-existence) of the object. Since the vehicle is unable to confirm the existence (or non-existence) of the object, a driver of the vehicle may be forced to decelerate (e.g., slow down) the vehicle in order to ensure that the vehicle does not collide with the object (e.g., which may be a ghost object that does not actually exist). This effect of causing the driver to perform unnecessary maneuvers (e.g., decelerating of the vehicle) can cause frustration to the driver and a disruption to traffic, which can result in a traffic jam and/or a vehicle collision.

FIG. **11** is a diagram illustrating an example of a vehicle configuration **1100** for a non-line of sight (NLoS) scenario, in accordance with some aspects of the present disclosure. In FIG. **11**, several equipped (e.g., V2X capable) network devices **1110a**, **1110b**, **1110c** in the form of trucks are shown. FIG. **11** also shows a non-equipped object (e.g., not V2X capable) object **1120** in the form of a pedestrian.

In FIG. **11**, the equipped network devices **1110a**, **1110b**, **1110c** (e.g., trucks) are all shown to be driving in the same direction on the road. In particular, equipped network device **1110a** (e.g., a truck) is shown to be driving in the direction of line **1130b**. The non-equipped object **1120** (e.g., pedestrian) is shown to be beginning to cross the road in the direction of line **1130a**. As such, paths of the equipped network device **1110a** (e.g., a truck) and the non-equipped object **1120** (e.g., pedestrian) should cross where the two lines **1130a**, **1130b** cross on the road.

The equipped network device **1110a** (e.g., a truck) may not be aware of the non-equipped object **1120** (e.g., pedestrian) because the non-equipped object **1120** (e.g., pedestrian) is not in the line of sight (NLoS) (e.g., not within the FoV) of sensors on the equipped network device **1110a** (e.g., truck). In some cases, an attacker (e.g., a vehicle) may transmit a vehicle-based message (e.g., a BSM or CAM) to the equipped network device **1110a** (e.g., a truck) notifying the equipped network device **1110a** (e.g., truck) of the non-equipped object **1120** (e.g., pedestrian) heading in the direction of the line **1130a**. However, since the non-equipped object **1120** (e.g., pedestrian) is located in the NLoS of the equipped network device **1110a** (e.g., a truck), the sensors of the equipped network device **1110a** (e.g., a truck) are not able to sense the non-equipped object **1120** (e.g., pedestrian) to verify whether the non-equipped object **1120** (e.g., pedestrian) has a physical existence or is merely a ghost object.

As previously mentioned, in V2X communication systems, information can be transmitted from vehicle sensors through wireless links to allow the information to be communicated to other vehicles, pedestrians, VRUs, and/or traffic infrastructure. The information may be transmitted using one or more vehicle-based messages, such as cellular-vehicle-to-everything (C-V2X) messages (e.g., vehicle-based messages), which can include Sensor Data Sharing Messages (SDSMs), Basic Safety Messages (BSMs), Cooperative Awareness Messages (CAMs), Collective Perception Messages (CPMs), Decentralized Environmental Messages (DENMs), and/or other types of vehicle-based messages.

Current vehicle-based messages (e.g., V2X messages) have their own purposes and limitations. For example, an equipped (e.g., V2X equipped) network device (e.g., vehicle) may transmit a CAM or BSM to share the location,

kinematic state, and/or dimensions of the equipped network device. A CAM can have the ability to depict (create) a ghost object. For another example, an equipped (e.g., V2X equipped) network device (e.g., vehicle) may transmit a CPM or an SDSM to share information regarding all objects perceived by sensors on the equipped network device. However, a CPM cannot contain an object (e.g., a V2X-capable object) that transmits V2X messages (e.g., BSMs and CAMs).

A CPM (e.g., sensor view) and a CAM/BSM (e.g., V2X object) do not have redundancy for detected objects. For example, a receiving equipped (e.g., V2X capable) network device (e.g., vehicle) can perceive an object based on a BSM received from the object. However, the receiving equipped network device may not use a CPM to verify if a vehicle has been detected by sensors of the an equipped (e.g., V2X capable) network device (e.g., vehicle) that transmitted the CPM (e.g., as shown in FIG. 12C).

Current vehicle-based messages (e.g., V2X messages) do not contain "fused objects." A "fused object" may be defined to be an equipped (e.g., V2X capable) object (e.g., vehicle or pedestrian with a smart phone) that is perceived by sensors of an ITS-S (e.g., as shown in FIG. 12D). A transmitting equipped (e.g., V2X capable) network device (e.g., vehicle) may transmit a fused object as a result of V2X-sensor data fusion. Equipped (e.g., V2X capable) network devices (e.g., vehicles) may not combine CPMs and BSMs for sensor-V2X misbehavior detection.

FIGS. 12A, 12B, 12C, and 12D show different views from a receiving equipped (e.g., V2X capable) network device's (e.g., vehicle's) perspective. In FIGS. 12A, 12B, 12C, and 12D a sensor FoV 1220 of sensors of a vehicle 1210 for different views is shown. In particular, FIG. 12A is a diagram illustrating an example of a ground truth view from a receiving equipped (e.g., V2X capable) network device's 1210 (e.g., vehicle's) perspective. In FIG. 12A, the sensor FoV 1220 is shown to include an equipped (e.g., V2X capable) object 1230a and a non-equipped (e.g., not V2X capable) object 1240a. Also shown in FIG. 12A, are an equipped (e.g., V2X capable) object 1230b and a non-equipped (e.g., not V2X capable) object 1240b that are not within the sensor FoV 1220.

FIG. 12B is a diagram illustrating an example of a Basic Safety Message (BSM) view from a receiving equipped (e.g., V2X capable) network device's 1210 (e.g., vehicle's) perspective. In FIG. 12B, the sensor FoV 1220 is shown to include the equipped (e.g., V2X capable) object 1230a. Also shown in FIG. 12B, is the equipped (e.g., V2X capable) object 1230b that is not within the sensor FoV 1220.

FIG. 12C is a diagram illustrating an example of a Collective Perception Message (CPM) view from a receiving equipped (e.g., V2X capable) network device's 1210 (e.g., vehicle's) perspective. In FIG. 12C, the sensor FoV 1220 is shown to include the non-equipped (e.g., not V2X capable) object 1240a. FIG. 12D is a diagram illustrating an example of a "fused object" view from a receiving equipped (e.g., V2X capable) network device's 1210 (e.g., vehicle's) perspective. In FIG. 12D, the sensor FoV 1220 is shown to include the equipped (e.g., V2X capable) object 1230a.

As previously mentioned, current detection systems can have issues with verifying an object's (e.g., a non-equipped object's or an equipped object's) physical existence in NLoS situations. Currently, detectors can be inefficient and/or insufficient to be able to detect an attack (e.g., in a NLoS situation). Current detectors may have a limited range, may not operate with V2X-capable objects that are not in the

sensorial line of sight (e.g., in non-line of sight), and/or may not be utilized by vehicles without sensors.

In one or more aspects, systems and techniques provide a misbehavior detection (MBD) service for sharing connected and sensed objects (e.g., a V2X security service for sharing V2X-sensor objects). The system can allow for the detection of NLoS V2X ghost objects using a disclosed techniques. For example, a neighboring ITS-S may have a location that is different from a location of an ego vehicle (e.g., an autonomous vehicle). A NLoS object (e.g., ghost) for the vehicle may be a LoS object (e.g., ghost object) for the neighboring ITS-S. The neighboring ITS-S can communicate to the vehicle whether the object exists or not (e.g., is a ghost object).

The disclosed system has multiple benefits. In one or more aspects, the system has the benefit of providing a solution for detecting a NLoS ghost object. This benefit can prevent a driver from braking (e.g., decelerating) and/or being immobilized in the presence of NLoS V2X objects. In some aspects, this solution may be incorporated as part of the European Telecommunications Standards Institute (ETSI) 103 759. The system has another benefit of providing a solution for a vehicle without sensors to use the V2X-sensor detector (sensors) from a neighboring ITS-S. Also, the system has the benefit of sharing "fused objects" with other ITS-Ss. For example, an ITS-S may not have an accurate location of NLoS objects. Neighboring ITS-Ss may have "top tier" (higher quality) sensors capable of making better measurements that could benefit other ITS-Ss with "low tier" (lower quality) sensors.

FIGS. 13, 14A, 14B, and 14C each illustrate an example system 1300, 1400, 1401, 1402 for a fused data sharing service (FDSS), which may be employed by V2X. FDSS allows for equipped network devices without sensors and/or in NLoS or out of FoV range with suspicious objects to verify whether the suspicious objects exist or are ghost objects.

FIG. 13 is a diagram illustrating an example of a system 1300 for a misbehavior detection service for sharing connected and sensed objects, where an equipped (e.g., V2X capable) network device A 1340 (e.g., traffic infrastructure, such as an ITS-s) with multiple sensors 1350a, 1350b provides visual confirmation of another equipped network device B 1310 (e.g., vehicle). In FIG. 13, an equipped (e.g., V2X capable) network device B 1310 (e.g., vehicle) is shown to be driving down a road bordered by obstructing buildings 1320a, 1320b and obstructions 1330a, 1330b. The sensor FoV 1370d of the equipped network device B 1310 is shown. An equipped (e.g., V2X capable) network device A 1340 (e.g., traffic infrastructure, such as an ITS-S, stop light, or an RSU) along with its associated sensors 1350a, 1350b are also shown in FIG. 13. Sensor FoVs 1370a, 1370b, 1370c for the equipped network device A 1340 and its associated sensors 1350a, 1350b are shown.

During operation of the system 1300, the equipped network device B 1310 (e.g., vehicle) may receive a BSM from an equipped (e.g., V2X capable) network device C 1360 (e.g., object), which is located within NLoS of equipped network device B 1310. The BSM may indicate a location of the equipped (e.g., V2X capable) network device C 1360 (e.g., object). Since the equipped (e.g., V2X capable) network device C 1360 (e.g., object) is located within NLoS of equipped network device B 1310, the equipped network device B 1310 cannot verify (e.g., with its sensors) the indicated location of the equipped (e.g., V2X capable) network device C 1360 (e.g., object) to determine whether

or not the equipped (e.g., V2X capable) network device C **1360** (e.g., object) exists at that indicated location.

However, the equipped (e.g., V2X capable) network device C **1360** (e.g., object) is located within the LoS and the sensor FoV **1370c** (e.g., within a LoS) of sensor **1350b** that is associated with the equipped (e.g., V2X capable) network device A **1340**. The equipped network device B **1310** (e.g., vehicle) may send (transmit) a fused data sharing message (FDSM) request to the equipped (e.g., V2X capable) network device A **1340** requesting the equipped (e.g., V2X capable) network device A **1340** to verify the location of the equipped (e.g., V2X capable) network device C **1360** (e.g., object). After receiving the FDSM request, the equipped (e.g., V2X capable) network device A **1340** may use its sensor **1350b** to verify the location of equipped (e.g., V2X capable) network device C **1360** (e.g., object). In response, the equipped (e.g., V2X capable) network device A **1340** may send (transmit) an FDSM response, including the verified location, to the equipped (e.g., V2X capable) network device B **1310**.

In one or more aspects, during operation of the FDSS, an equipped (e.g., V2X capable) network device (e.g., ITS-S, vehicle, RSU, pedestrian with smart phone, or drone) may provide feedback (e.g., regarding a ghost object) to another equipped (e.g., V2X capable) network device. In some aspects, during operation of the FDSS, a single equipped (e.g., V2X capable) network device (e.g., ITS-S, vehicle, RSU, pedestrian with smart phone, or drone), who may be acting as a platoon leader, may request feedback from another equipped (e.g., V2X capable) network device. FIGS. **14A**, **14B**, and **14C** show examples of operation of the FDSS where an equipped network device provides feedback to another equipped network device, and an equipped network device requests feedback from another equipped network device.

FIG. **14A** is a diagram illustrating an example of a system **1400** for a misbehavior detection service for sharing connected and sensed objects, where a platoon leader (e.g., equipped network device A **1410a**) provides visual confirmation of an equipped (e.g., V2X capable) network device C **1460**. In FIG. **14A**, equipped (e.g., V2X capable) network device A **1410a** (e.g., vehicle), equipped (e.g., V2X capable) network device B **1410c** (e.g., vehicle), and equipped (e.g., V2X capable) network device **1410c** (e.g., vehicle) are shown to be driving down a road. Equipped (e.g., V2X capable) network device A **1410a** (e.g., vehicle), equipped (e.g., V2X capable) network device B **1410c** (e.g., vehicle), and equipped (e.g., V2X capable) network device **1410c** (e.g., vehicle) may all be within a platoon, with equipped (e.g., V2X capable) network device A **1410a** (e.g., vehicle) operating as the platoon leader. The sensor FoV **1470b** of the equipped network device B **1410b**, and a sensor FoV **1470a** of the equipped network device A **1410a** are also shown in FIG. **14**.

During operation of the system **1400**, the equipped network device B **1410b** (e.g., vehicle) may receive a BSM from an equipped (e.g., V2X capable) network device C **1460** (e.g., object), which is located in NLoS of equipped network device B **1410b**. The BSM may indicate a location of the equipped (e.g., V2X capable) network device C **1460** (e.g., object). Since the equipped (e.g., V2X capable) network device C **1460** (e.g., object) is located in NLoS of equipped network device B **1410b**, the equipped network device B **1410b** cannot verify (e.g., with its sensors) the indicated location of the equipped (e.g., V2X capable) network device C **1460** (e.g., object) to determine whether

or not the equipped (e.g., V2X capable) network device C **1460** (e.g., object) exists at that indicated location.

However, the equipped (e.g., V2X capable) network device C **1460** (e.g., object) is located within LoS and the sensor FoV **1470a** (e.g., within a LoS) of the equipped (e.g., V2X capable) network device A **1410a** (e.g., vehicle). The equipped network device B **1410b** (e.g., vehicle) may send (transmit) an FDSM request to the equipped (e.g., V2X capable) network device A **1410a** requesting the equipped (e.g., V2X capable) network device A **1410a** to verify the location of the equipped (e.g., V2X capable) network device C **1460** (e.g., object). After receiving the FDSM request, the equipped (e.g., V2X capable) network device A **1410a** may use its sensors to verify the location of equipped (e.g., V2X capable) network device C **1460** (e.g., object). In response, the equipped (e.g., V2X capable) network device A **1410a** may send (transmit) an FDSM response, including the verified location, to the equipped (e.g., V2X capable) network device B **1410b**.

In one or more aspects, the platoon leader (e.g., equipped network device **1410a**) may send (transmit) an FDSM response without initially receiving an FDSM request, if a platoon member (e.g., equipped network device **1410b**) may benefit from the FDSS. In some aspects, system **1400** may include equipped network devices in the form of drones (and/or RSUs) that may send (transmit) FDSM responses without initially receiving FDSM requests.

FIG. **14B** is a diagram illustrating an example of a system **1401** for a misbehavior detection service for sharing connected and sensed objects, where an equipped (e.g., V2X capable) network device A **1441** (e.g., ITS-S) with a single sensor provides visual confirmation of another equipped network device C **1461** (e.g., an object). In FIG. **14B**, an equipped (e.g., V2X capable) network device B **1411** (e.g., vehicle) is shown to be driving down a road bordered by obstructing building **1421** and obstructions **1431a**, **1431b**. The sensor FoV **1471b** of the equipped network device B **1411** is shown. An equipped (e.g., V2X capable) network device A **1441a** (e.g., traffic infrastructure, such as an ITS-S, stop light, drone, or an RSU) is also shown in FIG. **14B**. A sensor FoV **1471a** for the equipped network device A **1441** is shown.

During operation of the system **1401**, the equipped network device B **1411** (e.g., vehicle) can receive a BSM from an equipped (e.g., V2X capable) network device C **1461** (e.g., object), which is located within NLoS of equipped network device B **1411**. The BSM can indicate a location (position) of the equipped (e.g., V2X capable) network device C **1461** (e.g., object). Since the equipped (e.g., V2X capable) network device C **1461** (e.g., object) is located within NLoS of equipped network device B **1411**, the equipped network device B **1411** may not verify (e.g., with its sensors) the indicated location (position) of the equipped (e.g., V2X capable) network device C **1461** (e.g., object) to determine whether or not the equipped (e.g., V2X capable) network device C **1461** (e.g., object) exists (or not exists) at that indicated location.

The equipped (e.g., V2X capable) network device C **1461** (e.g., object) is located within the LoS and the sensor FoV **1471a** (e.g., within a LoS) of the equipped (e.g., V2X capable) network device A **1441**. The equipped network device B **1411** (e.g., vehicle) may send (transmit) a FDSM request to the equipped (e.g., V2X capable) network device A **1441** requesting the equipped (e.g., V2X capable) network device A **1441** to verify the location of the equipped (e.g., V2X capable) network device C **1461** (e.g., object). After receiving the FDSM request, the equipped (e.g., V2X

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capable) network device A **1441** may use its sensors to verify the location of equipped (e.g., V2X capable) network device C **1461** (e.g., object). The equipped (e.g., V2X capable) network device A **1461** may send (transmit) an FDSM response, including the verified location (position), to the equipped (e.g., V2X capable) network device B **1411**.

FIG. **14C** is a diagram illustrating an example of a system **1402** for a misbehavior detection service for sharing connected and sensed objects, where an equipped (e.g., V2X capable) network device A **1442** (e.g., an RSU or drone) located in one range (e.g., range A) provides visual confirmation of another equipped network device C **1462** (e.g., an object, RSU, or drone) located within two ranges (e.g., ranges A and B). In FIG. **14C**, an equipped network device B **1412** (e.g., an RSU or drone) is shown to be located in one range (e.g., range B). A sensor FoV **1472b** of the equipped network device B **1412** (e.g., an RSU) is shown. An equipped (e.g., V2X capable) network device A **1442** (e.g., an RSU) is also shown in FIG. **14C**. A sensor FoV **1472a** for the equipped network device A **1442** is shown.

During operation of the system **1402**, the equipped network device B **1412** (e.g., RSU) may receive a BSM from an equipped (e.g., V2X capable) network device C **1462** (e.g., object) that is located outside of the sensor FoV **1472b** of equipped network device B **1412**. The BSM may indicate a location (position) of the equipped (e.g., V2X capable) network device C **1462** (e.g., object). Since the equipped (e.g., V2X capable) network device C **1462** (e.g., object) is located outside of the sensor FoV **1472b** of equipped network device B **1412**, the equipped network device B **1412** cannot verify (e.g., with its sensors) the indicated location (position) of the equipped (e.g., V2X capable) network device C **1462** (e.g., object) to determine whether the equipped (e.g., V2X capable) network device C **1462** (e.g., object) exists (or not exists) at that location.

The equipped (e.g., V2X capable) network device C **1462** (e.g., object) is located within the LoS and the sensor FoV **1472a** of the equipped (e.g., V2X capable) network device A **1442**. The equipped network device B **1412** (e.g., RSU) may send (transmit) a FDSM request to the equipped (e.g., V2X capable) network device A **1442** requesting the equipped (e.g., V2X capable) network device A **1442** to verify the location of the equipped (e.g., V2X capable) network device C **1462** (e.g., object). After receiving the FDSM request, the equipped (e.g., V2X capable) network device A **1442** may use its sensors to verify the location of equipped (e.g., V2X capable) network device C **1462** (e.g., object). The equipped (e.g., V2X capable) network device A **1462** may send (transmit) an FDSM response, including the verified location (position), to the equipped (e.g., V2X capable) network device B **1412**.

FIG. **15A** is a diagram illustrating an example of signaling **1500** for a feedback approach for FDSS. During operation of the feedback approach, equipped (e.g., V2X capable) network device B **1510** (e.g., an ITS-S) can send (transmit) **1515** a vehicle-based message (e.g., BSM), which may include a location for the network device B **1510**, to equipped (e.g., V2X capable) network device A **1540** (e.g., an ITS-S). An equipped (e.g., V2X capable) object C **1560** (e.g., object) may send (transmit) **1525** a vehicle-based message (e.g., BSM), which may include an expected (or purported) location for the object C **1560**, to the equipped (e.g., V2X capable) network device A **1540** (e.g., an ITS-S).

After the equipped (e.g., V2X capable) network device A **1540** (e.g., an ITS-S) receives the vehicle-based messages (e.g., BSMs), the equipped (e.g., V2X capable) network device A **1540** may determine (detect) that the vehicle-based

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message (e.g., BSM) sent from the equipped (e.g., V2X capable) object C **1560** (e.g., object) is suspicious **1535** (e.g., the expected or purported location of the object C **1560** in the vehicle-based message is suspicious because, for example, the object C **1560** is in NLoS of network device B **1510**). The equipped (e.g., V2X capable) network device A **1540** may then assess **1545** whether or not FDSS is useful for the equipped (e.g., V2X capable) network device B **1510** (e.g., an ITS-S), for example by determining whether the location of the object C **1560** may be relevant to the equipped (e.g., V2X capable) network device B **1510**.

If the equipped (e.g., V2X capable) network device A **1540** determines that FDSS is useful, the equipped (e.g., V2X capable) network device A **1540** may then generate **1555** a FDSM response, which may contain information regarding whether or not the equipped (e.g., V2X capable) network device A **1540** has determined that the object C **1560** exists at the expected location. Then, the equipped (e.g., V2X capable) network device A **1540** may send (transmit) **1565** the FDSM response to the equipped (e.g., V2X capable) network device B **1510**.

FIG. **15B** is a diagram illustrating an example of signaling **1501** for a request-response approach for FDSS. During operation of the request-response approach, equipped (e.g., V2X capable) network device A **1511** (e.g., an ITS-S) can send (transmit) **1516** a vehicle-based message (e.g., BSM), which may include a location for the network device A **1511** as well as a certificate, to equipped (e.g., V2X capable) network device B **1541** (e.g., an ITS-S). After the equipped network device B **1541** receives the vehicle-based message from the equipped network device A **1511**, the equipped network device B **1541** may check the certificate **1526** (e.g., within the vehicle-based message from equipped network device A **1511**) to determine whether the certificate indicates that the equipped network device A **1511** has the proper permissions (e.g., SSPs) for FDSS (e.g., network device A **1511** has permissions to send a FDSM response).

Then, an equipped (e.g., V2X capable) object C **1561** (e.g., object) may send (transmit) **1536** a vehicle-based message (e.g., BSM), which may include an expected or purported location for the equipped object C **1561**, to the equipped (e.g., V2X capable) network device B **1541** (e.g., an ITS-S). After the equipped network device B **1541** receives the vehicle-based message from the equipped object C **1561**, the equipped network device B **1541** may determine (detect) that the vehicle-based message (e.g., BSM) sent from the equipped (e.g., V2X capable) object C **1561** (e.g., object) is suspicious **1546** (e.g., the expected location of the object C **1561** in the vehicle-based message is suspicious because, for example, the object C **1561** is in NLoS of network device B **1541**). The equipped (e.g., V2X capable) network device B **1541** may then generate **1556** a FDSM request, which may request that equipped network device A **1511** determine whether or not the object C **1561** exists at the expected location. After the equipped network device B **1541** has generated the FDSM request, the equipped network device B **1541** may send (transmit) the FDSM request **1566** to the equipped network device A **1511**.

After the equipped network device A **1511** receives the FDSM request, the equipped network device A **1511** may generate **1576** a FDSM response, which may contain information regarding whether or not the equipped (e.g., V2X capable) network device A **1541** has determined that the object C **1561** exists at the expected location. Then, the equipped (e.g., V2X capable) network device A **1511** may send (transmit) **1586** the FDSM response to the equipped (e.g., V2X capable) network device B **1541**.

A fused data sharing protocol (FDSP) may be employed for communications for FDSS. In one or more aspects, if the ITS-S is an RSU, the ITS-S may broadcast the FDSM response, such that multiple equipped (e.g., V2X capable) network devices (e.g., ITS-Ss and/or vehicles) may receive the FDSM response. Other equipped network devices (e.g., ITS-Ss) may benefit from receiving the FDSM response. For example, all equipped network devices (e.g., vehicles) located behind an equipped network device (e.g., a vehicle), such as equipped network device B **1310** of FIG. **13**, will not be able to determine whether an equipped object, such as equipped object C **1360** in FIG. **13**, in NLoS of the equipped network device (e.g., a vehicle) is suspicious and, as such, these network devices (e.g., vehicles) will need to use FDSS.

In one or more aspects, ITS-S may communicate using a group-cast scheme, which is similar to MSCS (e.g., a platoon). In some aspects, the ITS-S may communicate using a uni-cast scheme, which is similar to the MSCS (e.g., for a one-to-one situation, where there are no vehicles located behind a vehicle in NLoS with an object). In one or more aspects, the uni-cast may be transmitted using various different communications methods including, but not limited to, DSRC, C-V2X, and/or visual light communication.

In one or more aspects, the protocol for FDSS can be utilized on an intelligent transportation system (ITS) stack (e.g., V2V). In some aspects, part of the protocol (e.g., exchange of FDSMs) can be accomplished through a transmission control protocol/internet protocol (TCP/IP) stack (e.g., for communications between RSUs over the network used by road operators).

Various conditions may be employed for triggering a FDSP. In one or more aspects, if a suspicious equipped (e.g., V2X capable) object is not visible by sensors of an equipped (e.g., V2X capable) network device (e.g., vehicle) located in NLoS with the equipped object, the suspicious equipped object is located outside of the FoV of the equipped network device; the suspicious equipped object is located in the FoV of the sensors of the equipped network device, but is not located within the LoS of the sensors of the equipped network device; the suspicious equipped object cannot be detected because the sensors of the equipped network device are broken or faulty, or the suspicious equipped object cannot be detected because the equipped network device does not have any sensors.

In one or more aspects, a trajectory (e.g., direction of movement) of a suspicious equipped (e.g., V2X capable) object may intersect with a trajectory of an equipped (e.g., V2X capable) network device (e.g., vehicle) located in NLoS with the suspicious equipped object, if the heading of the equipped network device and the heading of the suspicious equipped object intersect.

In one or more aspects, a fused data sharing detector (FDSD) may be part of the FDSS. In some aspects, the FDSD may include core data (e.g., a generation time and an identifier of the equipped network device, such as an ITS-S, that generates the FDSM). The FDSD may also include objects that have been detected by the sensors of equipped network device (e.g., ITS-S), but have not yet sent (transmitted) a vehicle-based message (e.g., a BSM). Allowed object types for the objects included in the FDSD may be all types of objects (e.g., vehicles and VRUs) that have been defined in vehicle-based message (e.g., BSM and CAM) standards. Unallowed object types for the objects included in the FDSD may be types of objects (e.g., trees and road signs) that have not been defined in the vehicle-based message standards.

In some aspects, the FDSD may include identifiers, such as a hashed certificate contained within a vehicle-based message (e.g., a BSM), as well as include identifiers of simulated equipped (e.g., V2X capable) objects identified by a sensor-V2X detector (e.g., an equipped network device, such as an ITS-S).

The FDSD may also include a listing of fused objects, which are objects that have had their location (existence) verified and have transmitted a vehicle-based message (e.g., a BSM). In one or more aspects, an object may be represented in the listing by an identifier, such as an identification (ID). In some aspects, an object may be represented in the listing by a set of fields, which may include, but are not limited to, an ID, a position, a type, a dimension, and/or kinematic data.

In one or more aspects, if an equipped object (which has transmitted a vehicle-based message) has not been confirmed by a sensor(s) of an equipped network device (e.g., ITS-S) and is located outside of the FoV of the sensor(s), an object identifier for that object cannot be inserted in a FDSM. In some aspects, if an equipped object (which has transmitted a vehicle-based message) has not been confirmed by a sensor(s) of an equipped network device (e.g., ITS-S) and is located within a FoV of at least one sensor, an object identifier for that object may be inserted in a FDSM (e.g., as a malicious object).

In one or more aspects, if an equipped object has not transmitted a vehicle-based message (e.g., a BSM) and the equipped object has an allowed object type, the equipped object can be inserted in a FDSM. In some aspects, if an equipped object has not transmitted a vehicle-based message (e.g., a BSM) and the equipped object has an allowed object type, the equipped object cannot be inserted in a FDSM.

In some aspects, if an equipped object has transmitted a vehicle-based message (e.g., BSM) and has been confirmed (e.g., the location of the equipped object has been verified) by at least one sensor of an equipped network device (e.g., ITS-S), an object identifier for the equipped object can be inserted in a FDSM (e.g., as a fused object).

FIGS. **16**, **17**, and **18** are flow charts for processes **1600**, **1700**, **1800** for a fused data sharing detector (FDSD). FIG. **16** is a flow chart illustrating an example of a process **1600** for a FDSD from an initial requester side perspective. During the process **1600**, a requester (e.g., an equipped network device, such as an ITS-S) may collect (fusion **1630**) all data detected by its sensors **1610**. The requester may receive a vehicle-based message (e.g., BSM) **1620** from an equipped (e.g., V2X capable) object (e.g., a suspicious object).

After the requester has collected all of the sensor data and has received a vehicle-based message from the equipped object, the requester may review the sensor data to check (determine) if at least one of the sensors detected the equipped object (e.g., the requester can run a sensor-V2X detector **1640**). After the requester has reviewed the sensor data, the requester can collect an ID for each suspicious equipped object **1650**, for a given time window of the sensing. In one or more examples, the ID collected may be the ID for the suspicious equipped object that was contained within the vehicle-based message (e.g., BSM) sent by the suspicious equipped object. In some examples, the duration of the time window may be a static value. In some examples, the duration of the time window may be based on the remaining time before making a decision (e.g., a driving decision, such as braking).

After the requester has collected the IDs for the suspicious equipped objects and the requester determines that it cannot

verify the location of at least one suspicious object, the requester may check (verify) if the requester is able to send (transmit) a request for the FDSS **1660**. In one or more examples, the decision of whether the requester can send a request can depend on the availability of surrounding equipped network devices (e.g., ITS-Ss) that can offer this service. In some examples, an equipped network device (e.g., ITS-S) may advertise this service in its certificate (e.g., which may be contained within a vehicle-based message, such as a BSM). The certificate may contain a set of permissions, such as service specific permissions (SSPs), that can define the actions that the certificate owner is allowed to perform within the FDSS. If available, the requester may use the sensor information contained in a vehicle-based message (e.g., CPM) to check if the equipped object is located within the FoV of the equipped network device (e.g., ITS-S).

If the requester has determined that the requester is able to send (transmit) a request for the FDSS, the requester can generate an FDSM **1670**. In one or more examples, the FDSM may request for the requester to use FDSS from a neighboring infrastructure. However, if the requester has determined that the requester is not able to send (transmit) a request for the FDSS, the requester may remain idle **1680**.

FIG. **17** is a flow chart illustrating an example of a process **1700** for a FDSM from a responder side perspective. During the process **1700**, for each received vehicle-based message (e.g., BSM) **1710** from an equipped (e.g., V2X capable) object, a responder (e.g., an equipped network device, such as an ITS-S) may run the sensor-V2X detector **1730** (e.g., using its sensors **1705** and fusion **1720**) to collect a list of fused objects **1735**, a list of simulated (fake) equipped (e.g., V2X capable) objects **1740**, and a list of sensor objects **1745**.

In one or more examples, a fused object may be an equipped (e.g., V2X) object that is perceived (detected) by the sensors of responder and has transmitted a vehicle-based message (e.g., BSM) received by the responder. In some examples, a fused object may be represented by an object ID. In some examples, a fused object may be represented by a set of fields, which may include, but are not limited to, an object ID, a position, a type, a dimension, and kinematic data for the equipped object.

In one or more examples, a simulated (fake) equipped object may be an equipped object that has sent a vehicle-based message, but cannot be detected by the sensors of the responder. The list for simulated (fake) equipped objects may include for each simulated (fake) equipped object a hashed certificate contained within the vehicle-based message (e.g., BSM). Identifiers of simulated (fake) equipped objects may be reported to the sensor-V2X detector.

In some examples, a sensor object may be an equipped object that has been detected by the sensors of the responder, but has not yet sent a vehicle-based message. The process filter **1750** may filter these simulated (fake) objects based on the ITS-S type (e.g., filter V2X objects **1750**). Allowed object types may be all types of objects (e.g., vehicles and VRUs) that have been defined in vehicle-based message (e.g., BSM and CAM) standards. Unallowed object types may be types of objects (e.g., trees and road signs) that have not been defined in the vehicle-based message standards.

After the responder has created the lists, the responder can determine whether the responder will send a FDSM response. The responder may send a FDSM response, if the responder has received a request (e.g., from an equipped network device) for FDSS, or if the responder has detected an equipped network device (e.g., a vehicle) that may need the FDSS. For example, an equipped network device (e.g.,

a vehicle) may need the FDSS, if the equipped network device (e.g., vehicle) is approaching a simulated (fake) equipped object, and the distance between the equipped network device (e.g., vehicle) and the simulated (fake) equipped object is less than a threshold distance value. In one or more examples, an RSU may send an FDSM to another RSU to confirm the presence of an equipped object.

As such, the responder can determine whether the responder has received a request for FDSS **1755**. If the responder determines that the responder has received a request for FDSS, the responder can generate a FDSM response **1765**. However, if the responder determines that the responder has not received a request for FDSS, the responder can then determine whether an equipped network device (e.g., a vehicle) may need the FDSS **1760**. If the responder determines that an equipped network device (e.g., a vehicle) may need the FDSS, the responder can generate a FDSM response **1765**. However, if the responder determines that an equipped network device (e.g., a vehicle) does not need the FDSS, the responder may remain idle **1770**.

FIG. **18** is a flow chart illustrating an example of a process **1800** for a FDSM from a final responder side perspective. During the process **1800**, at the reception of an FDSM **1820**, a requester (e.g., an equipped network device) can check **1830** (verify) if an equipped object is simulated (fake). In one or more examples, the requester can check (verify) if the suspicious equipped object (within a list of suspicious equipped objects **1810**) has its ID amongst the fused objects or simulated (fake) objects. Sensor objects cannot be used in this process **1800** because sensor objects do not have a vehicle-based message and, as such, do not have a certificate or an object ID.

In some examples, the requester can check (verify) for a match between the vehicle-based message (e.g., BSM) features and the fused and sensor objects' features. The vehicle-based message (e.g., BSM) of a real equipped object will match with content (e.g., position, speed, heading, and/or dimension) of an object contained in the FDSM. The vehicle-based message (e.g., BSM) of a simulated (fake) object can lead to an absence of a match with one of the objects contained within the FDSM.

After the requester has checked (determined) if an equipped object is simulated (fake), the requester can update a cooperative perception database **1840**. The requester may update the cooperative perception database to remove malicious tracks, if there are malicious equipped objects that have been reported in the FDSM response. The requester may update the cooperative perception database to update the correct tracks, if a fused object may contain more accurate data (e.g., because the responder may have better sensors).

After the requester has updated the cooperative perception database, the requester can generate a misbehavior report (MBR) **1850** for each suspected equipped object (e.g., which may be a vehicle). In one or more examples, a suspected equipped object may be represented as an identifier (e.g., ID) in the MBR. In some examples, a suspected equipped object may be represented as a set of fields (e.g., ID, position, type, dimension, and kinematic data) in the MBR.

In one or more aspects, the FDSS may be advertised. The FDSS may need to be advertised because, before sending a request for the FDSS, the request will need to know which equipped network device can provide the FDSS. As such, solutions are presented for how to communicate to the requesters which neighboring equipped network devices (e.g., ITS-Ss) can provide the FDSS.

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In one or more aspects, to advertise the FDSS, an additional permission bit (e.g., to advertise the FDSS) can be added to the digital certificate to indicate whether or not the network device can provide the FDSS. The permission bit can be a single bit that describes the availability of the FDSS in an equipped network device (e.g., ITS-S). For example, if the permission bit is set equal to zero (0), the FDSS is available for the equipped network device (e.g., ITS-S). If the permission bit is set equal to one (1), the FDSS is not available for the equipped network device (e.g., ITS-S).

The digital certificate may be contained within a vehicle-based message (e.g., BSM) received by a FDSS participant prior to the FDSM. In one or more examples, the emitter (e.g., transmitting equipped network device) of a FDSM may need to know if a neighboring equipped network device (e.g., ITS-S) supports the FDSS and, as such, is able to process the FDSM being sent. In one or more examples, a responder may need the digital certificate of the recipient for this response. In some examples, a requester may need the digital certificate of the responder before sending the request.

In some aspects, to advertise the FDSS, an additional service (e.g., to advertise the FDSS) can be added to the Wave Service Advertisement (WSA). The WSA approach may be used for an RSU. In one or more examples, an equipped network entity (e.g., ITS-S) responding to the request for the FDSS should be an RSU. In some examples, the RSU may broadcast a WSA every one (1) second. The WSA may contain "FDSS" among all of the services advertised by the RSU.

In one or more aspects, two message types may be employed for the FDSS. These two message types include a first message type (e.g., a fused data sharing message request **1910** of FIG. **19**) that is a FDSM used for requesting FDSS, and the second message type (e.g., a fused data sharing message response **2005** of FIG. **20**) that is a FDSM used for responding to a request for FDSS.

FIG. **19** is a diagram illustrating an example **1900** of a fused data sharing message (FDSM) request **1910**. In one or more aspects, the FDSM request **1910** of FIG. **19** may be employed for the request-response approach shown in FIG. **15B**. In FIG. **19**, the FDSM request **1910** is shown to include a plurality of fields, which may include a message (msg) timestamp field **1920** (e.g., containing a generation time for the FDSM), a source container field **1930** (e.g., including information related to the source), and a malicious objects field **1940**. The source container field **1930** may contain a source identification (ID) field **1950**, a three-dimensional (3D) position field **1960**, and a positional accuracy field **1970**. The malicious objects field **1940** may contain a plurality of object ID fields **1980a**, **1980b** (e.g., including a set of vehicle-based message, such as BSM, IDs). The FDSM request **1910** may be a V2X message. The fields **1920**, **1930**, **1940**, **1950**, **1960**, **1970**, **1980a**, **1980b** of the FDSM request **1910** may all be mandatory fields.

In one or more aspects, during operation, a transmitting equipped (e.g., V2X capable) network device may send (transmit) a FDSM request (e.g., FDSM request **1910**) to a receiving equipped (e.g., V2X capable) network device to notify the receiving equipped (e.g., V2X capable) network device of an expected or purported location for an object (e.g., ghost object) that transmitted a vehicle-based message (e.g., BSM) to the transmitting equipped (e.g., V2X capable) network device. After the receiving equipped (e.g., V2X capable) network device receives the FDSM request, since the receiving equipped (e.g., V2X capable) network device is now aware (e.g., from information contained within the

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FDSM request) of the supposed location of the object, the receiving equipped (e.g., V2X capable) network device can orient its sensor antenna beam(s) towards the expected location of the object to verify if the object exists (or does not exist) at that location.

FIG. **20** is a diagram illustrating an example **2000** of an FDSM response **2005**. In one or more aspects, the FDSM response **2005** of FIG. **20** may be employed for the feedback approach and the request-response approach shown in FIGS. **15A** and **15B**, respectively. In FIG. **20**, the FDSM response **2005** is shown to include a plurality of fields, which may include a message timestamp field **2010** (e.g., including a generation time for the FDSM), a source ID field **2015** (e.g., including an ID of a transmitting device) and object containers field **2020**. The object containers field **2020** may include a fused objects field **2025** (e.g., including a set of vehicle-based message, such as BSM, identifiers for objects that have been detected by sensors and that have sent vehicle-based messages), a malicious objects field **2030** (e.g., including a set of vehicle-based message, such as BSM, identifiers for verified ghost objects), and a sensor objects field **2035** (e.g., including a set of vehicle-based message, such as BSM, identifiers for objects detected by sensors and that have not yet sent vehicle-based messages).

The fused objects field **2025** may include a plurality of BSM object fields **2040a**, **2040b**. Each BSM object field **2040a**, **2040b** may include an object ID field **2070**, a kinematic data field **2080**, and a dimension data field **2090**. The malicious objects field **2030** may include a plurality of object ID fields **2050a**, **2050b**. The sensor objects field **2035** may include a plurality of CPM object fields **2060a**, **2060b**.

The FDSM response **2005** may be a V2X message. All of the fields may be mandatory fields, except for the kinematic data field **2080** and the dimension data field **2090**, which are both optional fields.

In one or more aspects, the logic for inserting objects into an FDSM is as follows. For a BSM object that has not been confirmed by a sensor of an ITS-S, if the object is located outside of the FoVs of all of the sensors, the object identifier for that object may not be inserted into the FDSM. If the object is in at least one sensor FoV, the object identifier for the object can be inserted in the FDSM (e.g., a ghost object).

For a sensor object that has not sent a BSM, if the object has an allowed object type, the object identifier for the object may be inserted in the FDSM. If the object has an unallowed object type, the object identifier for the object may not be inserted in the FDSM.

For a BSM object that has been confirmed by at least one sensor, the object identifier for the object may be inserted in the FDSM (e.g., fused objects).

FIG. **21** is a flow chart illustrating an example of a process **2100** for wireless communications. The process **2100** can be performed by a network device, such as a vehicle, a roadside unit (RSU), an intelligent transportation system-station (ITS-S), a UE (e.g., a mobile device such as a mobile phone, a drone or unmanned aerial vehicle (UAV), a wearable device such as a network-connected watch, an extended reality (XR) device such as a virtual reality (VR) or augmented reality (AR) headset or glasses, or other type of UE), or other type of network device or component, system, or apparatus (e.g., a chipset) thereof. The operations of the process **2100** may be implemented as software components that are executed and run on one or more processors (e.g., processor **2210** of FIG. **22** or other processor(s)) of the vehicle. Further, the transmission and reception of signals by the wireless communications device in the process **2100**

may be enabled, for example, by one or more antennas and/or one or more transceivers (e.g., wireless transceiver(s)) of the vehicle.

At block **2110**, the network device (or component, system, or apparatus thereof) may determine an estimated location of a second network device. In some cases, the first network device and the second network device are capable of Vehicle-to-Everything (V2X) communications (e.g., transmitting, receiving, and/or processing V2X messages). In some examples, the second network device is a vehicle, an ITS-S, an RSU, a UE, a drone or UAE, or other type of network device.

At block **2120**, the network device (or component, system, or apparatus thereof) may compare the estimated location with an expected location for the second network device. In some aspects, the network device (or component, system, or apparatus thereof) may receive a vehicle-based message from the second network device. In some cases, the vehicle-based message includes the expected location for the second network device. In some examples, the vehicle-based message is a Basic Safety Message (BSM), a Cooperative Awareness Message (CAM), a Collective Perception Message (CPM), a Sensor Data Sharing Message (SDSM), a Decentralized Environmental Message (DENM), or other type of vehicle-based message.

At block **2130**, the network device (or component, system, or apparatus thereof) may determine whether the second network device is a misbehaving device based on the comparing.

At block **2140**, the network device (or component, system, or apparatus thereof) may generate a report based on the determining of whether the second device is a misbehaving device. In some aspects, the network device (or component, system, or apparatus thereof) may output a response comprising the report for transmission to a third network device. In some cases, the second network device is not located within a field of view (FoV) of sensors of the third network device and/or is not located within a line-of-sight (LoS) of the third network device. In some examples, the response includes at least one of a message timestamp field, a source identification (ID) field, an object containers field, a fused objects field, a malicious objects field, or a sensor objects field.

In some aspects, the network device (or component, system, or apparatus thereof) may receive a request for data sharing from the third network device (or other network device). For instance, the request for data sharing may include a message timestamp field, a source field, a malicious objects field, or any combination thereof.

In some aspects, the network device (or component, system, or apparatus thereof) may generate a fused objects list, a sensor objects list, and a simulated objects list. The fused objects list, the sensor objects list, or the simulated objects list may include the second network device (e.g., an identifier or other information associated with the second network device). The fused objects list may include network devices that transmitted vehicle-based messages and are located within a FoV and/or a LoS of the third network device (or other network device). The sensor objects list may include network devices that have not yet transmitted any vehicle-based messages and are located within the FoV and/or the LoS of the third network device (or other network device). The simulated objects list may include network devices that are ghost objects.

FIG. 22 is a block diagram illustrating an example of a computing system **2200**, which may be employed by the disclosed system for a misbehavior detection service for

sharing connected and sensed objects, in accordance with some aspects of the present disclosure. In particular, FIG. 22 illustrates an example of computing system **2200**, which can be for example any computing device making up internal computing system, a remote computing system, a camera, or any component thereof in which the components of the system are in communication with each other using connection **2205**. Connection **2205** can be a physical connection using a bus, or a direct connection into processor **2210**, such as in a chipset architecture. Connection **2205** can also be a virtual connection, networked connection, or logical connection.

In some aspects, computing system **2200** is a distributed system in which the functions described in this disclosure can be distributed within a datacenter, multiple data centers, a peer network, etc. In some aspects, one or more of the described system components represents many such components each performing some or all of the function for which the component is described. In some aspects, the components can be physical or virtual devices.

Example system **2200** includes at least one processing unit (CPU or processor) **2210** and connection **2205** that communicatively couples various system components including system memory **2215**, such as read-only memory (ROM) **2220** and random access memory (RAM) **2225** to processor **2210**. Computing system **2200** can include a cache **2212** of high-speed memory connected directly with, in close proximity to, or integrated as part of processor **2210**.

Processor **2210** can include any general purpose processor and a hardware service or software service, such as services **2232**, **2234**, and **2236** stored in storage device **2230**, configured to control processor **2210** as well as a special-purpose processor where software instructions are incorporated into the actual processor design. Processor **2210** may essentially be a completely self-contained computing system, containing multiple cores or processors, a bus, memory controller, cache, etc. A multi-core processor may be symmetric or asymmetric.

To enable user interaction, computing system **2200** includes an input device **2245**, which can represent any number of input mechanisms, such as a microphone for speech, a touch-sensitive screen for gesture or graphical input, keyboard, mouse, motion input, speech, etc. Computing system **2200** can also include output device **2235**, which can be one or more of a number of output mechanisms. In some instances, multimodal systems can enable a user to provide multiple types of input/output to communicate with computing system **2200**.

Computing system **2200** can include communications interface **2240**, which can generally govern and manage the user input and system output. The communication interface may perform or facilitate receipt and/or transmission wired or wireless communications using wired and/or wireless transceivers, including those making use of an audio jack/plug, a microphone jack/plug, a universal serial bus (USB) port/plug, an Apple™ Lightning™ port/plug, an Ethernet port/plug, a fiber optic port/plug, a proprietary wired port/plug, 3G, 4G, 5G and/or other cellular data network wireless signal transfer, a Bluetooth™ wireless signal transfer, a Bluetooth™ low energy (BLE) wireless signal transfer, an IBEACON™ wireless signal transfer, a radio-frequency identification (RFID) wireless signal transfer, near-field communications (NFC) wireless signal transfer, dedicated short range communication (DSRC) wireless signal transfer, 802.11 Wi-Fi wireless signal transfer, wireless local area network (WLAN) signal transfer, Visible Light Communication (VLC), Worldwide Interoperability for Microwave

Access (WiMAX), Infrared (IR) communication wireless signal transfer, Public Switched Telephone Network (PSTN) signal transfer, Integrated Services Digital Network (ISDN) signal transfer, ad-hoc network signal transfer, radio wave signal transfer, microwave signal transfer, infrared signal transfer, visible light signal transfer, ultraviolet light signal transfer, wireless signal transfer along the electromagnetic spectrum, or some combination thereof.

The communications interface **2240** may also include one or more range sensors (e.g., LIDAR sensors, laser range finders, RF radars, ultrasonic sensors, and infrared (IR) sensors) configured to collect data and provide measurements to processor **2210**, whereby processor **2210** can be configured to perform determinations and calculations needed to obtain various measurements for the one or more range sensors. In some examples, the measurements can include time of flight, wavelengths, azimuth angle, elevation angle, range, linear velocity and/or angular velocity, or any combination thereof. The communications interface **2240** may also include one or more Global Navigation Satellite System (GNSS) receivers or transceivers that are used to determine a location of the computing system **2200** based on receipt of one or more signals from one or more satellites associated with one or more GNSS systems. GNSS systems include, but are not limited to, the US-based GPS, the Russia-based Global Navigation Satellite System (GLO-NASS), the China-based BeiDou Navigation Satellite System (BDS), and the Europe-based Galileo GNSS. There is no restriction on operating on any particular hardware arrangement, and therefore the basic features here may easily be substituted for improved hardware or firmware arrangements as they are developed.

Storage device **2230** can be a non-volatile and/or non-transitory and/or computer-readable memory device and can be a hard disk or other types of computer readable media which can store data that are accessible by a computer, such as magnetic cassettes, flash memory cards, solid state memory devices, digital versatile disks, cartridges, a floppy disk, a flexible disk, a hard disk, magnetic tape, a magnetic strip/stripe, any other magnetic storage medium, flash memory, memristor memory, any other solid-state memory, a compact disc read only memory (CD-ROM) optical disc, a rewritable compact disc (CD) optical disc, digital video disk (DVD) optical disc, a blu-ray disc (BDD) optical disc, a holographic optical disk, another optical medium, a secure digital (SD) card, a micro secure digital (microSD) card, a Memory Stick® card, a smartcard chip, a EMV chip, a subscriber identity module (SIM) card, a mini/micro/nano/pico SIM card, another integrated circuit (IC) chip/card, random access memory (RAM), static RAM (SRAM), dynamic RAM (DRAM), read-only memory (ROM), programmable read-only memory (PROM), erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), flash EPROM (FLASH EPROM), cache memory (e.g., Level 1 (L1) cache, Level 2 (L2) cache, Level 3 (L3) cache, Level 4 (L4) cache, Level 5 (L5) cache, or other (L #) cache), resistive random-access memory (RRAM/ReRAM), phase change memory (PCM), spin transfer torque RAM (STT-RAM), another memory chip or cartridge, and/or a combination thereof.

The storage device **2230** can include software services, servers, services, etc., that when the code that defines such software is executed by the processor **2210**, it causes the system to perform a function. In some aspects, a hardware service that performs a particular function can include the software component stored in a computer-readable medium

in connection with the necessary hardware components, such as processor **2210**, connection **2205**, output device **2235**, etc., to carry out the function. The term “computer-readable medium” includes, but is not limited to, portable or non-portable storage devices, optical storage devices, and various other mediums capable of storing, containing, or carrying instruction(s) and/or data. A computer-readable medium may include a non-transitory medium in which data can be stored and that does not include carrier waves and/or transitory electronic signals propagating wirelessly or over wired connections. Examples of a non-transitory medium may include, but are not limited to, a magnetic disk or tape, optical storage media such as compact disk (CD) or digital versatile disk (DVD), flash memory, memory or memory devices. A computer-readable medium may have stored thereon code and/or machine-executable instructions that may represent a procedure, a function, a subprogram, a program, a routine, a subroutine, a module, a software package, a class, or any combination of instructions, data structures, or program statements. A code segment may be coupled to another code segment or a hardware circuit by passing and/or receiving information, data, arguments, parameters, or memory contents. Information, arguments, parameters, data, etc. may be passed, forwarded, or transmitted via any suitable means including memory sharing, message passing, token passing, network transmission, or the like.

Specific details are provided in the description above to provide a thorough understanding of the aspects and examples provided herein, but those skilled in the art will recognize that the application is not limited thereto. Thus, while illustrative aspects of the application have been described in detail herein, it is to be understood that the inventive concepts may be otherwise variously embodied and employed, and that the appended claims are intended to be construed to include such variations, except as limited by the prior art. Various features and aspects of the above-described application may be used individually or jointly. Further, aspects can be utilized in any number of environments and applications beyond those described herein without departing from the broader scope of the specification. The specification and drawings are, accordingly, to be regarded as illustrative rather than restrictive. For the purposes of illustration, methods were described in a particular order. It should be appreciated that in alternate aspects, the methods may be performed in a different order than that described.

For clarity of explanation, in some instances the present technology may be presented as including individual functional blocks comprising devices, device components, steps or routines in a method embodied in software, or combinations of hardware and software. Additional components may be used other than those shown in the figures and/or described herein. For example, circuits, systems, networks, processes, and other components may be shown as components in block diagram form in order not to obscure the aspects in unnecessary detail. In other instances, well-known circuits, processes, algorithms, structures, and techniques may be shown without unnecessary detail in order to avoid obscuring the aspects.

Further, those of skill in the art will appreciate that the various illustrative logical blocks, modules, circuits, and algorithm steps described in connection with the aspects disclosed herein may be implemented as electronic hardware, computer software, or combinations of both. To clearly illustrate this interchangeability of hardware and software, various illustrative components, blocks, modules,

circuits, and steps have been described above generally in terms of their functionality. Whether such functionality is implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system. Skilled artisans may implement the described functionality in varying ways for each particular application, but such implementation decisions should not be interpreted as causing a departure from the scope of the present disclosure.

Individual aspects may be described above as a process or method which is depicted as a flowchart, a flow diagram, a data flow diagram, a structure diagram, or a block diagram. Although a flowchart may describe the operations as a sequential process, many of the operations can be performed in parallel or concurrently. In addition, the order of the operations may be re-arranged. A process is terminated when its operations are completed, but could have additional steps not included in a figure. A process may correspond to a method, a function, a procedure, a subroutine, a subprogram, etc. When a process corresponds to a function, its termination can correspond to a return of the function to the calling function or the main function.

Processes and methods according to the above-described examples can be implemented using computer-executable instructions that are stored or otherwise available from computer-readable media. Such instructions can include, for example, instructions and data which cause or otherwise configure a general purpose computer, special purpose computer, or a processing device to perform a certain function or group of functions. Portions of computer resources used can be accessible over a network. The computer executable instructions may be, for example, binaries, intermediate format instructions such as assembly language, firmware, source code. Examples of computer-readable media that may be used to store instructions, information used, and/or information created during methods according to described examples include magnetic or optical disks, flash memory, USB devices provided with non-volatile memory, networked storage devices, and so on.

In some aspects the computer-readable storage devices, mediums, and memories can include a cable or wireless signal containing a bitstream and the like. However, when mentioned, non-transitory computer-readable storage media expressly exclude media such as energy, carrier signals, electromagnetic waves, and signals per se.

Those of skill in the art will appreciate that information and signals may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the above description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof, in some cases depending in part on the particular application, in part on the desired design, in part on the corresponding technology, etc.

The various illustrative logical blocks, modules, and circuits described in connection with the aspects disclosed herein may be implemented or performed using hardware, software, firmware, middleware, microcode, hardware description languages, or any combination thereof, and can take any of a variety of form factors. When implemented in software, firmware, middleware, or microcode, the program code or code segments to perform the necessary tasks (e.g., a computer-program product) may be stored in a computer-readable or machine-readable medium. A processor(s) may perform the necessary tasks. Examples of form factors include laptops, smart phones, mobile phones, tablet devices

or other small form factor personal computers, personal digital assistants, rackmount devices, standalone devices, and so on. Functionality described herein also can be embodied in peripherals or add-in cards. Such functionality can also be implemented on a circuit board among different chips or different processes executing in a single device, by way of further example.

The instructions, media for conveying such instructions, computing resources for executing them, and other structures for supporting such computing resources are example means for providing the functions described in the disclosure.

The techniques described herein may also be implemented in electronic hardware, computer software, firmware, or any combination thereof. Such techniques may be implemented in any of a variety of devices such as general purposes computers, wireless communication device handsets, or integrated circuit devices having multiple uses including application in wireless communication device handsets and other devices. Any features described as modules or components may be implemented together in an integrated logic device or separately as discrete but interoperable logic devices. If implemented in software, the techniques may be realized at least in part by a computer-readable data storage medium comprising program code including instructions that, when executed, performs one or more of the methods, algorithms, and/or operations described above. The computer-readable data storage medium may form part of a computer program product, which may include packaging materials. The computer-readable medium may comprise memory or data storage media, such as random access memory (RAM) such as synchronous dynamic random access memory (SDRAM), read-only memory (ROM), non-volatile random access memory (NVRAM), electrically erasable programmable read-only memory (EEPROM), FLASH memory, magnetic or optical data storage media, and the like. The techniques additionally, or alternatively, may be realized at least in part by a computer-readable communication medium that carries or communicates program code in the form of instructions or data structures and that can be accessed, read, and/or executed by a computer, such as propagated signals or waves.

The program code may be executed by a processor, which may include one or more processors, such as one or more digital signal processors (DSPs), general purpose microprocessors, an application specific integrated circuits (ASICs), field programmable logic arrays (FPGAs), or other equivalent integrated or discrete logic circuitry. Such a processor may be configured to perform any of the techniques described in this disclosure. A general-purpose processor may be a microprocessor; but in the alternative, the processor may be any conventional processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices, e.g., a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration. Accordingly, the term "processor," as used herein may refer to any of the foregoing structure, any combination of the foregoing structure, or any other structure or apparatus suitable for implementation of the techniques described herein.

One of ordinary skill will appreciate that the less than (" $<$ ") and greater than (" $>$ ") symbols or terminology used herein can be replaced with less than or equal to (" $\leq$ ") and

greater than or equal to (“≥”) symbols, respectively, without departing from the scope of this description.

Where components are described as being “configured to” perform certain operations, such configuration can be accomplished, for example, by designing electronic circuits or other hardware to perform the operation, by programming programmable electronic circuits (e.g., microprocessors, or other suitable electronic circuits) to perform the operation, or any combination thereof.

The phrase “coupled to” or “communicatively coupled to” refers to any component that is physically connected to another component either directly or indirectly, and/or any component that is in communication with another component (e.g., connected to the other component over a wired or wireless connection, and/or other suitable communication interface) either directly or indirectly.

Claim language or other language reciting “at least one of” a set and/or “one or more” of a set indicates that one member of the set or multiple members of the set (in any combination) satisfy the claim. For example, claim language reciting “at least one of A and B” or “at least one of A or B” means A, B, or A and B. In another example, claim language reciting “at least one of A, B, and C” or “at least one of A, B, or C” means A, B, C, or A and B, or A and C, or B and C, or A and B and C. The language “at least one of” a set and/or “one or more” of a set does not limit the set to the items listed in the set. For example, claim language reciting “at least one of A and B” or “at least one of A or B” can mean A, B, or A and B, and can additionally include items not listed in the set of A and B.

Illustrative aspects of the disclosure include:

Aspect 1. A method of wireless communication performed at a first network device, the method comprising: determining, at the first network device, an estimated location of a second network device; comparing, at the first network device, the estimated location with an expected location for the second network device; determining, at the first network device, whether the second network device is a misbehaving device based on the comparing; and generating, at the first network device, a report based on the determining of whether the second device is a misbehaving device.

Aspect 2. The method of Aspect 1, further comprising receiving, at the first network device, a vehicle-based message from the second network device.

Aspect 3. The method of Aspect 2, wherein the vehicle-based message comprises the expected location for the second network device.

Aspect 4. The method of any of Aspects 2 or 3, wherein the vehicle-based message is one of a Basic Safety Message (BSM), a Cooperative Awareness Message (CAM), a Collective Perception Message (CPM), a Sensor Data Sharing Message (SDSM), or a Decentralized Environmental Message (DENM).

Aspect 5. The method of any of Aspects 1 to 4, further comprising transmitting, from the first network device, a response comprising the report to a third network device.

Aspect 6. The method of Aspect 5, wherein the response comprises at least one of a message timestamp field, a source identification (ID) field, an object containers field, a fused objects field, a malicious objects field, or a sensor objects field.

Aspect 7. The method of any of Aspects 5 or 6, wherein the second network device is at least one of not located within a field of view (FoV) of sensors of the third network device or not located within a line-of-sight (LoS) of the third network device.

Aspect 8. The method of any of Aspects 1 to 7, wherein the first network device and the second network device are capable of Vehicle-to-Everything (V2X) communications.

Aspect 9. The method of any of Aspects 1 to 8, wherein the first network device and the second network device are each one of an intelligent transportation system-station (ITS-S), a vehicle, a roadside unit (RSU), user equipment (UE), or a drone.

Aspect 10. The method of any of Aspects 1 to 9, further comprising receiving, at the first network device, a request for data sharing from a third network device.

Aspect 11. The method of Aspect 10, wherein the request for data sharing comprises at least one of a message timestamp field, a source field, or a malicious objects field.

Aspect 12. The method of any of Aspects 1 to 11, further comprising generating, at the first network device, a fused objects list, a sensor objects list, and a simulated objects list, wherein one of the fused objects list, the sensor objects list, or the simulated objects list comprises the second network device.

Aspect 13. The method of Aspect 12, wherein the fused objects list comprises network devices that transmitted vehicle-based messages and are located within at least one of a field of view (FoV) or line-of-sight (LoS) of a third network device.

Aspect 14. The method of any of Aspects 12 or 13, wherein the sensor objects list comprises network devices that have not yet transmitted any vehicle-based messages and are located within at least one of a field of view (FoV) or line-of-sight (LoS) of a third network device.

Aspect 15. The method of any of Aspects 12 to 14, wherein the simulated objects list comprises network devices that are ghost objects.

Aspect 16. A first network device for wireless communication, comprising: at least one memory; and at least one processor coupled to at least one memory and configured to: determine an estimated location of a second network device; compare the estimated location with an expected location for the second network device; determine whether the second network device is a misbehaving device based on the comparing; and generate a report based on the determining of whether the second device is a misbehaving device.

Aspect 17. The first network device of Aspect 16, wherein the at least one processor is configured to receive a vehicle-based message from the second network device.

Aspect 18. The first network device of Aspect 17, wherein the vehicle-based message comprises the expected location for the second network device.

Aspect 19. The first network device of any of Aspects 16 or 18, wherein the vehicle-based message is one of a Basic Safety Message (BSM), a Cooperative Awareness Message (CAM), a Collective Perception Message (CPM), a Sensor Data Sharing Message (SDSM), or a Decentralized Environmental Message (DENM).

Aspect 20. The first network device of any of Aspects 16 to 19, wherein the at least one processor is configured to output a response comprising the report for transmission to a third network device.

Aspect 21. The first network device of Aspect 20, wherein the response comprises at least one of a message timestamp field, a source identification (ID) field, an object containers field, a fused objects field, a malicious objects field, or a sensor objects field.

Aspect 22. The first network device of Aspect 20 or 21, wherein the second network device is at least one of not

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located within a field of view (FoV) of sensors of the third network device or not located within a line-of-sight (LoS) of the third network device.

Aspect 23. The first network device of any of Aspects 16 to 22, wherein the first network device and the second network device are capable of Vehicle-to-Everything (V2X) communications.

Aspect 24. The first network device of any of Aspects 16 to 23, wherein the first network device and the second network device are each one of an intelligent transportation system-station (ITS-S), a vehicle, a roadside unit (RSU), user equipment (UE), or a drone.

Aspect 25. The first network device of any of Aspects 16 to 24, wherein the at least one processor is configured to receive a request for data sharing from a third network device.

Aspect 26. The first network device of Aspect 25, wherein the request for data sharing comprises at least one of a message timestamp field, a source field, or a malicious objects field.

Aspect 27. The first network device of any of Aspects 16 to 26, wherein the at least one processor is configured to generate a fused objects list, a sensor objects list, and a simulated objects list, wherein one of the fused objects list, the sensor objects list, or the simulated objects list comprises the second network device.

Aspect 28. The first network device of Aspect 27, wherein the fused objects list comprises network devices that transmitted vehicle-based messages and are located within at least one of a field of view (FoV) or line-of-sight (LoS) of a third network device.

Aspect 29. The first network device of any of Aspects 27 or 28, wherein the sensor objects list comprises network devices that have not yet transmitted any vehicle-based messages and are located within at least one of a field of view (FoV) or line-of-sight (LoS) of a third network device.

Aspect 30. The first network device of any of Aspects 27 to 29, wherein the simulated objects list comprises network devices that are ghost objects.

Aspect 31. A non-transitory computer-readable medium having stored thereon instructions that, when executed by one or more processors, cause the one or more processors to perform operations according to any of Aspects 1 to 15.

Aspect 32. An apparatus for wireless communication, comprising: means for performing operations according to any of Aspects 1 to 15.

The previous description is provided to enable any person skilled in the art to practice the various aspects described herein. Various modifications to these aspects will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other aspects. Thus, the claims are not intended to be limited to the aspects shown herein, but is to be accorded the full scope consistent with the language claims, wherein reference to an element in the singular is not intended to mean "one and only one" unless specifically so stated, but rather "one or more."

What is claimed is:

1. A method of wireless communication performed at a first network device, the method comprising:
 - determining, at the first network device, an estimated location of a second network device;
 - comparing, at the first network device, the estimated location with an expected location for the second network device;
 - determining, at the first network device, whether the second network device is a misbehaving device based on the comparing;

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generating, at the first network device, a report based on the determining of whether the second device is a misbehaving device; and

transmitting, from the first network device, a response comprising the report to a third network device, wherein the second network device is at least one of not located within a field of view (FoV) of sensors of the third network device or not located within a line-of-sight (LoS) of the third network device.

2. The method of claim 1, further comprising receiving, at the first network device, a vehicle-based message from the second network device.

3. The method of claim 2, wherein the vehicle-based message comprises the expected location for the second network device.

4. The method of claim 2, wherein the vehicle-based message is one of a Basic Safety Message (BSM), a Cooperative Awareness Message (CAM), a Collective Perception Message (CPM), a Sensor Data Sharing Message (SDSM), or a Decentralized Environmental Message (DENM).

5. The method of claim 1, wherein the response comprises at least one of a message timestamp field, a source identification (ID) field, an object containers field, a fused objects field, a malicious objects field, or a sensor objects field.

6. The method of claim 1, wherein the first network device and the second network device are capable of Vehicle-to-Everything (V2X) communications.

7. The method of claim 1, wherein the first network device and the second network device are each one of an intelligent transportation system-station (ITS-S), a vehicle, a roadside unit (RSU), user equipment (UE), or a drone.

8. The method of claim 1, further comprising receiving, at the first network device, a request for data sharing from a third network device.

9. The method of claim 8, wherein the request for data sharing comprises at least one of a message timestamp field, a source field, or a malicious objects field.

10. The method of claim 1, further comprising generating, at the first network device, a fused objects list, a sensor objects list, and a simulated objects list, wherein one of the fused objects list, the sensor objects list, or the simulated objects list comprises the second network device.

11. The method of claim 10, wherein the fused objects list comprises network devices that transmitted vehicle-based messages and are located within at least one of a field of view (FoV) or line-of-sight (LoS) of a third network device.

12. The method of claim 10, wherein the sensor objects list comprises network devices that have not yet transmitted any vehicle-based messages and are located within at least one of a field of view (FoV) or line-of-sight (LoS) of a third network device.

13. The method of claim 10, wherein the simulated objects list comprises network devices that are ghost objects.

14. A first network device for wireless communication, comprising:

- at least one memory; and
- at least one processor coupled to at least one memory and configured to:
 - determine an estimated location of a second network device;
 - compare the estimated location with an expected location for the second network device;
 - determine whether the second network device is a misbehaving device based on the comparing;
 - generate a report based on the determining of whether the second device is a misbehaving device; and

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output a response comprising the report for transmission to a third network device, wherein the second network device is at least one of not located within a field of view (FoV) of sensors of the third network device or not located within a line-of-sight (LoS) of the third network device.

15. The first network device of claim 14, wherein the at least one processor is configured to receive a vehicle-based message from the second network device.

16. The first network device of claim 15, wherein the vehicle-based message comprises the expected location for the second network device.

17. The first network device of claim 15, wherein the vehicle-based message is one of a Basic Safety Message (BSM), a Cooperative Awareness Message (CAM), a Collective Perception Message (CPM), a Sensor Data Sharing Message (SDSM), or a Decentralized Environmental Message (DENM).

18. The first network device of claim 14, wherein the response comprises at least one of a message timestamp field, a source identification (ID) field, an object containers field, a fused objects field, a malicious objects field, or a sensor objects field.

19. The first network device of claim 14, wherein the first network device and the second network device are capable of Vehicle-to-Everything (V2X) communications.

20. The first network device of claim 14, wherein the first network device and the second network device are each one

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of an intelligent transportation system-station (ITS-S), a vehicle, a roadside unit (RSU), user equipment (UE), or a drone.

21. The first network device of claim 14, wherein the at least one processor is configured to receive a request for data sharing from a third network device.

22. The first network device of claim 21, wherein the request for data sharing comprises at least one of a message timestamp field, a source field, or a malicious objects field.

23. The first network device of claim 14, wherein the at least one processor is configured to generate a fused objects list, a sensor objects list, and a simulated objects list, wherein one of the fused objects list, the sensor objects list, or the simulated objects list comprises the second network device.

24. The first network device of claim 23, wherein the fused objects list comprises network devices that transmitted vehicle-based messages and are located within at least one of a field of view (FoV) or line-of-sight (LoS) of a third network device.

25. The first network device of claim 23, wherein the sensor objects list comprises network devices that have not yet transmitted any vehicle-based messages and are located within at least one of a field of view (FoV) or line-of-sight (LoS) of a third network device.

26. The first network device of claim 23, wherein the simulated objects list comprises network devices that are ghost objects.

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